

Structure Theory of Finite 677 Magmas

A working paper: verified by-products of an unfinished campaign

AutoMath Collaboration

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Abstract

Let $E677$ denote the law $x = y * (x * ((y * x) * y))$ and $E255$ the law $x = ((x * x) * x) * x$. Whether every *finite* $E677$ magma satisfies $E255$ is the last open finite implication of the Equational Theories Project. We do not settle it. We do develop the structure theory of finite $E677$ magmas far enough to record a coherent body of results, all of them either proved here or machine-verified, and all of them by-products of an ongoing automated campaign. We prove a pointwise criterion identifying $E255$ at y with the existence of a left unit for y , and derive from it a cluster of equivalent witness criteria, a cycle-length obstruction $m(y) \notin \{2, 3\}$, a row-code minimum-distance bound $\geq \lceil n/2 \rceil$, and the unconditional counting identity $\sum_{y,w} |\text{Fix}(L_y R_w)| = n^2$. We classify the affine models with commuting coefficients completely: for F, G commuting, $x * y = Fx + Gy + c$ on an abelian group A satisfies $E677$ iff $F(G + G^3) = 1$, $P(G) = \Phi_{10}(G) \cdot (G^4 + G^3 + 2G^2 + 2G + 1) = 0$ and $(G^2F + G^2 + G + 1)c = 0$, so the affine orders realizable with commuting coefficients are exactly the orders of finite $\mathbb{Z}[u]/(P(u))$ -modules; the Φ_{10} branch consists precisely of the Alexander quandles. (For general F, G the criterion is a pair of noncommutative equations, and whether every affine model has commuting coefficients is open.) In particular 2 is inert of residue degree 4 in $\mathbb{Z}[\zeta_5]$, whence such an affine model of even order forces $16 \mid n$, and 16 is attained by an $\mathbb{F}_{16}$ quandle. This refutes an odd-order conjecture that an earlier round of the campaign had proposed: since models of even order exist, no argument can conclude that n is odd,

***Disclosure of the production pipeline.** This paper reports by-products of an autonomous, still-running attack on the last open finite implication of the Equational Theories Project, namely whether every finite magma satisfying $E677$ satisfies $E255$. *That question is not resolved here, and this paper does not claim to resolve it.* Every mathematical statement below was produced by automated systems: an orchestrating agent (Claude Fable 5) that designed the attack families, audited the returning reports, and drafted this document; subordinate exploration agents (Claude Fable 5 instances, OpenAI GPT-5.6 driven through the `codex` command-line interface, and GPT-5.6 at “Pro” effort through the web interface) that carried out the individual lines of attack; and machine checkers (`kissat`, purpose-written C and Python enumerators, and independent from-scratch Python re-verifiers). No human supplied a definition, a lemma, a proof idea, or a correction. Every numbered *assertion* — theorem, lemma, proposition, corollary — carries an explicit verification marker; numbered definitions, problems and the single conjecture assert nothing and carry none. **[machine-verified: path]** means a script at that path in the campaign repository checks the statement (usually on explicit models, exhaustively over the stated finite range); **[proved in X , root-verified]** means a human-readable proof exists in the cited internal report and the orchestrating agent independently re-derived or spot-checked it; **[computational, partial: ...]** means an incomplete search whose exact coverage is stated and which must *not* be read as a classification; and **[computational: ...]** means a completed computation of exactly the stated scope, which — when the computation is a *failed* search — is evidence and not a proof of impossibility. Several of the campaign’s own working conjectures were refuted by its own later rounds (Sections 3 and 4); those self-refutations are recorded rather than quietly deleted. Responsibility for what follows rests with the reader who re-runs the scripts.

which removes the parity, sign and 2-adic routes the campaign had been pursuing. We then exhibit and machine-verify a finite $E677$ magma of order $176 = 11 \times 16$ which is *not* right-cancellative; it is smaller than the order-496 example of the project blueprint, which is the only such magma our literature reconnaissance could find recorded anywhere. It satisfies $E255$, so it is not a counterexample; what it does refute is a family of proposed proof routes, including every constraint of the form $N(t, v) + N(v, t) \geq 1$ off the diagonal and the uniform isoperimetric bound $e(O) \leq |O||O^c|$ on which the averaging route of §5.7 rests. Finally we map the obstructions. Equational and quasi-equational derivations of $E255$ from $E677$ plus left cancellation resisted every saturation experiment we could run — which is computational evidence, not a proof of underderivability; the natural state map is provably circular; local amplification of the row-distance bound is capped at $\lfloor n/2 \rfloor$; the transport tree has branching factor 1; affine fibre extensions are forced to satisfy $E255$ by a $\{1, 3\}$ subscript collision; and idempotent bases in the extension framework are circular, leaving exactly one habitat for a counterexample built that way. We give the complete classification of $E677$ magmas of order ≤ 9 (orders 2, 3, 4, 6, 8 are empty; orders 5, 7, 9 carry only affine quasigroups), a deliberately partial order-11 census, and a list of open problems and closed routes. A final section reports a sixth round: the general-magma analogue of the $\{1, 3\}$ collision is identified and turns out to be the problem itself one level down; two $E677$ magmas of order 77 are constructed — the first non-idempotent non-right-cancellative example, certified in LEAN 4, and the first extension that is not translation-invariant; a new counting identity gives the upper-bound criterion $\sum_x N(x \setminus x, x) \leq \sum_x N(x, x) \Rightarrow E255$, which no term witness can prove; every idempotent-free *affine* model is shown to have order in $31 \cdot S$, where $31 = \text{Res}(\Phi_{10}, Q)$; and a conjecture of that same round is refuted twice over, which exposes a systematic bias in the campaign’s own benchmark set that we document as a result in its own right. Two further sections report Rounds 7 and 8: the local configuration that decides $E255$ is shown to be the same over every base (idempotent levels are circular, non-idempotent levels carry a three-instance pattern), a cocycle theorem protects the entire separable family, and two structure theorems say that a minimal counterexample must be simple and can have no idempotent quotient. Congruences are then identified with block systems of the left-multiplication group satisfying a kernel condition, which reduces “simple $\Rightarrow$ right-cancellative” to three exhaustively verified finite statements, one of which — a transport law — would close the last surviving construction route. None of the three is first-order derivable, and we prove two independent barriers explaining why: one for existential statements and one, new, for injectivity statements. A final section refutes that programme. The object its own failure analysis had specified — a magma whose fibre operation degenerates at one fibre point and is injective at another — is constructed, twice and independently, at orders 77 and 385; it satisfies $E677$, $E255$ and every genuine first-order consequence of $E677$, and it refutes the transport law and six further statements, including two we discovered in the course of writing this. We carry out the retraction in full, marking rather than deleting the voided results, since the record of which hypotheses died — all of them supported by a model set whose members shared one accidental property — is the most transferable thing the campaign produced.

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1 Introduction

1.1 The problem

The Equational Theories Project [1] determined, for a list of 4694 magma laws, all 4694^2 implications between them, in the process producing a very large formalized corpus. A single *finite* implication survived the effort. Writing $*$ for the magma operation, the two laws in question are

$$\begin{aligned}
 (E677) \quad & x = y * (x * ((y * x) * y)), \\
 (E255) \quad & x = ((x * x) * x) * x.
 \end{aligned}$$

$E677$ does not imply $E255$ in general: the free one-generated 677 magma violates $E255$, and there is an infinite greedy counterexample [2, Ch. 13]. The question is whether the implication holds when the carrier is finite. The project’s own paper tentatively conjectures that it does *not* [1], and the DeepMind `formal-conjectures` corpus [3] carries the same expectation as a marginal note; but no finite counterexample is known, and every concrete finite model on record satisfies $E255$ [4].

Problem 1.1 (the open implication). Does every finite magma satisfying $E677$ satisfy $E255$?

This paper does not answer Problem 1.1. What it does is report the part of the structure theory that has been established and independently checked in the course of an automated attack on it, together with an explicit map of which attack routes are now closed — some by proof, some by an explicit counterexample, and some only in the weaker sense that they were exhausted under a stated computational budget. The three strengths are kept apart throughout and never conflated. Negative results are a substantial fraction of the content, and they are stated as results rather than as apologies: several of them refute proposals that looked, from inside the campaign, like the natural next step.

1.2 Why the finite case is delicate

The single structural consequence of $E677$ is that every left translation is surjective: $E677$ exhibits $x * ((y * x) * y)$ as an L_y -preimage of x . On a finite carrier surjectivity upgrades to bijectivity, so a finite $E677$ magma is a *left quasigroup*, and this — as far as anything we have found goes — is the only thing finiteness gives directly. Everything in this paper is a consequence of $E677$ together with the injectivity of every L_y .

The delicacy is that the corresponding right-hand statement is exactly what is missing. Theorem 2.7 below shows that $E255$ at y is equivalent to y possessing a *left unit*, i.e. to the value y appearing in column y of the Cayley table; and the sets $\text{Fix}(L_w)$ are always pairwise disjoint (Lemma 2.6), so $\sum_w |\text{Fix}(L_w)| \leq n$ always, with $E255$ equivalent to equality. The whole problem is therefore a covering statement: n pairwise disjoint sets, and one must show they exhaust. By contrast the mirror statement — that the sets $\text{Fix}(R_u)$ partition M — is unconditionally true (Proposition 2.11). That asymmetry is the problem.

1.3 What is proved here

1. **Section 2: the basic lemma package.** The KEY identity $(y * x) * y = x \setminus (y \setminus x)$, which is equivalent to $E677$ over a left quasigroup and is the working form throughout; the left-unit criterion (Theorem 2.7), which coincides with [2, Lemma 13.1(ii)] and was rediscovered independently by three of the campaign’s attack families; a cluster of equivalent witness criteria with unique candidate witnesses (Theorem 2.14); the cycle-length obstruction $m(y) \notin \{2, 3\}$ (Theorem 2.15); the row-code bound $|\{x : p * x = q * x\}| \leq \lfloor n/2 \rfloor$ (Theorem 2.17); the counting identity $\sum_{y,w} |\text{Fix}(L_y R_w)| = n^2$ (Theorem 2.20); and the two single-variable bijectivity reductions (Proposition 2.21). We also record two genuine linking constraints on the multiplicity matrix N : its support digraph is strongly connected (Theorem 2.22) and satisfies an edge-expansion bound (Theorem 2.23).
2. **Section 3: the affine criterion and the commuting spectrum.** The general criterion, a pair of noncommutative equations in $F, G \in \text{End}(A)$, and its commuting form $F(G +$

$G^3) = 1$, $P(G) = \Phi_{10}(G)Q(G) = 0$ with $Q(u) = u^4 + u^3 + 2u^2 + 2u + 1$ (Theorem 3.1); the module-theoretic spectrum of the commuting family (Theorem 3.2); the identification of the Φ_{10} branch with the Alexander quandles (Proposition 3.3); the *even-order spectrum theorem* (Theorem 3.4): a commuting affine model of even order forces $16 \mid n$, because 2 is inert of residue degree 4 in $\mathbb{Q}(\zeta_5)$, and 16 is realized by an $\mathbb{F}_{16}$ quandle. Whether an affine $E677$ magma can have F, G non-commuting is **open** (Problem 11.3), so “complete classification” below always means *complete within the commuting/ R -module family*. This refutes the campaign’s own working parity conjecture; since $E677$ magmas of even order exist, no parity, sign, 2-adic or Sylow-2 argument can conclude that n is odd (Section 5.6). We also record that all commuting-coefficient affine models are medial, and that affine models and *affine* fibre extensions are forced to satisfy $E255$.

3. **Section 4: an order-176 model.** A finite $E677$ magma on $\mathbb{F}_{11} \times \mathbb{F}_{16}$ that is not right-cancellative (Theorem 4.1), verified by an independent from-scratch re-implementation. At order 176 it is smaller than the order-496 example of [2, §13.1], which was the only non-right-cancellative finite $E677$ magma previously recorded anywhere (Section 7 brings this down to 77). It satisfies $E255$. As a witness it refutes (Corollary 4.2) the statement (S-off): $N(t, v) + N(v, t) \geq 1$ for $t \neq v$, and — for every size class $2 \leq s \leq n - 2$ — the specific uniform isoperimetric upper bound $e(O) \leq |O||O^c|$ on which the R3-C averaging route rests (§5.7). Section 4.3 gives the structure theory of the translation-invariant extension family with *affine fibres* in which it lives, including the exact criterion for the failure, and the exact reason (Theorem 4.8) why the same mechanism can never produce a failure *on the diagonal*, i.e. can never break $E255$, when the fibre operations are affine. General (non-affine) fibre tables are not covered and remain open.
4. **Section 5: the obstruction map.** Seven routes are mapped, each with its exact strength: equational/quasi-equational derivation (*computational evidence only* — bounded saturation did not collapse, which does not prove underivability); surjectivity of the state map (provably circular); pure marginal counting on N (explicit certificate); local amplification of the row-distance bound (exhausted for words of length ≤ 3); the transport tree and the pair rotation (proved); parity and sign (refuted by an explicit model); and the off-diagonal/isoperimetric programme in the form in which it was proposed (refuted by an explicit model). In the extension framework we record the *idempotent-base circularity*, which leaves the exceptional base $x*y = 5x - 4y + c$ on $\mathbb{F}_{31}$ as the only translation-invariant habitat for a counterexample built that way.
5. **Section 6: the small-order landscape.** A complete classification for $n \leq 9$ (Theorem 6.1), due to a single implementation and corroborated over smaller ranges by three others (§6.1): orders 2, 3, 4, 6, 8 carry no $E677$ magma at all; orders 1, 5, 7, 9 carry only affine quasigroups, and hence only models satisfying $E255$. A deliberately partial order-11 census (22 of 87 pointed cycle types, all UNSAT; four affine models found, the exclusion of a fifth unfinished) and the near-miss data.
6. **Section 7: Round 6.** The general-magma analogue of the $\{1, 3\}$ collision, identified exactly — it is Problem 1.1 one level down (Corollary 7.8); a new family of counting identities yielding the *upper-bound* criterion $\sum_x N(x \setminus x, x) \leq \sum_x N(x, x) \Rightarrow E255$ (Corollary 7.14), together with a proof that no term witness for it exists (Theorem 7.15); two $E677$ magmas of order 77 (Theorems 7.18, 7.20) — the first non-idempotent non-right-cancellative

example, certified in LEAN 4, and the first extension that is not translation-invariant; the refutation, twice over, of the round’s own conjecture that $e(x) = x \setminus x$ is an automorphism (§7.4); and the spectrum theorem $S_0^{\text{aff}} = 31 \cdot S$ for idempotent-free affine magmas (Theorem 7.34), resting on $\text{Res}(\Phi_{10}, Q) = 31$. The round also produced a methodological finding, §7.6: four separate conjectures in this campaign were supported only by an artefact of the model set they were tested on.

7. **Section 8: Round 7.** The occurrence trichotomy (Theorem 8.1): at an idempotent level the extension equation collapses four-fold and is circular, at every non-idempotent level the $E255$ -deciding operation occurs in exactly three instances with a $\{1, 3\}$ collision — and this is *independent of the base*, so changing the base cannot help. On the last surviving base, $\mathbb{F}_7(4x+3y)$: a strengthening of [2, Lemma 13.4] needing only *one* separable operation over *any* group (Corollary 8.6), a rigidity theorem needing only three affine neighbours (Theorem 8.8), and the proof that the $\{1, 3\}$ collision is *not* a protection there. A cocycle protection theorem (Theorem 8.9) kills the entire separable family. Two structure theorems about minimal counterexamples: every quotient is a blueprint extension, so “a minimal counterexample is simple” is equivalent to the extension statement (Theorem 8.12); and no minimal counterexample has an idempotent quotient (Theorem 8.13).
8. **Section 9: Round 8.** A dictionary putting congruences of an $E677$ magma inside permutation group theory — θ is a congruence iff it is an $\langle L \rangle$ -block system satisfying a kernel condition, whence $\langle L \rangle$ primitive $\Rightarrow M$ simple (Theorem 9.1). An exact reduction of “simple $\Rightarrow$ right-cancellative” to three finite statements, of which one is a *transport law* (T) verified on 17.3 million instances (Theorem 9.7); conditionally on it, a dichotomy (Theorem 9.12) that appeared to kill the last surviving base of Round 7. And two independent proofs that the remaining statements are out of reach of first-order reasoning (§9.7) — the second of which shows that even plain *injectivity* statements about the collapse structure require finiteness.
9. **Section 10: Round 9, and a retraction.** The object those blueprints specified exists: an order-77 magma M_{77}^D with a non-affine fibre operation whose degeneracy depends on the fibre point (Theorem 10.5). It refutes (T), (T*), (W), (D), (A), (S’), both falsification filters, the closure of Idem, and “ θ_κ is a congruence” — while satisfying $E677$, $E255$ and every genuine first-order consequence of $E677$. A second, independent object of order 385 separates four strata (Theorem 10.8) and shows the filters have a blind spot and that (A) and (T) are decoupled. §10.4 carries out the resulting retraction chain in full — nothing is deleted, everything is marked — and §10.5 states the corrected skeleton $(P) \Leftarrow [(NTS)] + [(B)] + [(Prop)] + [(Q) \Rightarrow E255]$, in which (B) has newly lost its proof. The positive residue is the machinery (Theorems 10.1–10.3) that makes non-affine models constructible at will.

1.4 Verification conventions

Because this is an automated campaign’s output, the epistemic status of each statement is recorded explicitly and is not uniform. Four markers are used.

- [machine-verified: path] — a script at that path in the campaign repository checks the statement. For statements about specific models this is exhaustive over the model; for statements about ranges it is exhaustive over the stated range and over nothing else.

- [proved in report, root-verified] — a human-readable proof appears in the cited internal report, and the orchestrating agent re-derived or spot-checked it independently. Where the proof is short it is reproduced here in full.
- [computational, partial: scope] — an *incomplete* computation. The stated scope is the exact coverage. Such a statement is never a classification, and we flag the two places (the order-11 census in §6.3, and the minimality range of the order-176 model in §4.5) where it would be easiest, and most wrong, to read one as if it were.
- [computational: scope] — a completed computation of the stated scope which is not a re-runnable single script in the repository (typically a solver run whose log is archived). It is evidence of exactly the stated scope and of nothing beyond it; in particular a *failed* bounded search is never a proof of impossibility, and where one appears (Proposition 5.1) the text says so explicitly.

Markers are attached to every numbered assertion; definitions, open problems and the single conjecture carry none, since they assert nothing. Where a marker names a script, that script checks the statement it is attached to and *only* that statement: after the round-1 adversarial review, three markers whose scripts did not in fact compute the quantities claimed were replaced by new verifiers written for the purpose (`isoperim_check.py`, `struct_check.py`, `term_full_check.py`).

Where a statement is due to the project blueprint [2] rather than to this campaign we say so at the statement. Several results below were obtained independently here and only afterwards matched against the blueprint; we record both, since the independent derivations are what the campaign actually used, and the coincidence is evidence that the transcription is faithful.

2 Preliminaries: the basic lemma package

2.1 Notation

Throughout, $(M, *)$ is a finite magma of order $n = |M|$ satisfying E677, unless the text says otherwise. We write

$$L_y(z) := y * z, \quad R_y(z) := z * y, \quad S(y) := y * y, \quad \delta(y) := (y * y) * y.$$

Lemma 2.1 (left quasigroup; known, [2, Lemma 13.1(i)]). *[proved in [2, Lemma 13.1(i)]; proof reproduced below, root-verified] In any magma satisfying E677, every L_y is surjective, with $L_y^{-1}(x) = x * ((y * x) * y)$. If M is finite, every L_y is a bijection.*

Proof. E677 reads $L_y(x * ((y * x) * y)) = x$, which exhibits a preimage of an arbitrary x ; finiteness turns surjectivity into bijectivity, and then the displayed formula computes the inverse. $\square$

Accordingly we write $\Lambda_y := L_y^{-1}$ and $y \setminus z := \Lambda_y(z)$, so that $y * (y \setminus z) = z$. We also write

$$y_k := \Lambda_y^k(y) \quad (k \in \mathbb{Z}), \quad m(y) := \min\{k > 0 : y_k = y\},$$

so $L_y(y_k) = y_{k-1}$, $y * y = y_{-1}$, and $m(y)$ is the length of the L_y -cycle through y . Finally $e(y) := y_1 = \Lambda_y(y)$ is the unique *right* unit of y (indeed $y * e(y) = y$, and $e(y)$ is unique because L_y is injective).

Finiteness enters this paper only through the injectivity of the maps L_y . Every statement below that does not explicitly invoke a counting argument therefore holds in every E677 left quasigroup, finite or not; we flag this where it matters, because it delimits exactly what a proof of Problem 1.1 may not do.

2.2 The KEY identity

Lemma 2.2 (KEY). *[proved in F5, §1 and R2, root-verified] In a left quasigroup, E677 is equivalent to*

$$(y * x) * y = x \backslash (y \backslash x) = \Lambda_x \Lambda_y(x) \quad \text{for all } x, y. \quad (\text{KEY})$$

*Equivalently, writing $\Theta_t := R_t \circ L_t$, one has $\Theta_t(x) = (t * x) * t = \Lambda_x(\Lambda_t(x))$.*

Proof. E677 says $L_y L_x((y * x) * y) = x$; apply Λ_y and then Λ_x . Every step is reversible in a left quasigroup. $\square$

KEY is the working form used throughout. It makes visible that E677 carries exactly n^2 bits of information and that the right translation R is a *derived* operation, determined by the left division:

Lemma 2.3 (R is determined by $\backslash$). *[proved in F1, Lemma B', root-verified] For all v, y ,*

$$v * y = \Lambda_{\Lambda_y(v)}(\Lambda_y^2(v)) = (y \backslash v) \backslash (y \backslash (y \backslash v)). \quad (2.1)$$

Consequently a finite E677 magma is completely determined by its left-division left quasigroup $(M, \backslash)$, and E677 is equivalent to the single law $u \backslash ((y \backslash u) \backslash (y \backslash (y \backslash u))) = y$ in $(M, \backslash)$.

Proof. Substituting $x := y \backslash u$ into KEY (so that $y * x = u$) gives the *master form*

$$(y \backslash u) * (u * y) = \Lambda_y^2(u) \quad \text{for all } u, y, \quad (2.2)$$

i.e. $L_{\Lambda_y(u)}(R_y(u)) = \Lambda_y^2(u)$. Solving (2.2) for $u * y$ using injectivity of $L_{\Lambda_y(u)}$ gives (2.1). $\square$

Identity (2.2) is used so often below — it is the source of every “transport” argument in the campaign — that we give it a name: the *master identity*. In the form

$$(\Lambda_t x) * (x * t) = \Lambda_t^2(x) \quad (2.3)$$

it is the *transport identity*; (2.3) is equivalent to KEY, hence to E677.

Lemma 2.4 (diagonal). *[proved in F1 Lemma D; F4 (4a); F5 Lemma B; R3-B L-a, root-verified] $\delta(y) = (y * y) * y = \Lambda_y^2(y) = y_2$, equivalently $L_y^2(\delta(y)) = y$.*

Proof. E677 at $x = y$ reads $L_y^2((y * y) * y) = y$. $\square$

So the first three points of the R_y -orbit of y are y_0, y_{-1}, y_2 , and E255 at y is precisely $R_y^3(y) = y$, i.e. $R_y(y_2) = y$.

2.3 Left units: the pointwise criterion

Lemma 2.5. *[proved in F1 Lemma F, root-verified] If $y * v = v$ then $v * y = e(v)$; in particular $v * y$ does not depend on y .*

Proof. $y * v = v$ says $\Lambda_y(v) = v$, hence also $\Lambda_y^2(v) = v$. Then (2.1) gives $v * y = \Lambda_{\Lambda_y(v)}(\Lambda_y^2(v)) = \Lambda_v(v) = e(v)$. $\square$

Lemma 2.6 (uniqueness of left units). *[proved in F1 Lemma G; F5 L2; R3-B L-b, root-verified] For every $v \in M$ there is at most one y with $y * v = v$. Equivalently, the fixed-point sets $\text{Fix}(L_w)$, $w \in M$, are pairwise disjoint.*

Proof. By Lemma 2.5 every such y satisfies $L_v(y) = e(v)$, and L_v is injective. $\square$

Theorem 2.7 (left-unit criterion). *[proved in F1 Thm 1; F4 Lemma 2; F5 T1; blueprint Lemma 13.1(ii), root-verified] Let M be a finite E677 magma and $y \in M$. The following are equivalent.*

1. $((y * y) * y) * y = y$, i.e. E255 holds at y ;
2. there exists w with $w * y = y$ (“ y has a left unit”);
3. $y \in \text{Im}(R_y)$, i.e. the value y occurs in column y of the Cayley table.

Moreover the w of (2) is unique and equals $\delta(y) = (y * y) * y$.

Proof. (1) $\Rightarrow$ (2): take $w = \delta(y)$. (2) $\Rightarrow$ (1): given $w * y = y$, Lemma 2.5 gives $y * w = e(y) = \Lambda_y(y)$; but also $y * \delta(y) = y * \Lambda_y^2(y) = \Lambda_y(y) = e(y)$ by Lemma 2.4. Injectivity of L_y forces $w = \delta(y)$, whence $\delta(y) * y = y$, which is E255 at y . (2) $\Leftrightarrow$ (3) is a restatement. $\square$

Remark. Theorem 2.7 was obtained independently by three of the campaign’s attack families before the blueprint was consulted; it is [2, Lemma 13.1(ii)]. Its proof uses finiteness *only* to upgrade surjectivity of L_y to injectivity, so it holds in every E677 left quasigroup, infinite ones included. This locates the entire finite/infinite gap precisely:

$$y \notin \text{Im}(R_y) \implies R_y \text{ not surjective} \implies \text{finite } R_y \text{ not injective} \implies \exists a \neq b : a * y = b * y.$$

So a finite counterexample must have two distinct rows of its Cayley table agreeing in some column, and Lemma 2.6 says they may not agree at a diagonal position. Only the off-diagonal case is open.

Corollary 2.8. *[proved in F1 Cor 1.1; F5 T2, root-verified] Every finite E677 magma which is a quasigroup — equivalently, whose Cayley table is a Latin square, equivalently for which every R_y is injective — satisfies E255.*

Corollary 2.9 (partition criterion). *[proved in F1 Cor 1.3; F5 T1, root-verified] $\#\{y : \text{E255 holds at } y\} = \sum_{w \in M} |\text{Fix}(L_w)| \leq n$, with equality iff E255 holds. Thus E255 holds iff the sets $\text{Fix}(L_w)$ partition M . In particular, if every L_w has a fixed point then E255 holds.*

Corollary 2.10 (dichotomy). *[proved in F1 Cor 1.2, root-verified] For every y : either $R_y^3(y) = y$, so y lies on an R_y -cycle of length dividing 3, or y has no R_y -preimage at all. There is no intermediate behaviour — y can never sit on a longer R_y -cycle, nor on a tail at positive distance from a cycle.*

Proposition 2.11 (the right-hand dual, unconditional). *[proved in F1 Prop 2, root-verified]* $\text{Fix}(R_u) = e^{-1}(u)$, because $x * u = x \iff L_x(u) = x \iff u = \Lambda_x(x) = e(x)$. Hence the sets $\text{Fix}(R_u)$, $u \in M$, always partition M , and $\sum_u |\text{Fix}(R_u)| = n$ exactly.

The contrast between Corollary 2.9 and Proposition 2.11 is the whole problem: the right translations' fixed-point sets always partition M ; the left translations' fixed-point sets are always disjoint, and E255 says exactly that they also cover.

2.4 The reformulations (P) and (Q)

Definition 2.12. For finite E677 magmas consider

- (P) every y satisfies $y \in \text{Im}(R_y)$ (equivalently: every element has a left unit);
- (Q) every R_t is injective: $a * t = b * t \implies a = b$.

By Theorem 2.7, (P) is Problem 1.1: it is a reformulation, not a strengthening. By Corollary 2.8, (Q) $\implies$ (P). During Rounds 1–3 the campaign pursued (Q) as the target, on the evidence that every model then known was a quasigroup. That was a mistake, and it is worth recording as such:

Proposition 2.13 ((Q) is false). *[machine-verified: problems/etp677/R5C_scripts/verify176.py (order 176), m496.py (order 496)]* (Q) fails: there are finite E677 magmas that are not right-cancellative. The blueprint's order-496 example [2, §13.1] is one; the order-176 magma of Theorem 4.1 is a smaller one. Both satisfy E255.

So (Q) is strictly stronger than (P) and is false, while (P) remains open. Any future work must target (P), or one of the seven equivalent pointwise conditions of [2, Lemma 13.2], and not (Q).

2.5 Witness criteria with unique candidate witnesses

Theorem 2.14 (E255 criteria). *[proved in F1 Thm 3, root-verified]* For y in a finite E677 magma the following are equivalent, and in each case the witness is unique and is the displayed element.

1. E255 holds at y ;
2. $\exists w : w * y = y \setminus w$, i.e. $\text{Fix}(L_y R_y) \neq \emptyset$ — the only possible w is $y_3 = \Lambda_y^3(y)$;
3. $\exists c : c * c = y \setminus c$ — the only possible c is $y_4 = \Lambda_y^4(y)$;
4. $\exists c : y * (c * c) = c$, i.e. $c \mapsto y * (c * c)$ has a fixed point;
5. $y_4 * y_4 = y_5$.

Proof. (1) $\Leftrightarrow$ (2): if $v * y = y$, put $w := \Lambda_y(v)$; (2.2) gives $w * (v * y) = \Lambda_y(w)$, i.e. $w * y = \Lambda_y(w)$. Conversely from $w * y = \Lambda_y(w)$ put $v := L_y(w)$; (2.2) gives $w * (v * y) = \Lambda_y(w) = w * y$, so $v * y = y$ by injectivity of L_w . Uniqueness of $v = \delta(y) = y_2$ (Theorem 2.7) gives $w = \Lambda_y(y_2) = y_3$. (2) $\Leftrightarrow$ (3): by (2.1), $w * y = \Lambda_{\Lambda_y(w)}(\Lambda_y^2(w))$; with $c := \Lambda_y(w)$ the condition $w * y = \Lambda_y(w)$ becomes $\Lambda_c(\Lambda_y(c)) = c$, i.e. $c * c = \Lambda_y(c)$; and $c = \Lambda_y(y_3) = y_4$. (3) $\Leftrightarrow$ (4) is $\Lambda_y(c) = c * c \iff y * (c * c) = c$. (3) $\Leftrightarrow$ (5) since $\Lambda_y(y_4) = y_5$. $\square$

Criterion (5) is worth isolating: *E255 at y is equivalent to a single equation about the fourth and fifth iterates of Λ_y at y .* Criterion (3) is the global version: *E255 holds iff, for every y , the squaring map $S(c) = c * c$ agrees with the permutation Λ_y somewhere.* Equivalently, in a counterexample the squaring map must avoid the entire graph of Λ_y — n simultaneous disequalities, and the sharpest single pointwise obstruction available.

Theorem 2.15 (cycle-length obstruction). *[proved in F1 Thm 2; independent proof R3-B L-c, root-verified] For every y in a finite E677 magma:*

- (a) $m(y) \notin \{2, 3\}$;
- (b) if E255 holds at y then $m(y) \notin \{2, 3, 4\}$; equivalently, $m(y) = 4$ forces E255 to fail at y ;
- (c) $m(y) = 1$ (i.e. y idempotent) implies E255 at y .

Proof. (c) $y * y = y$ gives $((y * y) * y) * y = y$.

(a) If $m(y) = 2$: write $y * y = a \neq y$, $y * a = y$, so $y_2 = y$. Lemma 2.4 gives $(y * y) * y = y_2 = y$, i.e. $a * y = y$, so a is a left unit of y ; by Theorem 2.7, $a = \delta(y) = y_2 = y$, a contradiction.

If $m(y) = 3$: write $y * y = a$, $y * a = b$, $y * b = y$ with y, a, b distinct; then $y_1 = b$ and $y_2 = a$. Lemma 2.4 gives $a * y = a$. Applying (2.2) with $u := y_1$, $y := y$ gives $y_1 * (y * y) = y_2$, i.e. $b * a = a$. Now KEY with $y := a$, $x := y$ reads $y * ((a * y) * a) = \Lambda_a(y)$; since $a * y = a$ this is $y * (a * a) = \Lambda_a(y)$. From $b * a = a$ and Theorem 2.7, b is the left unit of a , so $a * b = e(a)$ by Lemma 2.5; independently $a * y = a$ says $y = e(a)$, so $a * b = y$, i.e. $\Lambda_a(y) = b$. Hence $y * (a * a) = b = y * a$, so $a * a = a$ by injectivity of L_y . But then $e(a) = a$, contradicting $e(a) = y \neq a$.

(b) Assume E255 at y , i.e. $y_2 * y = y$. Applying (2.2) with $u := y_3$, $y := y$ gives $y_3 * (y_2 * y) = y_4$, hence $y_3 * y = y_4$. If $m \mid 4$ then $y_4 = y$, so $y_3 * y = y$, making y_3 a left unit of y ; Theorem 2.7 forces $y_3 = \delta(y) = y_2$, i.e. $m \mid 1$, so $m = 1$. If $m \mid 3$ then $y_{-1} = y_2$, and Lemma 2.4 ($y_{-1} * y = y_2$) reads $y_2 * y = y_2$; with $y_2 * y = y$ this gives $y_2 = y$, i.e. $m \mid 2$, so together $m = 1$. $\square$

Remark. An earlier report [8, L3] recorded the weaker statement “ $m(y) \in \{2, 3\} \Rightarrow$ E255 fails at y ”. Theorem 2.15(a) shows those cycle lengths do not occur at all, so that statement, while true, is vacuous. Empirically the observed values of m across all models the campaign enumerated are 1, 6, 7, 8, 9, 12, 15, 18, 36, 42; $m \in \{4, 5\}$ has never been observed, and is open. [computational: F1 §5, R3-B mvals.py]

2.6 The row code and the multiplicity matrix

Lemma 2.16. *[proved in F5 L1, L4, root-verified] In a finite E677 magma:*

- (L1) $z \mapsto L_z$ is injective; hence M embeds in $\text{Sym}(M)$ as a set Λ of n distinct permutations.
- (L4) $p * x = q * x$ and $p \setminus x = q \setminus x$ together imply $p = q$.

Proof. (L4): KEY at (p, x) and (q, x) gives, with $u := p * x = q * x$, $u * p = \Lambda_x(\Lambda_p(x))$ and $u * q = \Lambda_x(\Lambda_q(x))$. If $\Lambda_p(x) = \Lambda_q(x)$ the right sides agree, so $L_u(p) = L_u(q)$ and $p = q$.

(L1): assume $L_p = L_q$ and put $u := p * x = q * x$ for arbitrary x . E677 at (p, x) and at (q, x) gives, after applying $\Lambda_p = \Lambda_q$, that $x * (u * p) = p \setminus x$ and $x * (u * q) = q \setminus x = p \setminus x$. Cancel L_x to get $u * p = u * q$, then cancel L_u . $\square$

Theorem 2.17 (row-code distance). *[proved in F5 T3; two-line proof R3-B L-d, root-verified] Let $p \neq q$ and $F_{pq} := \{x : p * x = q * x\} = \text{Fix}(L_p^{-1}L_q)$. Then $x \in F_{pq} \Rightarrow L_p^{-1}(x) \notin F_{pq}$; consequently F_{pq} meets every L_p -cycle C in at most $\lfloor |C|/2 \rfloor$ points, so*

$$|F_{pq}| \leq \lfloor n/2 \rfloor.$$

Equivalently, the n rows of the Cayley table form an n -ary code of length n with minimum Hamming distance $\geq \lceil n/2 \rceil$.

Proof. Let $x \in F_{pq}$ and $f := L_p^{-1}(x)$, so $p * f = x$. If also $f \in F_{pq}$ then $q * f = p * f = x$, i.e. $p \setminus x = f = q \setminus x$; with $p * x = q * x$, (L4) gives $p = q$. $\square$

Theorem 2.17 is a genuine counting obstruction of the right type, and it is quantitatively insufficient: the Latin property needs minimum distance n , and no Plotkin- or Singleton-type bound bites at distance $n/2$ with only n codewords. Closing that gap was the campaign's route R3-A; §5.5 records that it is capped.

Definition 2.18 (the multiplicity matrix and its support sizes). $N(t, v) := \#\{a : a * t = v\}$, the multiplicity of the value v in column t . We also fix here, once and for all, the two support sizes used from §4 onwards:

$$r_t := |\text{Im}(R_t)| = \#\{v : N(t, v) > 0\}, \quad c_v := \#\{t : v \in \text{Im}(R_t)\} = \#\{t : N(t, v) > 0\},$$

i.e. r_t is the number of *distinct values* occurring in column t , and c_v is the number of *columns* in which the value v occurs at all. Both equal n exactly when (Q) holds; Lemma 6.8 records the two inequalities they satisfy in general.

Lemma 2.19 (N counts fixed points). *[proved in R2 (13); R3-C Lemma 1.1, root-verified] For all t, v the map $a \mapsto \Lambda_t(a)$ is a bijection $\{a : a * t = v\} \rightarrow \text{Fix}(L_t R_v) = \{x : t * (x * v) = x\}$. Hence $N(t, v) = |\text{Fix}(L_t R_v)|$, and also $N(t, v) = |\Theta_t^{-1}(v)|$ where $\Theta_t = R_t \circ L_t$.*

Proof. Put $x := \Lambda_t(a)$, so $a = t * x$. By KEY, $a * t = (t * x) * t = \Lambda_x(\Lambda_t x)$. Hence $a * t = v \iff \Lambda_x(\Lambda_t x) = v \iff \Lambda_t x = x * v \iff t * (x * v) = x$, and Λ_t is a bijection. $\square$

Theorem 2.20 (the counting identity). *[proved in orchestrating agent, R2; equivalent form R2 (12)+(14), root-verified] In every finite E677 magma, all row sums and all column sums of N equal n . Hence, unconditionally,*

$$\sum_{y, w \in M} |\text{Fix}(L_y R_w)| = n^2. \quad (2.4)$$

Moreover $N(v, v) \leq 1$ for all v (Lemma 2.6), and

$$\begin{aligned} E255 &\iff \text{tr } N = n \iff \text{every } \Theta_t \text{ has a fixed point,} \\ (Q) &\iff N \equiv 1 \iff \text{every } L_v R_u \text{ has a fixed point.} \end{aligned}$$

Proof. For fixed t , every x lies in exactly one fibre $\Theta_t^{-1}(v)$, so $\sum_v N(t, v) = n$. For fixed v , every row of the Cayley table contains v exactly once (rows are permutations), so $\sum_t N(t, v) = n$. Summing either way gives (2.4) through Lemma 2.19. The diagonal statement is Lemma 2.6 restated, and $E255 \iff \text{tr } N = n$ is Theorem 2.7 restated. For (Q): if every $L_t R_v$ has a fixed point then $N(t, v) \geq 1$ for all t, v , and each row of N has n entries summing to n , so every entry equals 1; conversely $N \equiv 1$ trivially gives fixed points. Thus mere *existence* already forces uniqueness. $\square$

Identity (2.4) is the reason *E255* can be described as a *diagonal slice* of (Q): the total mass is fixed at n^2 unconditionally, the diagonal is capped at 1 per entry, and *E255* asks that the diagonal be saturated.

Proposition 2.21 (single-variable reductions). *[proved in R3-B L-g, L-h, root-verified] Fix $t, x \in M$.*

(L-g) *Column t of the Cayley table is a permutation $\iff$ the map $\Theta_t : z \mapsto \Lambda_z(\Lambda_t(z))$ is a bijection of M . Hence (Q) $\iff \Theta_t$ is bijective for every t .*

(L-h) *$\beta_x(y) := (y * x) * y$ is a bijection $\iff N(t, x) = 1$ for every $t \iff$ the value x occurs exactly once in every column.*

Proof. (L-g) $\Theta_t = R_t \circ L_t$ with L_t a permutation, so Θ_t is bijective iff R_t is; and $\Theta_t(z) = \Lambda_z \Lambda_t(z)$ by KEY. (L-h) By KEY, $\beta_x(y) = \Theta_y(x) = \Lambda_x(\Lambda_y(x)) = \Lambda_x(\alpha_x(y))$ with $\alpha_x(y) := \Lambda_y(x)$; since Λ_x is a bijection, β_x is injective iff α_x is. And $\alpha_x(y) = \alpha_x(y') = z$ means $y * z = x = y' * z$, so α_x is injective iff $N(z, x) \leq 1$ for all z ; the column sums $\sum_t N(t, x) = n$ then force $N(\cdot, x) \equiv 1$. $\square$

(L-g) and (L-h) are the two “orthogonal” halves of (Q) — columns versus values — each expressed as bijectivity of an explicit *single-variable* map built from KEY, rather than as a condition on the $n \times n$ matrix N .

2.7 Two linking constraints on N

Pure marginal information about N (row sums, column sums, diagonal cap, support bound) is provably insufficient: §5.3 exhibits a numerical matrix satisfying all of it with $N \not\equiv 1$. The following two theorems are constraints that genuinely *link* entries of N , and they are not implied by the marginal package: a block-diagonal N with row and column sums n and diagonal ≤ 1 satisfies the marginals and violates Theorem 2.22.

Theorem 2.22 (strong connectivity). *[proved in R3-C Thm A, root-verified] Let D_N be the digraph on M with an arc $a \rightarrow b$ iff $N(a, b) > 0$ (equivalently $b \in \text{Im}(R_a)$). Then D_N is strongly connected. Equivalently, there is no $\emptyset \neq O \subsetneq M$ with $M * O \subseteq O$, and $\langle L_y : y \in M \rangle$ acts transitively on M .*

Proof. Let O be the set reachable from some $a \in M$ in D_N (so $a \in O$). Then O is forward closed: $L_y(O) \subseteq O$ for every y , hence $L_y(O) = O$ and $\Lambda_y(O) = O$ by finiteness and injectivity.

Step 1 (O is right closed). Let $x \in O$ and $t \in M$. Then $\Lambda_t x \in O$ and therefore $\Lambda_x(\Lambda_t x) \in O$. By KEY, $(t * x) * t = \Lambda_x(\Lambda_t x) \in O$. As x runs over O , $u := t * x$ runs over $L_t(O) = O$. Hence

$$u * t \in O \quad \text{for every } u \in O \text{ and every } t \in M. \quad (2.5)$$

Step 2 (counting). For $a \in O$, all n products $y * a$ lie in O , so $\sum_{b \in O} N(a, b) = n$; summing over $a \in O$ gives $\sum_{a, b \in O} N(a, b) = n|O|$. On the other hand each $b \in O$ has full column sum n , so $\sum_{b \in O} \sum_{a \in M} N(a, b) = n|O|$. Subtracting,

$$\sum_{a \notin O, b \in O} N(a, b) = 0, \quad \text{i.e. } u * t \notin O \text{ whenever } t \notin O. \quad (2.6)$$

Step 3. If $O \neq M$ pick $t \notin O$ and $u \in O$; then (2.5) gives $u * t \in O$ and (2.6) gives $u * t \notin O$. Hence $O = M$; as a was arbitrary, D_N is strongly connected. $\square$

Implication (2.5) — *any set closed under all left translations is closed under all right translations* — is the only place in the whole campaign where KEY produced genuinely global, non-tautological information. A localized or weighted version of it remains the most promising unexplored mechanism.

Theorem 2.23 (edge expansion). *[proved in R3-C Thm B, root-verified] For $\emptyset \neq O \subsetneq M$ put $e(O) := \sum_{a \in O, b \notin O} N(a, b) = \#\{(y, a) : a \in O, y * a \notin O\}$ and $m := \min(|O|, |O^c|)$. Then*

$$m \leq e(O) + \sqrt{e(O)}, \quad \text{i.e.} \quad e(O) \geq m + \frac{1}{2} - \sqrt{m + \frac{1}{4}}.$$

Proof. Write $\varepsilon := e(O)$, $s := |O|$. Step 2 of Theorem 2.22 generalizes to the flow balance $\sum_{a \notin O, b \in O} N(a, b) = ns - \sum_{a, b \in O} N(a, b) = \varepsilon$, since $\sum_{a \in O, b \in M} N(a, b) = ns = \sum_{b \in O, a \in M} N(a, b)$. In particular $\#\{(u, t) : t \notin O, u \in M, u * t \in O\} = \varepsilon$.

Let $\delta_y := |L_y(O) \setminus O|$; then $\sum_y \delta_y = \varepsilon$, so $Y := \{y : L_y(O) = O\}$ has $|Y| \geq n - \varepsilon$, and for $y \in Y$ also $\Lambda_y(O) = O$. Fix $t \in Y \setminus O$ and $x \in O \cap Y$. Then $\Lambda_t x \in O$ and, since $\Lambda_x(O) = O$, also $\Lambda_x(\Lambda_t x) \in O$; by KEY, $(t * x) * t \in O$, while $u := t * x \in L_t(O) = O$. Each such pair (x, t) therefore produces a pair (u, t) with $u \in O$, $t \notin O$, $u * t \in O$, and $x \mapsto t * x$ is injective for fixed t . Hence $\varepsilon \geq |Y \setminus O| \cdot |O \cap Y| \geq (|O^c| - \varepsilon)(|O| - \varepsilon)$. If $\varepsilon \geq m$ the claim is immediate; otherwise both factors are $\geq m - \varepsilon > 0$, so $\varepsilon \geq (m - \varepsilon)^2$, i.e. $m - \varepsilon \leq \sqrt{\varepsilon}$. $\square$

Theorem 2.22 is the case $m \geq 1$ of Theorem 2.23 ($\varepsilon = 0$ would give $1 \leq 0$). Both are *lower* bounds on $e(O)$. Section 5.7 explains why this direction is the wrong one, and Section 4 shows that the right one is unavailable.

3 The affine models: criterion and commuting spectrum

Every finite *E677* magma known to us — to the campaign, to the blueprint, and to the public model database [4] — is affine over an abelian group, or a fibred extension of one. This section gives the exact criterion for an affine magma to satisfy *E677*, and classifies completely those affine models whose two coefficients *commute*; every known affine model is of that kind. *The converse — that an affine E677 magma must have commuting coefficients — is not proved here and is open* (Problem 11.3), so “the affine spectrum” below always means the spectrum of the commuting family. The classification is the source of the campaign’s most useful pieces of information and of one of its most instructive mistakes.

3.1 The affine criterion

Theorem 3.1 (affine criterion). *[proved in R3-B Thm 1, root-verified] [machine-verified: `problems/etp677/R3B_scripts/final_verify.py`, `constc.py` (scalar, hence commuting, models over cyclic groups only)] Let A be an abelian group and $F, G \in \text{End}(A)$, and define $x * y := Fx + Gy$.*

(a) $(A, *)$ satisfies *E677* iff

$$(i) \quad GF + G^2FG = \text{id}_A \quad \text{and} \quad (ii) \quad F + G^2F^2 + G^3 = 0.$$

(b) If F and G commute, (i)–(ii) read $F(G + G^3) = 1$ and $F + F^2G^2 + G^3 = 0$.

(c) Suppose $G + G^3 \in \text{Aut}(A)$ and put $F := (G + G^3)^{-1}$ (so F, G commute). Then (i)–(ii) hold iff

$$P(G) = 0, \quad \text{where} \quad P(u) := u^8 + 2u^6 + u^4 + u^2 + u + 1 = \Phi_{10}(u) \cdot Q(u),$$

$$\text{with } \Phi_{10}(u) = u^4 - u^3 + u^2 - u + 1 \text{ and } Q(u) = u^4 + u^3 + 2u^2 + 2u + 1.$$

(d) Any such $(A, *)$ is a quasigroup (both F and G are units), hence satisfies (Q) and therefore E255.

(e) With a constant, $x*y = Fx + Gy + c$ satisfies E677 iff (i), (ii) hold and $(G^2F + G^2 + G + 1)c = 0$.

Proof. (a) Expanding, and keeping the order of composition,

$$\begin{aligned} y * x &= Fy + Gx, \\ (y * x) * y &= F^2y + FGx + Gy, \\ x * ((y * x) * y) &= Fx + GF^2y + GFGx + G^2y, \\ y * (x * ((y * x) * y)) &= (GF + G^2FG)x + (F + G^2F^2 + G^3)y. \end{aligned}$$

Setting this equal to x for all x, y gives exactly (i)–(ii) (take $y = 0$ and $x = 0$ respectively); conversely (i)–(ii) plainly suffice. (b) is immediate.

(c) From (i), $G + G^3 = G(1 + G^2)$ is invertible, hence so is G . Multiplying (ii) by $(G + G^3)^2$ and using $F(G + G^3) = 1$,

$$F(G + G^3)^2 + F^2G^2(G + G^3)^2 + G^3(G + G^3)^2 = (G + G^3) + G^2 + G^3(G^2 + 2G^4 + G^6) = G \cdot P(G).$$

Since G and $(G + G^3)^2$ are units, (ii) $\iff P(G) = 0$. Conversely if $P(G) = 0$ and $G + G^3$ is invertible, set $F := (G + G^3)^{-1}$; the displayed identity gives $(F + F^2G^2 + G^3)(G + G^3)^2 = GP(G) = 0$, hence (ii).

(d) $F = (G + G^3)^{-1}$ is a unit, so $a \mapsto a * t = Fa + Gt$ is injective: the table is a Latin square, so (Q) holds and Corollary 2.8 applies.

(e) The same expansion with the constant produces the extra term $(G^2F + G^2 + G + 1)c$. $\square$

Note that (i) already forces F and G to commute *once one assumes* $F = (G + G^3)^{-1}$, and conversely commuting F, G satisfying (i) must equal $(G + G^3)^{-1}$; so parts (b), (c) describe exactly the commuting solutions. Nothing below rules out a noncommuting solution of (i)–(ii), and we know of none; see Problem 11.3.

The side condition in (c) is in fact automatic:

Theorem 3.2 (module description of the commuting affine spectrum). [*proved in R3-B Thm 1', root-verified*] Let $R := \mathbb{Z}[u]/(P(u))$. In R the element u is a unit (from $P(u) = 0$ one has $u \cdot (u^7 + 2u^5 + u^3 + u + 1) = -1$) and $1 + u^2$ is a unit (since $P(u) \equiv u \pmod{(1 + u^2)}$), so $u + u^3 = u(1 + u^2)$ is automatically a unit of R . Consequently, for every finite R -module A , letting G be the action of u and $F := (G + G^3)^{-1}$, the magma $(A, x * y = Fx + Gy)$ is an E677 magma of order $|A|$, and it is a quasigroup satisfying E255 and (Q).

Hence: the orders realized by affine E677 magmas with commuting F, G are exactly the orders of finite $\mathbb{Z}[u]/(P(u))$ -modules. Concretely, for a prime p let $d(p)$ be the multiset of degrees of the distinct irreducible factors of $P \pmod{p}$; the achievable exponents of p form the numerical semigroup generated by $d(p)$.

Remark (exactly what is and is not classified). The “ $\supseteq$ ” half — every finite R -module yields a model of its own order — is proved above. The “ $\subseteq$ ” half is proved only for solutions with F, G commuting, equivalently with $F = (G + G^3)^{-1}$, since only then does $P(G) = 0$ follow. A hypothetical affine model with noncommuting F, G would satisfy neither hypothesis and could in principle realize an order outside the R -module spectrum. Every affine model exhibited anywhere in this paper, in the campaign, in [2] or in [4] is commuting (indeed scalar or a matrix power series in G), and the verifier `R3B_scripts/final_verify.py` tests scalar models over cyclic groups; so the noncommuting case is unexplored rather than excluded.

Proposition 3.3 (the Φ_{10} branch is exactly the Alexander quandles). *[proved in R3-B Prop 2, root-verified] For commuting $F, G \in \text{End}(A)$ the following are equivalent:*

1. $\Phi_{10}(G) = 0$ and $F = 1 - G$;
2. $F = 1 - G$ and $(A, x * y = Fx + Gy)$ satisfies E677;
3. $(A, x * y = Fx + Gy)$ satisfies E677 and is idempotent.

*In that case $(A, *)$ is a quandle: idempotent, left-distributive, with all L_x bijective; i.e. it is the Alexander quandle $x * y = (1 - G)x + Gy$ with G a primitive 10th root of unity in $\text{End}(A)$.*

Proof. If $\Phi_{10}(G) = 0$, i.e. $G^4 - G^3 + G^2 - G = -1$, then $(1 - G)(G + G^3) = G + G^3 - G^2 - G^4 = -(G^4 - G^3 + G^2 - G) = 1$, so $F := 1 - G = (G + G^3)^{-1}$, and $P(G) = \Phi_{10}(G)Q(G) = 0$, so Theorem 3.1(c) gives E677. Conversely if $F = 1 - G$ and E677 holds, then $F(G + G^3) = 1$ forces $(1 - G)(G + G^3) = 1$, which unwinds to $\Phi_{10}(G) = 0$. Idempotency: $x * x = (F + G)x$, so $x * x = x$ for all x iff $F + G = 1$. Left distributivity for an affine magma: $x * (y * z) = Fx + GFy + G^2z$ while $(x * y) * (x * z) = (F^2 + GF)x + FGy + G^2z$; with commuting F, G these agree for all x, y, z iff $F = F(F + G)$, i.e. iff $F + G = 1$ (F being a unit). $\square$

The order-5 dihedral (Takasaki) quandle $x * y = 2x - y$ over $\mathbb{Z}_5$ is the case $A = \mathbb{Z}_5, G = -1$ (since $\Phi_{10}(-1) = 5 \equiv 0$). The Q -branch is not idempotent in general: the two order-7 models $x * y = 4x + y$ ($G = 1, Q(1) = 7$) and $x * y = 4x + 3y$ ($G = 3, Q(3) = 133 = 7 \cdot 19$) are Q -branch, and the first has a left identity. In particular E677 does not force idempotency, even finitely.

Remark. Theorem 3.1 was derived independently by two of the campaign’s families — first as the cyclic-modulus statement $P(G) \equiv 0 \pmod{m}$, $F = (G + G^3)^{-1}$, $c \cdot (G^2F + G^2 + G + 1) = 0$ over $\mathbb{Z}/m$ [8], then in the endomorphism form above [11] — and it matches the blueprint’s Type 1/Type 2 dichotomy exactly: Type 1 is the Φ_{10} factor, Type 2 the quartic factor Q . This is a cross-check of a transcription, not an independent theorem.

3.2 The even-order spectrum theorem

The following was the campaign’s sharpest self-correction. Rounds 1–2 had recorded, on the strength of the cyclic case plus the exhaustive emptiness of orders 2, 4, 6, 8, a working *parity conjecture*: every finite E677 magma has odd order. It is false.

Theorem 3.4 (even-order spectrum, commuting case). *[proved in R3-B Thm 3, Thm 4, root-verified] [machine-verified: `problems/etp677/R3B_scripts/final_verify.py, extra.py, paranoia.py`] Throughout, “affine” means $x * y = Fx + Gy + c$ with F, G commuting, equivalently $F = (G + G^3)^{-1}$ (Theorem 3.1(c)); this is what makes $P(G) = 0$ available.*

- (a) (the cyclic case, which is all the earlier claim covered) *If $A \cong \mathbb{Z}_m$ is cyclic then no affine E677 magma of even order exists on A : indeed $P(G) \equiv G^8 + G^4 + G^2 + G + 1 \equiv 1 \pmod{2}$ for every integer G , so $P(G)$ is always odd and $P(G) \equiv 0 \pmod{m}$ is unsolvable for even m .*
- (b) (general abelian A) *If A carries an affine E677 magma structure and $|A|$ is even, then $16 \mid |A|$; more precisely the 2-primary part A_2 satisfies $|A_2| = 16^k$ for some $k \geq 1$.*
- (c) *16 is attained. Let $\mathbb{F}_{16} = \mathbb{F}_2[z]/(z^4 + z^3 + z^2 + z + 1)$, so $z^5 = 1$ and $z \neq 1$, and define on $M = \mathbb{F}_{16}$*

$$x * y = (1 + z)x + zy.$$

*Then $(M, *)$ is an E677 magma of order 16; it satisfies E255, its table is a Latin square (so (Q) holds in it), and it is a quandle.*

Hence the minimal even order of a commuting affine model is exactly 16.

Proof. (a) is the displayed congruence.

(b) $P \bmod 2$ factors over $\mathbb{F}_2$ as $P \equiv (u^4 + u^3 + u^2 + u + 1)(u^4 + u^3 + 1) = \Phi_5 \cdot f$, a product of two *distinct* irreducible quartics. Let $A_2 \neq 0$ be the 2-primary part (which is G -invariant). For each i , $V_i := 2^i A_2 / 2^{i+1} A_2$ is an $\mathbb{F}_2$ -vector space carrying an induced $\bar{G}$ with $P(\bar{G}) = 0$, i.e. V_i is a module over $\mathbb{F}_2[u]/(\Phi_5 f) \cong \mathbb{F}_2[u]/(\Phi_5) \times \mathbb{F}_2[u]/(f) \cong \mathbb{F}_{16} \times \mathbb{F}_{16}$ by CRT. A module over a product of fields is a direct sum of vector spaces over each factor, so $\dim_{\mathbb{F}_2} V_i \equiv 0 \pmod{4}$ and $|V_i| = 16^{k_i}$. Multiplying over the filtration, $|A_2| = 16^{\sum k_i}$, and $A_2 \neq 0$ forces some $k_i \geq 1$.

(c) In characteristic 2, $\Phi_{10}(u) = u^4 + u^3 + u^2 + u + 1 = \Phi_5(u)$, and z is a primitive 5th root of unity, so $\Phi_{10}(z) = 0$; also $1 - z = 1 + z$. Now apply Proposition 3.3 with $G = \text{multiplication by } z$ and $F = 1 + z$, and Theorem 3.1(d). Finally Φ_5 is irreducible over $\mathbb{F}_2$ (the order of 2 mod 5 is 4), so $\mathbb{F}_2[z]$ really is a field of 16 elements, and z exists there because $5 \mid 15 = |\mathbb{F}_{16}^\times|$. $\square$

The Cayley table of the order-16 quandle was re-parsed from printed text and the raw E677 term $y * (x * ((y * x) * y))$ evaluated on all 256 pairs: 0 violations; E255: 0 violations [machine-verified: `problems/etp677/R3B_scripts/paranoia.py`]. There is a second family at order 16: taking $G \in \mathbb{F}_{16}$ a root of the other factor $u^4 + u^3 + 1$ (a primitive 15th root of unity) and $F = (G + G^3)^{-1}$ gives a Latin, E255, *non-idempotent* model. Scanning all $G \in \mathbb{F}_{16}$ with $P(G) = 0$ gives exactly 8 values, splitting into two Frobenius orbits, and a backtracking isomorphism search shows these 8 magmas form exactly *two* isomorphism classes [machine-verified: `problems/etp677/R3B_scripts/extra.py`]:

G (multiplicative order)	F	quandle?	# idempotents	cycle type of every L_y
5 (Φ_{10} branch)	$1 + G$	yes	16	(5, 5, 5, 1)
15 (Q branch)	$(G + G^3)^{-1}$	no	1	(15, 1)

Corollary 3.5 (the structural reason there is no parity obstruction). *[proved in R3-B §6.3, root-verified] The only “mod 2” phenomenon in this problem is that $P \bmod 2 = \Phi_5 \cdot (u^4 + u^3 + 1)$ has no root in $\mathbb{F}_2$ — which is exactly why $\mathbb{Z}_m$ -linear even-order models do not exist. But both factors are irreducible of degree 4, so they do have roots in $\mathbb{F}_{16}$. The would-be parity obstruction is really a residue-degree statement: the prime 2 has residue degree 4 in $\mathbb{Z}[u]/(P)$, i.e. 2 is inert in $\mathbb{Q}(\zeta_5)$ (because 2 has order 4 mod 5). That is an arithmetic fact about the number 5, not a parity fact about n ; it forces $16 \mid n$, never n odd. Any future global mechanism for Problem 1.1 must be insensitive to the parity of n .*

3.3 The affine spectrum

Computing the factorizations of P modulo small primes [machine-verified: `problems/etp677/R3B_scripts/final_verify.py`]:

p	irreducible factors of $P \bmod p$	minimal p -part
2	$u^4 + u^3 + 1, \quad u^4 + u^3 + u^2 + u + 1$	$2^4 = 16$
3	$(u^2 - u - 1)^2, \quad u^4 - u^3 + u^2 - u + 1$	$3^2 = 9$
5	$(u + 1)^4, \quad Q$	5
7	$u - 1, \quad u - 3, \quad u^2 - 2u - 2, \quad \Phi_{10}$	7
11, 13, 19, 31, 37, 41, 43	have a linear factor	p
17, 23, 29, 53, 59	minimal degree 2	p^2
47	irreducible quartics only	p^4

The orders ≤ 200 achievable by a commuting affine model are

1, 5, 7, 9, 11, 13, **16**, 19, 25, 31, 35, 37, 41, 43, 45, 49, 55, 61, 63, 65, 67, 71, 73, 77, **80**, 81, 91, 95, 97, 99, 101, 103, 109, **112**, 117, 121, 125, 131, 133, 139, 143, **144**, 151, 155, 157, 163, 169, 171, 175, **176**, 181, 185, 191, 193.

The even orders ≤ 1000 are 16, 80, 112, 144, 176, 208, 256, 304, 400, 496, 560, 592, 656, 688, 720, 784, 880, 976 — all divisible by 16, as Theorem 3.4(b) requires. Since $E677$ is an identity, the class of $E677$ magmas is closed under direct products, so $16m$ is realizable for every realizable odd m ; there are infinitely many even orders.

Remark (a superseded list). An earlier round recorded the model orders as “1, 5, 7, 9, 11, 13, 19, 31, 35, 37, 41, 43”, obtained by scanning $x * y = ax + by \bmod n$ for $n \leq 39$. That list is *not* exhaustive even among affine models: e.g. 25 is realizable by $A = (\mathbb{Z}/5)^2$, $G = \begin{pmatrix} 4 & 1 \\ 0 & 4 \end{pmatrix} = -I + N$ with $N^2 = 0$ (then $\Phi_{10}(u) \equiv (u + 1)^4 \bmod 5$, so $\Phi_{10}(G) = N^4 = 0$, and $G + G^3 = -2I + 4N$ is invertible). The correct statement is Theorem 3.2. [machine-verified: `problems/etp677/R3B_scripts/extra.py`]

3.4 Mediality, and why affine models cannot be counterexamples

Proposition 3.6 (commuting affine models are medial). [proved in *R3-F*, root-verified] [machine-verified: `problems/etp677/r3f_enum.py`] *Every affine $E677$ model with commuting coefficients ($FG = GF$) is medial: $(x * y) * (z * w) = (x * z) * (y * w)$. Conversely, direct expansion shows an affine operation is medial iff $FG = GF$, so mediality of all affine models is equivalent to the open commutation problem (Problem 11.3).*

Proof. Expanding both sides of mediality for $x * y = Fx + Gy + c$ shows they differ only by interchanging $FGb + GFc$ with $FGc + GFb$; under $FG = GF$ the two sides agree. In the commuting setting $F = (G + G^3)^{-1}$ automatically commutes with G . $\square$

Mediality was verified exhaustively over all quadruples for the named models $(\mathbb{Z}_5, 2x + 4y)$, $(\mathbb{Z}_7, 4x + y)$, $(\mathbb{Z}_7, 4x + 3y)$, $(\mathbb{F}_9, x + (t + 2)y)$ with $t^2 + 1 = 0$, and $(\mathbb{F}_{16}, (1 + z)x + zy)$ — all medial, all $E255$, all Latin.

Three separate obstructions say that no counterexample can be affine, or even affinely fibred:

Proposition 3.7 (no affine counterexamples). [markers are attached to the individual items (a)–(d)]

- (a) [proved in Theorem 3.1(d); [2, Lemma 13.3], root-verified] Any finite E677 magma of the form $x * y = \alpha x + \beta y + c$ on an abelian group satisfies E255.
- (b) [proved in F4 Prop 5, root-verified] Let A be any abelian group (finite or not), $h : A \rightarrow A$ arbitrary — not assumed linear — and $y * x := x + h(y)$. If this satisfies E677 then it satisfies E255.
- (c) [proved in F4 Prop 6, root-verified] For every $n \geq 1$ and every $(\alpha, \beta, \gamma) \in (\mathbb{Z}/n)^3$, the E677 conditions for $y * x = \alpha y + \beta x + \gamma$ imply the E255 conditions.
- (d) [proved in [2, Lemma 13.4], external, root-verified] If $G \times M$ satisfies E677, where G is an E677 magma satisfying E255, M is an abelian group, and $(x, s) * (y, t) = (x * y, \alpha_{x,y}s + \beta_{x,y}t + c_{x,y})$ with α, β endomorphisms, then $G \times M$ satisfies E255.

Proof of (b). Every row of $y * x = x + h(y)$ is a translation, hence a permutation. Direct expansion shows E677 holds exactly when

$$h(x + h(y)) + h(x) + h(y) + y = x \quad (x, y \in A). \quad (3.1)$$

Put $y = x$ in (3.1): $h(x + h(x)) = -2h(x)$. Now put $y = x + h(x)$ in (3.1) and use the previous display; the remaining terms cancel to $h(x - 2h(x)) = 0$. The three right multiplications by x are therefore

$$x * x = x + h(x), \quad (x * x) * x = x + h(x + h(x)) = x - 2h(x), \quad ((x * x) * x) * x = x + h(x - 2h(x)) = x.$$

□

Proof of (c). Expansion of E677 for $y * x = \alpha y + \beta x + \gamma$ gives the exact necessary and sufficient congruences mod n

$$(A1) \alpha\beta(1 + \beta^2) = 1, \quad (A2) \alpha + \alpha^2\beta^2 + \beta^3 = 0, \quad (A3) (\beta^2(\alpha + 1) + \beta + 1)\gamma = 0,$$

while E255 is exactly

$$(B1) \alpha^3 + \alpha^2\beta + \alpha\beta + \beta = 1, \quad (B2) (\alpha^2 + \alpha + 1)\gamma = 0.$$

Set $u := 1 + \beta^2$. By (A1), α, β, u are units and $\alpha = (\beta u)^{-1}$. Multiplying (A2) by the unit βu^2 turns it into $F := u + \beta + \beta^4 u^2 = 0$. Multiplying the left side of (B1) minus 1 by the unit $\beta^3 u^3$ gives $N = 1 + \beta^2 u + \beta^3 u^2 + \beta^4 u^3 - \beta^3 u^3$, and using only $u = 1 + \beta^2$, direct expansion gives the polynomial identity $N = (1 - \beta + \beta^2)F$; thus (A2) implies (B1). For the constant term put $H := \beta + u(\beta^2 + \beta + 1)$ and $J := 1 + \beta u + \beta^2 u^2$; the coefficient in (A3) is H/u , while $\alpha^2 + \alpha + 1 = J/(\beta^2 u^2)$, and direct expansion gives $J = (1 - \beta + \beta^2)H$. Hence $(\alpha^2 + \alpha + 1)\gamma = \frac{1 - \beta + \beta^2}{\beta^2 u}(\beta^2(\alpha + 1) + \beta + 1)\gamma = 0$, which is (B2). □

Part (b) is worth emphasizing, since it goes beyond the blueprint's linear obstruction: the row-shift function h is completely arbitrary, so *no* choice of nonlinear row shifts and no modulus can turn the natural periodic-successor construction into a counterexample. The obstruction is the two-step use of (3.1), not a failed parameter search. As an independent check, exhaustive enumeration of all (α, β, γ) modulo every $n = 11, \dots, 40$ found affine E677 models exactly at $n \in \{11, 13, 19, 31, 35, 37\}$ (with 4, 13, 38, 65, 14, 74 solutions respectively) and zero violations of (B1)–(B2) in every one; and exhaustive enumeration of all functions $h : \mathbb{Z}/n \rightarrow \mathbb{Z}/n$ for $n \leq 7$ found no violation in the family (3.1) [computational: [7], §3–§4].

4 An $E677$ magma of order 176 that is not right-cancellative

4.1 Context

Until this campaign, exactly one finite $E677$ magma failing right cancellation was recorded anywhere: the order-496 example of [2, §13.1], built as a fibred extension of an $\mathbb{F}_{31}$ base by three $\mathbb{F}_{16}$ fibre operations. It satisfies $E255$. Its existence is what makes (Q) of Definition 2.12 false, and hence what forced the campaign to abandon the route it had been pursuing for two rounds.

The natural next target, proposed at the end of Round 3 as “the smallest currently unclaimed step in the right direction”, was the following weakening of (Q):

$$(S\text{-off}) \quad N(t, v) + N(v, t) \geq 1 \quad \text{for all } t \neq v.$$

Its appeal is that setting $t = v$ turns it into $N(v, v) \geq 1$, which is exactly $E255$; so (S-off) is the “weaker off-diagonal sibling” of the target, and whatever technique proved it would have been informative about the diagonal. It is false, and the rest of this section is an expanded and re-checked account of [16].

4.2 The model

Theorem 4.1. *[machine-verified: `problems/etp677/R5C_scripts/verify176.py`, `m176.json`] Let $M := \mathbb{F}_{11} \times \mathbb{F}_{16}$ with $\mathbb{F}_{16} = \mathbb{F}_2[z]/(z^4 + z^3 + 1)$ (elements encoded as integers 0..15, bit i the coefficient of z^i). Define, for $x, y \in \mathbb{F}_{11}$ and $s, t \in \mathbb{F}_{16}$,*

$$(x, s) * (y, t) := (4x + 8y \bmod 11, \quad a_{y-x} s + b_{y-x} t),$$

where the fibre coefficients $(a_d, b_d) \in \mathbb{F}_{16}^2$, indexed by $d = (y - x) \bmod 11$, are

d	0	1	2	3	4	5	6	7	8	9	10
(a_d, b_d)	(14, 15)	(0, 1)	(11, 10)	(10, 11)	(10, 11)	(11, 10)	(10, 11)	(11, 10)	(11, 10)	(10, 11)	(0, 1)

Then $|M| = 176$ and:

- (a) $(M, *)$ satisfies $E677$ (zero violations over all 176^2 pairs);
- (b) every L_y is a bijection; all row sums and all column sums of N equal 176;
- (c) $(M, *)$ satisfies $E255$: $N(v, v) = 1$ for every v ;
- (d) $(M, *)$ is not right-cancellative: the entries of N take the values $\{0, 1, 16\}$;
- (e) there are 2640 unordered pairs $\{t, v\}$ with $t \neq v$ and $N(t, v) = N(v, t) = 0$. An explicit witness is $t = (0, 0)$, $v = (4, 1)$: there is no a with $a * t = v$ and no b with $b * v = t$.

The off-diagonal distribution of $N(t, v) + N(v, t)$ over the $\binom{176}{2} = 15400$ unordered pairs is $\{0 : 2640, 2 : 12584, 32 : 176\}$. Moreover, writing r_t and c_v for the support sizes of Definition 2.18, N has 5280 zero entries and $r_t = c_v = 146$ for all t, v [machine-verified: `problems/etp677/R5C_scripts/struct_check.py`].

The key feature is $a_1 = a_{10} = 0$: the fibre operations at offsets $d = \pm 1$ are the projections $s \diamond t = t$, which are not injective in their left argument.

Verification. The script `R5C_scripts/verify176.py` rebuilds $\text{GF}(16)$, the base $x \diamond y = 4x + 8y$ on $\mathbb{F}_{11}$, and the whole 176×176 Cayley table from scratch, importing nothing from the search code, and evaluates the raw $E677$ term on all 176^2 pairs together with all the counts quoted above; its archived output `verify176.out` reports 0 violations. The table itself is stored as `R5C_scripts/m176.json`. The construction was found by SAT (`R5C_scripts/bp_sat.py`, log `bp_theta_sat.out`) inside the extension framework of §4.3, and verified twice — once through the extension equation (4.1), once by brute force on the product magma. $\square$

Corollary 4.2. *[machine-verified: `problems/etp677/R5C_scripts/verify176.py` for (a)–(b); `isoperim_check.py` for (c)]*

- (a) (S-off) is false. A fortiori so are $N + N^\top = 2J$ and $N + N^\top \geq 2J - I$.
- (b) The order-176 magma is a non-right-cancellative finite $E677$ magma of order smaller than 496. As far as the campaign’s literature reconnaissance [14] could determine, no smaller one had been recorded at the time. (Round 6 later produced one of order 77, Theorem 7.18; it satisfies (S-off) and so does not replace the order-176 witness for (a) and (c).)
- (c) With $e(O)$ as in Theorem 2.23, write $\text{exc}(O) := e(O) - |O||O^c|$; since all row sums of N equal n one has identically $\text{exc}(O) = |O|^2 - \sum_{a,b \in O} N(a,b)$. In the order-176 magma $\text{exc}(O) > 0$ for at least one O of every size $2 \leq s \leq n-2$: the script `isoperim_check.py` exhibits such an O for each of the 173 size classes and evaluates $e(O)$ on the stored Cayley table. In particular for the (S-off)-violating pair $O = \{(0,0), (4,1)\}$ one has $e(O) = 350 > 348 = |O||O^c|$. Full enumeration gives the exact maxima $\max_{|O|=s} \text{exc}(O) = 0, +2, +4$ at $s = 1, 2, 3$; for the middle size classes only lower bounds are established here, and they are large (e.g. an explicit O with $|O| = 88$ has $\text{exc}(O) = 1152$).

Remark (a corrected number). [16], §4.2 printed a table of purported extremal excesses (+20 at $s = 8$, +282 at $s = 88$, +264 at $s = 120$, ...). Those values are *not* maxima: `isoperim_check.py` exhibits sets with $\text{exc} = +32, +1152$ and $+768$ at those sizes. Exact maxima are verified only at $s \in \{1, 2, 3\}$ (and, by complementation, $s \in \{173, 174, 175\}$); at $s = 4, 5$ the witness family reproduces the reported values without an exactness proof. Nothing in the paper depends on the extremal values — only on the sign of exc — but the table should not be quoted.

Part (c) closes the programme it was aimed at, and needs to be read carefully. Round 3 had proved [12] that if $e(O) \leq |O||O^c|$ holds for all O of one *single* size s , then $E255$ follows — an averaging argument, since $\sum_{|O|=s} e(O) = (n^2 - \text{tr } N) \binom{n-2}{s-1}$ and $s(n-s) \binom{n}{s} = (n^2 - n) \binom{n-2}{s-1}$. The implication is correct. What Corollary 4.2(c) shows is that its *premise* is false for every size class s that is not already equivalent to $E255$ — and false in a magma that *does* satisfy $E255$. So that premise cannot be proved, and this particular route is closed, not merely unproven. Any proof of $E255$ along these lines must bound $e(O)$ for a *restricted* family of sets O (for instance only $|O| = 1$, which is $E255$ itself), not uniformly over a size class. This says nothing about other conceivable off-diagonal or isoperimetric quantities; what is refuted is the specific threshold $e(O) \leq |O||O^c|$.

Proposition 4.3 (what survives in the order-176 magma). *[machine-verified: `problems/etp677/R5C_scripts/struct_check.py`, `verify176.py`] The following all hold in the order-176 magma, and are therefore consistent with the failure of the Latin property and cannot by themselves give $E255$: Theorem 2.22 (strong connectivity of $\text{supp } N$); the diameter-2 property (for all t, z there is u with $N(t, u) > 0$ and $N(u, z) > 0$, namely $u = f_t(z) = (t * (t * z)) * t$); row and column*

$\text{sums} = n$; $\text{tr } N = n$; and the counting condition $\#\text{zeros}(N) = 5280 \leq \binom{176}{2} = 15400$ that (S-off) necessarily implies. In addition $Q := \sum_{t,v} N(t,v)^2 = 115456$, so $Q - n^2 = \sum_{p \neq q} |F_{pq}| = 84480$; here $F_{pq} \neq \emptyset$ exactly for the 2640 ordered pairs $p = (x, s)$, $q = (x, s')$ with $s \neq s'$, and then $|F_{pq}| = 32$.

Remark. For comparison, in the order-496 magma the same count gives $|F_{pq}| = 240$ against the bound $\lfloor n/2 \rfloor = 248$ of Theorem 2.17 — i.e. the row-code bound is essentially *tight* there. That is why the Cauchy–Schwarz route to a lower bound on $\sum_t r_t$ is hopeless: it gives $\sum_t r_t \geq n^4/Q = 29772$ against the required 123 256 [machine-verified: `problems/etp677/R5C_scripts/m496.py`].

4.3 The translation-invariant extension family with affine fibres

The order-176 magma is an instance of the extension framework of [2, Ch. 13]: given an *E677* magma G , an abelian group M , and a family $\diamond_{x,y}$ of operations on M satisfying

$$s = t \diamond_{y, L_y^{-1}x} \left(s \diamond_{x, (y \diamond x) \diamond y} ((t \diamond_{y,x} s) \diamond_{y \diamond x, y} t) \right) \quad \text{for all } x, y \in G, \quad (4.1)$$

the product $(x, s) * (y, t) := (x \diamond y, s \diamond_{x,y} t)$ satisfies *E677*. This subsection gives the complete structure theory of the *translation-invariant* (TI) sub-family *with affine fibre operations*, in which $\diamond_{x,y}$ depends only on $y - x$ and each $\diamond_d$ is affine. **Both restrictions are in force throughout §4.3 and §4.4**; “complete” always means complete for that sub-family, and general (non-affine) fibre tables are *not* covered by anything below (see the scope remark after Theorem 4.8 and Problem 11.8). The restriction is what made the search tractable and what makes the diagonal analysis in §4.4 sharp.

Setting. Base $x \diamond y = Ax + By + c$ on $\mathbb{F}_p$ with $A + B = 1$ and $A, B \neq 0$ (so the base is translation-invariant and Latin; *E677* for it is checked directly). Fibre operations $s \diamond_d t = a_d s + b_d t + c_d$ are M -linear with b_d invertible (needed so that rows are permutations), and the operation attached to the pair (x, y) is $\diamond_{y-x}$.

Lemma 4.4 ((4.1) in coordinates). [*proved in R5-C Lemma 2.1; adversarially re-derived in R5-A, root-verified*] [machine-verified: `problems/etp677/r5a_search.py selftest`, `R5C_scripts/blueprint_search.py`] *Put*

$$D_1 = -\frac{d+c}{B}, \quad D_2 = (A^2 + B)d + (A+1)c, \quad D_3 = -d, \quad D_4 = Bd - c.$$

Then (4.1) at parameter $d = y - x$ is equivalent to the three scalar equations in the operations at indices $D_1, \dots, D_4$:

$$(E1) \ b_1(a_2 + b_2 a_4 b_3) = 1, \quad (E2) \ a_1 + b_1 b_2(a_4 a_3 + b_4) = 0, \quad (E3) \ b_1(b_2(a_4 c_3 + c_4) + c_2) + c_1 = 0,$$

where index j abbreviates index D_j .

Proof. Expand $t \diamond_{D_1} (s \diamond_{D_2} ((t \diamond_{D_3} s) \diamond_{D_4} t))$ as an affine function of (s, t) and equate to s ; the coefficients of s , of t , and the constant give (E1), (E2), (E3). The index formulas are obtained by specializing the four base pairs of (4.1): solving $y \diamond z = x$ gives $z - y = -(d + c)/B$; $(y \diamond x) \diamond y = ABx + (A^2 + B)y + (A + 1)c$, whose difference from x is $(A^2 + B)d + (A + 1)c$ using $A + B = 1$; and the last two differences are $x - y = -d$ and $y - (y \diamond x) = Bd - c$. $\square$

Remark. The index formulas were subjected to an adversarial re-derivation by an independent agent, which confirmed them together with the general (non-TI) pair subscripts and the reduction of the $E255$ -violation constraint; a self-test reconstructs the blueprint order-496 model with zero violations of (4.1), zero $E677$ violations on the full order-496 magma, and zero $E255$ failures [machine-verified: `problems/etp677/r5a_search.py selftest, r5a_audit.py`].

Lemma 4.5 (the N -matrix of a TI extension). *[proved in R5-C Lemma 2.3, root-verified]* Let the base be Latin. For $T = (y, t)$ and $V = (v_1, v_2)$ put $\delta := (y - v_1 + c)/A$. Then

$$N(T, V) = \begin{cases} 1 & \text{if } a_\delta \neq 0, \\ |M| & \text{if } a_\delta = 0 \text{ and } b_\delta t + c_\delta = v_2, \\ 0 & \text{if } a_\delta = 0 \text{ and } b_\delta t + c_\delta \neq v_2. \end{cases}$$

Proof. x is uniquely determined by $Ax + By + c = v_1$, and then $y - x = (y - v_1 + c)/A = \delta$; the remaining equation is $a_\delta s + b_\delta t + c_\delta = v_2$, which is affine in s . $\square$

Corollary 4.6 (exact criterion for the failure of (S-off)). *[proved in R5-C Cor 2.4, root-verified]* For $T = (y, t)$ and $V = (v_1, v_2)$ the reverse pair has offset $\delta' = (v_1 - y + c)/A = \rho(\delta)$ where $\rho(\delta) := 2c/A - \delta$. Writing $A_0 := \{d : a_d = 0\}$ for the degenerate set, in this family:

- (a) if (S-off) fails then $A_0 \cap \rho(A_0) \neq \emptyset$ (no hypothesis on $|M|$);
- (b) provided $|M| \geq 3$, the converse holds as well, so

$$(\text{S-off}) \text{ holds} \iff A_0 \cap \rho(A_0) = \emptyset.$$

For $c = 0$ this reads $A_0 \cap (-A_0) = \emptyset$: the degenerate set must be antisymmetric.

Proof. (a) A failure at the pair (T, V) needs $N(T, V) = N(V, T) = 0$, and by Lemma 4.5 each vanishing forces the corresponding offset to lie in A_0 ; those offsets are δ and $\rho(\delta)$. (b) Given $\delta \in A_0 \cap \rho(A_0)$, fix base coordinates y, v_1 realizing δ ; these satisfy $y \neq v_1$ unless $\delta = d^* := c/A$, and $d^* \notin A_0$ by Theorem 4.8 below, so $y \neq v_1$ and hence $T \neq V$. By Lemma 4.5 the pair $(t, v_2) \in M^2$ fails to give a violation only if $b_\delta t + c_\delta = v_2$ or $b_{\rho(\delta)} v_2 + c_{\rho(\delta)} = t$; each condition excludes exactly $|M|$ of the $|M|^2$ pairs, so at most $2|M|$ are excluded, and $|M|^2 > 2|M|$ exactly when $|M| \geq 3$. $\square$

The hypothesis $|M| \geq 3$ in (b) is necessary as stated: at $|M| = 2$ one has $|M|^2 = 2|M|$ and the two hit conditions can cover all four pairs, so the index condition need not produce a violating pair. The order-176 magma has $|M| = 16$, and the whole discussion below is for $|M| \geq 3$; see also Proposition 5.9, where $|M| = 2$ is disposed of on other grounds.

Lemma 4.7 (position conflicts). *[proved in R5-C Lemma 2.5, root-verified] [machine-verified: `problems/etp677/R5C_scripts/conflict_check.py`]* In (E1)–(E2) with all $b_j \neq 0$:

$$a_1 = 0 \implies a_3 \neq 0 \text{ and } a_4 \neq 0, \quad a_2 = 0 \implies a_4 \neq 0,$$

and no other implication holds: exactly the eight degeneracy patterns

$$\emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{2, 3\}, \{3, 4\}$$

are realizable. Equivalently, the forbidden position pairs are $\{1, 3\}, \{1, 4\}, \{2, 4\}$.

Proof. If $a_1 = 0$, (E2) gives $a_4 a_3 = -b_4 \neq 0$, so $a_3, a_4 \neq 0$. If $a_2 = 0$, (E1) gives $b_1 b_2 a_4 b_3 = 1$, so $a_4 \neq 0$. Realizability of the remaining patterns is exhibited by direct solution; exhaustive enumeration over $\text{GF}(3)$, $\text{GF}(4)$, $\text{GF}(5)$, $\text{GF}(8)$ prints exactly those eight patterns and no conflict violation. $\square$

4.4 Why the diagonal survives: the $\{1, 3\}$ subscript collision

This is the sharpest structural statement the campaign produced, and it is a complete answer for the family in which the counterexample lives. Setting $t = v$ in (S-off) gives $N(v, v) \geq 1$, i.e. E255. Both cases are governed by the *same* criterion of Corollary 4.6, but at *different* indices:

- *off-diagonal* (base coordinates $y \neq v_1$): the two offsets are δ and $\rho(\delta) = 2c/A - \delta$, two *distinct* indices; both must be degenerate for a failure;
- *diagonal* ($y = v_1$): here $\delta = \rho(\delta) = c/A$, and a failure would need the *single* index $d^* := c/A$ to be degenerate.

Theorem 4.8 (diagonal protection). *[proved in R5-C Thm 5.1, root-verified] [machine-verified: problems/etp677/R5C_scripts/conflict_check.py diag_check] For every translation-invariant base ($A + B = 1$, $A, B \neq 0$, any c) and every field of fibre scalars, if the fibre operations are affine $s \diamond_d t = a_d s + b_d t + c_d$ then $a_{c/A} \neq 0$. Consequently E255 holds in the entire translation-invariant blueprint family with affine fibres, however those fibre operations are chosen.*

Proof. Take the instance of (4.1) at parameter $d = -c/A$. Then

$$D_3(-c/A) = c/A = d^*, \quad D_1(-c/A) = -\frac{-c/A + c}{B} = -\frac{c(A-1)}{AB} = \frac{c}{A} = d^* \quad (\text{using } A-1 = -B).$$

So positions 1 and 3 carry the *same* index d^* . If $a_{d^*} = 0$ then $a_1 = a_3 = 0$, contradicting the forbidden pair $\{1, 3\}$ of Lemma 4.7. The index coincidence was machine-checked for every TI base with $p \in \{5, 11, 31, 41, 61, 71\}$. $\square$

Remark (scope — an internal correction). An intermediate report summary claimed that “the entire TI blueprint family satisfies E255” without qualification. That is *not* what is proved: Theorem 4.8 assumes affine fibre operations (§4.3, Lemma 4.4 is stated for affine fibres). A general-table TI fibre assignment, or a shift ansatz $T_\delta(s, t) = s + g_\delta(t - s)$ with arbitrary g_δ , is *not* covered by Theorem 4.8. Searches over those remain open (§5.8). The unqualified reading is recorded here only to be retracted.

Theorem 4.9 (the exact off-diagonal gap). *[proved in R5-C Thm 5.2, root-verified] For an off-diagonal pair the relevant index pair is $(\delta, \rho(\delta))$ with $\rho(\delta) = 2c/A - \delta$, and (4.1) forbids simultaneous degeneracy only when ρ coincides with one of the three conflict maps*

$$\sigma_1(e) = \frac{e - c}{B}, \quad \sigma_2(f) = -\frac{f + c(1 + B)}{B^2}, \quad \sigma_3(f) = \frac{(A^2 + B)(f + c)}{B} + (A + 1)c$$

(arising from the forbidden pairs $\{1, 3\}, \{1, 4\}, \{2, 4\}$ respectively), or with one of their inverses. For $c = 0$ this happens iff $-1 \in \{1/B, -1/B^2, (A^2 + B)/B\}^{\pm 1}$.

Corollary 4.10 (why 496 works and 176 does not). *[proved in R5-C §2.4, root-verified] [machine-verified: problems/etp677/R5C_scripts/m496.py] For the blueprint base $A = 1 + \beta$, $B = -\beta$, $c = 0$ on $\mathbb{F}_{31}$ with $\beta = 2$ a primitive 5th root of unity, one has $A_0 = \{\text{nonzero squares}\}$ and $\rho(\delta) = -\delta$; the blueprint’s own hypothesis “ -1 is a non-square mod 31” is precisely $A_0 \cap (-A_0) = \emptyset$. So (S-off) holds in the order-496 magma for a number-theoretic accident of $\mathbb{F}_{31}$, not for a structural reason. For the base $A = 4$, $B = 8$, $c = 0$ on $\mathbb{F}_{11}$ the three conflict maps of Lemma 4.7 become multiplication by 7, 6, 3 (with inverses 8, 2, 4), and $-1 = 10$ is not among them; so the two-element degenerate set $A_0 = \{1, -1\} = \{1, 10\}$ is admissible, and a SAT search produced a consistent operation family realizing exactly that A_0 — the order-176 magma.*

Summary of the $t = v$ mechanism. The diagonal is protected by an *index collision* ($D_1 = D_3$ at one specific parameter) which forces the $\{1, 3\}$ conflict to bite; the off-diagonal case involves two generically distinct indices and is protected only by an arithmetic accident of the base field. So the technique does not “fail at $t = v$ ”: it works there and *only* there. In this family the diagonal is strictly *easier* than the off-diagonal — the opposite of the working hypothesis that motivated the search.

What generalizes, and what does not. In the general (non-translation-invariant, general-table) framework the *subscript* half of the argument survives verbatim: positions 1 and 3 of (4.1) carry the pairs $(y, \Lambda_y x)$ and (y, x) , and they coincide exactly when $y \diamond x = x$, i.e. exactly at the pair that would have to be degenerate for $E255$ to fail at x . The *conclusion* does not. Lemma 4.7 is a statement about the scalars a_j of *affine* fibre operations: there degeneracy is the single scalar condition $a_d = 0$, which promotes a one-point collision to a collision of whole rows, and it is whole-row collisions that (E1)–(E2) forbid at positions $\{1, 3\}$. For a general fibre table, breaking $E255$ requires only a *one-point* collision, and no analogue of the $\{1, 3\}$ conflict lemma is known. In particular the statement

for every y, x with $y \diamond x = x$, the fibre operation $\diamond_{y,x}$ is left-injective

is proved here only for affine fibre operations (where it is the mechanism behind [2, Lemma 13.4]); for general tables it was the open R5-A question (Problem 11.8). **Note added:** Round 6 settled that question negatively — see Corollary 7.8. The general-table statement is false, and the correctly generalized mechanism (Theorems 7.3–7.6) is Problem 1.1 itself for a smaller magma. The recommendation that stood here in earlier drafts — “look for the general-magma analogue of this self-referential instance, not for counting bounds” — is therefore withdrawn in its literal form; what remains of it is Problem 11.8(b).

4.5 Minimality: exactly what was searched

Proposition 4.11. *[computational, partial: θ -family TI extensions with $p \in \{5, 11\}$, $q \leq 32$] [machine-verified: `problems/etp677/R5C_scripts/minimal_scan.py`] Within the translation-invariant blueprint family with θ -form fibre operations $s \diamond t = s - \theta(t - s)$, base prime $p \in \{5, 11\}$ and fibre field size $q \leq 32$, the smallest order at which non-right-cancellativity occurs is 176, and the smallest order at which (S-off) fails is 176. Representative rows of the scan:*

order	p	base (A, B, c)	q	non-cancellative	(S-off) violation
80	5	(2, 4, 0)	16	<i>no</i>	<i>none</i>
125	5	(2, 4, 0)	25	<i>no</i>	<i>none</i>
176	11	(4, 8, 0)	16	yes	(1, 10)
176	11	(5, 7, 0)	16	<i>no</i>	<i>none</i>
176	11	(6, 6, 0)	16	<i>yes</i>	<i>none</i>
176	11	(10, 2, 0)	16	<i>yes</i>	<i>none</i>
275	11	(4, 8, 0)	25	<i>no</i>	<i>none</i>

This is not a minimality theorem. It states minimality *within the scanned range and within one ansatz*. Whether a smaller non-right-cancellative finite $E677$ magma exists at all is open. In particular a *non-translation-invariant* fibre assignment over $\mathbb{F}_5$ would give order 80, and was not searched. Independently, a direct small-order SAT search for any $E677$ magma of order n carrying an (S-off)-violating pair returned UNSAT for $n = 3, 4, 5, 6, 7$; $n = 8$ ran 39 minutes without a verdict and was stopped; $n = 9, 10, 11$ were never run [machine-verified: `problems/etp677/R5C_scripts/soff_sat.py`].

Proposition 4.12 (no equational witness for (S-off) was ever possible). *[proved in R5-C §4.4, root-verified] [machine-verified: `problems/etp677/R5C_scripts/freemagma.py`] In the free 677 magma of [2, §13.2], let $t \neq v$ be two distinct generators. Since t is not a pair, the collapsing clause can never fire with right argument t , so $a \diamond t = (a, t)$ for every a , which is a pair and hence never equal to the generator v . Thus $N(t, v) = 0$, and symmetrically $N(v, t) = 0$. Hence (S-off) fails in the free 677 magma already, and no derivation of it from the E677 identities alone can exist; any proof had to use finiteness.*

Honest caveat. The free 677 magma has L_y surjective but not necessarily injective, so it is a 677 magma but need not be a left quasigroup. Proposition 4.12 therefore excludes proofs in the language $\{*\}$; it does not by itself exclude proofs in $\{*, \backslash\}$. The order-176 magma excludes those too.

Remark (a single-model coincidence, recorded so that nobody chases it). In the order-496 magma, on all 111 600 ordered pairs where the direction is forced ($N(v, t) = 0$, so (t, v) must carry the witness, and there $N(t, v) = 1$), the unique witness equals $w(t, v) = ((t \backslash v) * (t * v)) * v$ [machine-verified: `problems/etp677/R5C_scripts/term_full_check.py`]. (The term was *found* by `term_search.py`, which searches on a deterministic random sample of 1200 of those pairs; the all-pairs statement is the separate full-domain check `term_full_check.py`, which also reports that the predicate $w(t, v) * t = v$ is false on all 111 600 pairs forced in the other direction.) This is a single-model coincidence; Proposition 4.12 shows no term identity of this kind can be a theorem. And had the *conditional* form “ $N(t, v) = 0 \Rightarrow w(v, t) * v = t$ ” been a theorem, at $t = v$ it would read “ $N(v, v) = 0 \Rightarrow N(v, v) \geq 1$ ”, i.e. E255 — so the conditional-witness route is not easier than the main problem.

5 The obstruction map: routes that are closed

A campaign of this kind produces more closed routes than open theorems, and the closed routes are the more reusable output: each one is a statement about what a proof of Problem 1.1 may not do. This section collects them. They are stated with their exact strength — “provably circular”, “refuted by an explicit model”, and “exhausted under a stated computational budget” are three different things and are not conflated.

5.1 Equational and quasi-equational derivations

Proposition 5.1 (the equational meta-theorem). *[computational: F5 §4, scripts `saturate.py`, `saturate2.py`, `saturate3.py`] Consider the theory $T := \{\text{left quasigroup}\} + \text{E677}$ in the signature $\{*, \backslash\}$, with congruence closure under sound merges only, so that any derived equality is a theorem of T .*

- (a) *Depth-bounded full saturation — all instances, all terms up to depth d — is complete for $d \leq 4$ (1693 classes, 27 merges) and derives neither $c(a) = a$ nor $a * a = a$. At $d = 5, 6$ the closure passes 400 000 classes without deriving anything.*
- (b) *Pool-restricted saturation with unbounded term depth, instantiating E677 and KEY at all pairs from $\{\Lambda_a^{-k}(a) : |k| \leq K\} \cup \{S, \delta, c\}$ for $K \leq 6$ and iterated to a fixed point, leaves the pool at $2K + 1$ distinct classes (no collapse) and does not derive $c(a) = a$.*

Since the free 677 magma violates E255 [2, §13.2], any proof must use finiteness, and the only first-order consequence finiteness supplies is left cancellation. Item (b) is significant because the empirically minimal set of E677-instances whose failure destroys E255 (§6.4) lies inside the saturated pool.

Reading — and what this does *not* say. Proposition 5.1 reports depth-bounded and pool-restricted saturation experiments. Failure of a bounded search to derive E255 or (Q) is *not* a proof that no equational or quasi-equational derivation exists; unbounded saturation is not a decision procedure here, and no underivability certificate was produced. What the experiments support is the weaker, heuristic statement that the natural first-order strengthening does not collapse under the strongest saturation we could run, which is evidence — not proof — that a proof of Problem 1.1 will have to use a global counting or finiteness argument rather than an equational manipulation. A genuine theorem of this shape does exist for the *sibling* statement, but only in the language $\{*\}$: Proposition 4.12 proves that (S-off) has no derivation from the E677 identities alone, and even that argument does not reach the left-quasigroup (quasi-equational) setting, as the caveat after it records.

A separate bounded search for an equational derivation of *mediality* from E677 plus the left-division equations reached 168 182 distinct terms per side with at most 9 operation nodes and rewrite depth ≤ 3 , without the two closures meeting; a larger run with at most 17 operation nodes hit an explicit cap of 250 000 states per side, and is truncated evidence only [computational, partial: R3-F, r3f_saturation.py]. This does not show mediality is underivable.

5.2 The state map is circular

Iterating the master identity (2.2) produces a canonical dynamical system on the witness set, and it is the most natural finite-pumping mechanism available. It was pursued as the campaign’s Round-2 main line [9], and it is provably circular.

Theorem 5.2 (circularity of the state map). *[proved in R2 §2, (10); recoordinated in R3-C, root-verified] Let $\mathcal{E} := \{(t, v, x) : x \in \text{Fix}(L_t R_v)\}$, so $|\mathcal{E}| = n^2$ by Theorem 2.20. Then*

$$T(t, v, x) := (v, \Lambda_t x, \Lambda_v x)$$

is a well-defined self-map of $\mathcal{E}$, and

$$T \text{ is injective on } \mathcal{E} \iff T \text{ is surjective on } \mathcal{E} \iff (Q).$$

Proof. Well-definedness: let $x \in F(t, v)$ and set $p := \Lambda_t x = x * v$, $c := \Lambda_v x$; (2.2) with $y = v$, $u = x$ yields $c * p = \Lambda_v c = \Lambda_v^2 x$, so $c \in F(v, p)$. For the equivalence: given a target (v, p, c) , set $x := L_v c$; the target equation and (2.2) imply $p = x * v$, and a preimage under T is obtained by choosing t with $\Lambda_t x = p$, equivalently $t * p = x$. So the number of preimages of (v, p, c) is exactly $|\{t : t * p = x\}| = N(p, x)$, and T is bijective iff all such fibres are singletons, which is (Q). $\square$

Writing the iteration as $t_{k+1} = v_k$, $v_{k+1} = \Lambda_{t_k} x_k$, $x_{k+1} = \Lambda_{v_k} x_k$ makes the failure mode explicit: finiteness guarantees that every forward orbit is eventually periodic, but not that the *initial point* is periodic. A non-injective T can have tails feeding cycles, and excluding those tails is exactly the injectivity assertion. Moreover the subscripts of successive Λ ’s change along the orbit, so the finite order of any single permutation Λ_y does not close it.

The same circularity afflicts a variant proposed as an “easier foothold”: with $P := \{(t, v, A) : A * v = \Lambda_t(A)\}$ (of size n^2) and $\Xi(t, v, A) := (v, \Lambda_t A, \Lambda_v A)$, one has Ξ injective $\iff (Q)$, while surjectivity of Ξ would suffice; but surjectivity is Theorem 5.2 again.

5.3 Pure marginal counting on N

Proposition 5.3 (marginal insufficiency). *[proved in R2 §3, (17), root-verified] The constraints available on N from margins alone — all row sums and all column sums equal n , diagonal entries ≤ 1 , and the support bound $N(t, v) \leq s(v) := |\{p : N(v, p) > 0\}|$ — do not force $N \equiv 1$. The matrix*

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 2 & 1 & 0 \end{pmatrix}$$

has all row and column sums 3, diagonal entries ≤ 1 , and satisfies the support bound (the row support sizes are 2, 3, 2), yet is not all-ones. A constraint linking distinct entries of N is required.

The matrix above is not asserted to arise from an $E677$ magma; it demonstrates only that the current counting inequalities are insufficient. Theorems 2.22 and 2.23 are genuine linking constraints of the required kind — and they are still not enough:

Proposition 5.4 ($A + B +$ the transport are still insufficient). *[computational: R3-C §2.4, machine-checked certificate] There is a 5×5 nonnegative integer matrix N with row and column sums 5, diagonal $\equiv 1$ (so $E255$ “holds”), strongly connected support (Theorem 2.22 holds, with minimal Theorem 2.23 slack 4.0 over all 30 proper subsets), realizable transport data, and $N \not\equiv 1$:*

$$N = \begin{pmatrix} 1 & 0 & 2 & 2 & 0 \\ 0 & 1 & 2 & 2 & 0 \\ 2 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 0 & 1 \end{pmatrix}, \quad \begin{aligned} f_0 &= (2, 2, 0, 3, 3), & f_1 &= (2, 1, 2, 3, 3), & f_2 &= (2, 1, 0, 0, 4), \\ f_3 &= (4, 4, 1, 3, 4), & f_4 &= (0, 1, 1, 0, 4). \end{aligned}$$

Machine-checked: $|f_t^{-1}(v)| = N(t, v)$ for all t, v ; $N(f_t(u), u) > 0$ for all t, u ; row and column sums; $\text{tr } N = n$; strong connectivity.

5.4 The transport tree and the pair rotation

Two mechanisms were proposed in Round 3 for producing the missing linking constraint. Both are dead, and the reasons are worth recording because they are the kind of error that looks like progress.

Proposition 5.5 (the transport tree has branching factor 1). *[proved in R3-C §2, root-verified] Write $\tau(a, t) := (\Lambda_t a, a * t)$ for the transport map on M^2 ; it is a two-coordinate map and must not be confused with the three-coordinate state map T of Theorem 5.2. Let $a \in F(t, v)$ (i.e. $a * t = v$). Then $\tau(a, t) = (\Lambda_t a, v)$ is a witness of the fibre $(v, \Lambda_t^2(a))$, so the map $F(t, v) \rightarrow \text{supp}(\text{row } v \text{ of } N)$, $a \mapsto \Lambda_t^2(a)$, is injective and $N(t, v) \leq r_v$ (Definition 2.18). But the iteration adds nothing further:*

- (a) *Each point $(a, t) \in M^2$ has exactly one image $\tau(a, t)$: the transport is a function, not a correspondence. “A fibre of size k spawns k children” means the k points have k images which lie in k distinct fibres. It is the same k objects, redistributed. The “tree” is simply the functional graph of τ , with out-degree 1 at every node and total mass exactly n^2 at every depth. Mass is conserved identically at every level, so no pigeonhole on n^2 and no forced row-sum overflow can ever arise.*

- (b) *Distinctness survives to depth 2 and no further: at depth 1 the images lie in k distinct fibres, at depth 2 they lie in k distinct rows of N , and at depth ≥ 3 two paths merge exactly when a τ -collision occurs — which is a right-collision, i.e. the failure of (Q) itself.*
- (c) *The full combinatorial content of the transport is: for each t there is a function $f_t : M \rightarrow M$ (explicitly $f_t = R_t L_t^2$, i.e. $f_t(u) = (t * (t * u)) * t$) with fibre sizes $|f_t^{-1}(v)| = N(t, v)$ and $N(f_t(u), u) > 0$ for every u . The last condition is a tautology: for any s, u one has $s * (\Lambda_s u) = u$, so $N(\Lambda_s u, u) \geq 1$ automatically, with no algebra used.*

Proposition 5.6 (the pair rotation carries no E677 information). *[proved in R3-C §3, root-verified] [computational: cycle data on ten affine models] $P(r, s) := (r * s, r)$ is a bijection of M^2 for every left quasigroup, with $P^{-1}(u, r) = (r, \Lambda_r u)$; E677 is nowhere used. Its iterates satisfy $x_{k+1} = x_k * x_{k-1}$, and the E677 side condition $x_{k-1} = x_k * (x_{k-1} * x_{k+2})$ is literally the transport identity (2.3) again. Moreover P 's cycle type is wildly model-dependent — two affine models of the same order 11 have identical P -type but different τ -type, and two others have τ -type equal to P -type — so there is no cycle-type invariant to count two ways, and τ is never a power of P . The dictionary is $\tau(x_k, x_{k-1}) = (x_{k-2}, x_{k+1})$: the transport is one P -step backwards in the first coordinate and one P -step forwards in the second, which is exactly why it mixes distinct P -orbits and why P -orbit bookkeeping cannot see it. (Here τ is the transport map of Proposition 5.5, not the state map T of Theorem 5.2.)*

5.5 Local amplification of the row-distance bound is capped

Theorem 2.17 gives $|F_{pq}| \leq \lfloor n/2 \rfloor$; the Latin property needs 0. The obvious plan is to amplify by finding more maps that move F_{pq} off itself. It amplifies cleanly and does not amplify numerically [10].

Theorem 5.7 (amplification cap). *[proved in R3-A, root-verified] [machine-verified: `problems/etp677/r3a_search.py`] Fix $p \neq q$ and let $F := F_{pq}$.*

- (a) *For every $x \in F$ all four immediate neighbours $L_p x, L_q x, \Lambda_p x, \Lambda_q x$ lie outside F .*
- (b) *But $L_p x = L_q x$ on F , so the three image sets are $B := L_p(F) = L_q(F)$, $C := \Lambda_p(F)$, $D := \Lambda_q(F)$, each of size $|F|$, and the proved disjointness is only $A \cap B = A \cap C = A \cap D = \emptyset$ where $A := F$. Explicit five-element partial-KEY tables satisfying global (L4) realize each of $B \cap C, B \cap D, C \cap D, C \cap C'$ nonempty. Hence $n \geq |A \cup B| = 2|F|$ and one cannot replace $2|F|$ by $3|F|$ or $4|F|$.*
- (c) *All 84 nonempty words of length ≤ 3 in $\{L_p, L_q, \Lambda_p, \Lambda_q\}$ were tested against a local SAT model (five distinct elements, every row a permutation, global (L4), $x, x' \in F$ distinct, the four KEY instances at $(p, x), (q, x), (p, x'), (q, x')$, and $x' = w(x)$). Exactly 50 are UNSAT and 34 are SAT; and every UNSAT case is either a one-step exclusion from (a) or a reduction to x or to a one-step exclusion. The 34 locally feasible words include $\Lambda_p \Lambda_q, \Lambda_p^2, L_p^2, L_p L_q$ and their mixed analogues. The nonlinear candidate $x \mapsto x * v$ (with $v := p * x = q * x$) is also locally feasible, and its injectivity would already require (Q).*

Consequently the bound remains $|F_{pq}| \leq \lfloor n/2 \rfloor$: the four-instance KEY package is exhausted.

Proof of (a). Let $a := \Lambda_p x$. If $a \in F$ then $p * a = x = q * a$, hence $\Lambda_p x = a = \Lambda_q x$; combined with $p * x = q * x$, (L4) gives $p = q$. Same for $\Lambda_q x$. For $L_p x = L_q x = v$: if $v \in F$ then $p * v = q * v$,

while $p * x = q * x = v$ gives $\Lambda_p v = x = \Lambda_q v$, so (L4) at v gives $p = q$. (As a degenerate subcase, $v = x$ is impossible for the same reason.) $\square$

Exact remaining obstruction. The four KEY instances keep the x - and x' -packages separated unless a proposed word identifies an immediate predecessor/successor, and those identifications are precisely the (L4) proofs above. Longer words introduce fresh table entries rather than a second product-and-division equality at a common point. Any continuation must propagate KEY to those fresh entries and use finiteness to make two chains meet; listing more short words provably cannot improve the bound.

5.6 Parity, sign, 2-adic and Sylow-2 routes

Proposition 5.8 (the sign identity is true and empty). *[proved in R3-B Prop 5, root-verified] [machine-verified: `problems/etp677/R3B_scripts/analysis.py`] For any finite left quasigroup $(M, *)$ of order n , the bijection $P(r, s) = (r * s, r)$ satisfies*

$$\text{sign}(P) = (-1)^{n(n-1)/2} \cdot \prod_{y \in M} \text{sign}(L_y).$$

But it carries no parity information: on genuine E677 models both $\text{sign}(P)$ and $\prod_y \text{sign}(L_y)$ take both values already at $n = 7$. Hence no identity of the shape “ $\text{sign}(P) = \varepsilon(n)$ ” or “ $\prod_y \text{sign}(L_y) = \varepsilon(n)$ ” can hold, and no sign invariant of P can force n odd.

Proof. Let $D(r, s) := (r, r * s)$, which acts on each fibre $\{r\} \times M$ by L_r , so $\text{sign}(D) = \prod_r \text{sign}(L_r)$. Let $\text{sw}(a, b) := (b, a)$ (the coordinate swap — unrelated to the transport map τ of Proposition 5.5); sw fixes the n diagonal points and swaps the $n(n-1)/2$ unordered off-diagonal pairs, so $\text{sign}(\text{sw}) = (-1)^{n(n-1)/2}$. Finally $P = \text{sw} \circ D$. $\square$

Combined with Theorem 3.4 and Corollary 3.5, this closes the whole family: fixed-point parity and odd-cycle counting fail because the two order-16 models have completely different cycle data (L_y of type $(5, 5, 5, 1)$ for all y versus $(15, 1)$ for all y) at the same even order; the cycle-length obstruction $m(y) \notin \{2, 3\}$ is genuine but does not interact with parity; and even order forces nothing about idempotents (16 idempotents in one order-16 model, 1 in the other). **Any global mechanism for Problem 1.1 must be insensitive to the parity of n .**

5.7 The off-diagonal and isoperimetric programme

This is Section 4 restated as a closure. The order-176 magma refutes:

- any constraint of the form $N(t, v) + N(v, t) \geq k$ for $t \neq v$ with $k \geq 1$;
- $N + N^\top = 2J$ and $N + N^\top \geq 2J - I$;
- the isoperimetric threshold $e(O) \leq |O||O^c|$, uniformly over any one size class $2 \leq s \leq n-2$ — which is precisely the premise of the R3-C averaging implication.

N ’s off-diagonal support carries no usable information: it can be arbitrarily holey in both directions simultaneously while E255 holds.

Scope. The third item refutes one specific threshold, not “every isoperimetric upper bound”. Uniform upper bounds on $e(O)$ certainly exist — $e(O) \leq n|O|$ is trivial — and the order-176

magma says nothing about thresholds other than $|O||O^c|$, nor about bounds restricted to a family of sets that is not a full size class. What is closed is the R3-C averaging route in the form in which it was proposed.

The diagnosis is a *direction mismatch*, and it was visible before the counterexample. Theorems 2.22 and 2.23 are *lower* bounds on $e(O)$, obtained by propagating KEY through an invariant set; the transport gives only support information ($N(t, v) \leq r_v$ is an “enough room” statement); E255 needs an *upper* bound on $e(O)$ at the threshold $|O||O^c|$. No mechanism in the campaign produced such an upper bound, and Corollary 4.2(c) shows that no bound at that threshold can hold uniformly over a size class.

5.8 The extension framework: where a counterexample could still live

Within the blueprint extension framework of §4.3, three obstructions carve out the remaining habitat.

Proposition 5.9 (affine fibres are blocked; and $m = 2$ is always affine). *[proved in [2, Lemma 13.4] and Theorem 4.8; the $m = 2$ statement is proved below, root-verified] Theorem 4.8 (equivalently [2, Lemma 13.4]) forces E255 for every affine fibre extension of a base satisfying E255. The mechanism is exactly the $\{1, 3\}$ subscript collision of §4.4: affinity turns a would-be single-point collision into a whole-row collision, and [2, Lemma 13.4] only excludes whole-row collisions, whereas violating E255 needs only a single point. A general (non-affine) fibre table does not force the promotion. Consequently fibre size $m = 2$ cannot produce a counterexample, and $m \geq 3$ is the survival region.*

Proof of the $m = 2$ statement. For the product $(x, s) * (y, t) = (x \diamond y, s \diamond_{x,y} t)$ to have all rows permutations, each fibre operation $\diamond_{x,y}$ must have all its rows permutations: $t \mapsto s \diamond_{x,y} t$ is a bijection of M for every s . If $|M| = 2$, say $M = \mathbb{F}_2$, the only bijections of M are $t \mapsto t$ and $t \mapsto t + 1$, so $s \diamond_{x,y} t = t + \varepsilon(s)$ for some function $\varepsilon : \mathbb{F}_2 \rightarrow \mathbb{F}_2$; and every function $\mathbb{F}_2 \rightarrow \mathbb{F}_2$ is affine, $\varepsilon(s) = \alpha s + \gamma$ with $\alpha = \varepsilon(1) + \varepsilon(0)$, $\gamma = \varepsilon(0)$. Hence every admissible two-element fibre operation is of the affine shape $\alpha s + \beta t + \gamma$ with $\beta = 1$, and [2, Lemma 13.4] (Proposition 3.7(d)) applies: the extension satisfies E255. There is no genuinely non-affine two-element fibre. $\square$

Proposition 5.10 (idempotent-base circularity). *[proved in orchestrating agent, Round 5; recorded in the campaign registry, root-verified] Let the base G be a translation-invariant E677 model possessing an idempotent, and build an extension by fibres of size m . Then the base’s idempotent forces $\delta^* = 0$, and the $d = 0$ instance of (4.1) forces the fibre operation $\diamond_0$ to be itself an m -element E677 magma, while the E255-violation clause demands that $\diamond_0$ violate E255. This is circular: a counterexample of order $|G| \cdot m$ built this way requires a counterexample of order m . The same argument applies to non-translation-invariant fibre assignments: at an instance (a, a) the operation $\text{op}_{(a,a)}$ is forced to be an m -element E677 magma required to violate E255.*

*Consequently, within the translation-invariant blueprint framework, a counterexample can only be built over a base with no idempotent at all. The unique such translation-invariant base is the blueprint’s exceptional Type 2 model $x * y = 5x - 4y + c$ on $\mathbb{F}_{31}$ with $c \neq 0$, for which $\delta^* = c/5 \neq 0$. (Theorem 7.27 later shows it is the unique such base over any field.)*

Note the direction: what forces $\diamond_0$ to satisfy E677 here is the hypothesis (4.1) imposed on the fibre family, read at its $d = 0$ instance — not the assumption that the product magma satisfies E677. The distinction is the subject of Remark 7.1, and it is what makes this proposition a genuine obstruction rather than a circular one: the circularity it exhibits is a property of the construction being attempted, not of the argument.

Proposition 5.11 (no class-partition ansatz over the exceptional base). *[proved in orchestrating agent, Round 5; recorded in the campaign registry, root-verified] For the exceptional base $x*y = 5x-4y+1$ on $\mathbb{F}_{31}$, the four subscript maps $d \mapsto D_1, \dots, D_4$ of Lemma 4.4 generate the full affine group $\text{AGL}(1, 31)$ (21 is a primitive root of $\mathbb{F}_{31}$, and the maps have no common fixed point). Since 31 is prime, this action is primitive and admits no nontrivial block system. Hence the blueprint’s $\{0, \text{QR}, \text{QNR}\}$ trichotomy, available for the $c = 0$ base, has no analogue at $c \neq 0$: a translation-invariant search over the exceptional base has no shortcut, and a non-constant solution, if it exists, must be genuinely inhomogeneous.*

The associated SAT searches were discontinued on 2026-08-17 (compute-budget decision); only the runs that terminated are reported, each with its exact status [computational, partial: R5-A, discontinued]: translation-invariant fibre size $m = 3$ over the exceptional base: UNSAT, and (crucially) UNSAT *also without* the violation clause, so the $m = 3$ result says the extension does not exist at all, not that it satisfies *E255*; non-translation-invariant $m = 3$: UNSAT both with and without the violation clause, giving the machine theorem *the exceptional base admits no 3-element fibre extension whatsoever*; for $m = 4$: the translation-invariant search *with* the violation clause is UNSAT (kissat, ≈ 30 CPU-minutes), while the two existence questions (translation-invariant without violation, and pair-indexed) remain open at the time of writing; and $m = 5$ is the smallest fibre size *known* to have a nonempty solution space — the constant family $\diamond_\delta \equiv 2x - y$ over $\mathbb{F}_5$ was verified to give zero violations of (4.1), hence an order-155 product model. At $m = 8$ a nonlinear five-operation partial assignment (“gadget”) on $\mathbb{F}_2^3$, proposed by an independent reasoning model and re-verified in our parametrization, satisfies both fully determined instances ($d = 0$ and $d = 6$) while destroying every left unit in $\diamond_{25}$; however, fixing this particular gadget (one concrete choice of its parameters (S, h)) and searching the remaining 26 operations is UNSAT (kissat, under one minute), so that instantiation does not extend. The $m = 4$, $m = 5$, $m = 7$ and $m = 8$ classifications are *not* complete, and no claim is made about them. A dedicated backtracking solver and an independent adversarial re-derivation of the encoding are recorded in [15].

5.9 Summary table: the routes and their exact status

route	status	why
Equational / quasi-equational derivation of $E255$ or (Q) from $E677$ + left cancellation	<i>not</i> closed: computational evidence only	bounded saturation does not collapse (Prop. 5.1); this does not prove undervability. The free-magma argument (Prop. 4.12) is a genuine theorem, but only for the sibling statement (S-off) and only in the language $\{*\}$
Injectivity/surjectivity of the state map T on $\mathcal{E}$	closed (proved circular)	Theorem 5.2
Pure marginal counting on N	closed (counterexample matrix)	Prop. 5.3, Prop. 5.4
Iterated transport tree; pair rotation P	closed (proved)	branching factor 1 (Prop. 5.5); P is $E677$ -free (Prop. 5.6)
Local amplification of $ F_{pq} \leq \lfloor n/2 \rfloor$	closed (exhausted, local)	Theorem 5.7: all words of length ≤ 3
Parity / sign / 2-adic / Sylow-2	closed (refuted)	Theorem 3.4, Prop. 5.8
Off-diagonal strengthenings $N(t, v) + N(v, t) \geq k$; the uniform threshold $e(O) \leq O O^c $ over a size class	closed (refuted)	Corollary 4.2, order-176 magma. Only this threshold is refuted; other isoperimetric quantities are untouched
Affine (linear) models and affine fibre extensions as counterexamples	closed (proved)	Prop. 3.7, Theorem 4.8
Extension over an idempotent-bearing base	closed (proved circular)	Prop. 5.10
$677\text{-quasigroup} \Rightarrow \text{medial} \Rightarrow \text{Toyoda}$	<i>moot</i>	requires (Q), which is false (Prop. 2.13)

Remark (third-party corroboration). An independent project on the same problem [5], working in purely combinatorial/graph-theoretic language rather than the quandle/affine framework used here, independently established that all rows of a finite $E677$ magma are permutations and is likewise stalled at localized obstructions. That two methodologically disjoint efforts both report “local mechanisms are systematically insufficient” is the strongest evidence available that the remaining step is global.

6 The small-order landscape

6.1 Complete classification up to order 9

Theorem 6.1 (classification for $n \leq 9$). [*computational: F1 §5.1 (cp677.c) for the full range $n \leq 9$; corroborated over smaller ranges by F5 §5.3 ($n \leq 7$), R2 §5 ($n \leq 6$) and R3-B search.c (emptiness/nonemptiness only, $n \leq 9$) — see §6.1*] *The finite $E677$ magmas of order ≤ 9 are exactly the following.*

n	# labelled	up to iso.	E255?	quasigroup?	description
1	1	1	yes	yes	trivial
2	0	—	—	—	no model
3	0	—	—	—	no model
4	0	—	—	—	no model
5	6	1	yes	yes	dihedral (Takasaki) quandle $x * y = 2x - y$ over $\mathbb{Z}_5$; $ \text{Aut} = 20$
6	0	—	—	—	no model
7	1680	2	yes	yes	$x * y = 4x + y$ (has a left identity, not idempotent) and $x * y = 4x + 3y$, both over $\mathbb{Z}_7$
8	0	—	—	—	no model (45 canonical row-0 classes, all UNSAT)
9	—	1	yes	yes	affine over $\mathbb{F}_9 = \mathbb{F}_3[t]/(t^2 + 1)$, not over $\mathbb{Z}/9$

In particular every E677 magma of order ≤ 9 is an affine quasigroup and satisfies E255, so there is no counterexample to Problem 1.1 of order ≤ 9 .

Cross-validation, and its exact extent. The complete classification through order 9 — counts of labelled models, isomorphism classes, and the identification of the order-9 model — rests on implementation (i) alone. The other three implementations agree with it, but over strictly smaller or weaker ranges; we spell this out because “cross-validated four ways” would overstate it.

- (i) A constraint-propagation DFS over Cayley-table cells with row `alldifferent` and the derived lemmas as propagators [machine-verified: `problems/etp677/F1_tools/cp677.c`], cross-checked at $n \leq 6$ against naive brute force and at $n = 5, 7$ with and without the derived-lemma propagators. *This is the source of the complete statement through $n = 9$* , including the 45 canonical row-0 classes at $n = 8$ and the 67 at $n = 9$.
- (ii) A propagating backtracking enumerator plus `kissat` encodings [machine-verified: `problems/etp677/F5_scripts/enum677.py`, `sat677.py`]: complete classification through $n = 7$ (orders 2, 3, 4, 6 empty; one class at 5; two at 7). In the report of [8] order 8 was still undecided and orders 9, 10 were not reached.
- (iii) A division-row CSP with arc consistency, complete for orders ≤ 6 only, cross-checked by direct enumeration of all n -tuples of row permutations at $n \leq 4$ [proved in R2 §5, root-verified]. Its order-7 run was started but not finished, and [9] makes no order-7 claim.
- (iv) A C enumerator with KEY propagation and the proved prunes of Lemmas 2.6, 2.15, 2.17 plus isomorphism reduction on row 0 [machine-verified: `problems/etp677/R3B_scripts/search.c`]: independently confirms that $n = 2, 3, 4, 6, 8$ are *empty* and that $n = 5, 7, 9$ are *nonempty*, with the models re-verified in Python. It does not report the order-9 isomorphism count.

So emptiness at 2, 3, 4, 6, 8 is confirmed twice independently (i)+(iv), the orders ≤ 7 three or four times, and the order-9 enumeration *once*.

On the order-9 entry. The search finds exactly 2 of the 67 canonical row-0 classes satisfiable, one solution each; these are the same magma relabelled, affine over $\mathbb{F}_9$ with $(a, b) = (1, t + 2)$ and $(1, 2t + 2)$ — the two parameters are Frobenius conjugates. We record this as *one* isomorphism class, following [6]; a separate script [11] refers to “two $\mathbb{F}_9$ models”, meaning the two parameter pairs. *This is flagged as a point where the two internal reports use “model” with different scopes; the mathematics is not in dispute.*

6.2 Orders 10 and beyond: exactly what is settled

Proposition 6.2. *[computational, partial: see below]*

- (a) *There is no commuting affine E677 magma of order 10, 12 or 14: their 2-parts are 2, 4, 2, all smaller than 16 (Theorem 3.4(b)).*
- (b) *At order 10, the sub-case $m(0) = 4$ is fully exhausted (all 11 canonical row-0 classes, $\approx 4 \cdot 10^8$ nodes, UNSAT). Since $m(y) = 4$ forces E255 to fail at y (Theorem 2.15(b)), that sub-case is thereby settled at order 10. The remaining classes $m(0) = 5, \dots, 10$ (19 subproblems) were still running when the underlying report was written, as was an independent C search ($> 1.9 \cdot 10^8$ nodes, no model found); **all of these were subsequently discontinued without a verdict** (compute-budget decision, 2026-08-17), so nothing is claimed about $m(0) \geq 5$ at order 10, and in particular $m(y) = 5$ is not excluded at order 10.*
- (c) *Whether a non-affine E677 magma exists at order 10, 12 or 14 is **open**. All known models of all orders are affine, so the expected answer is “no”, making 16 the expected smallest even order — but this is an expectation, not a theorem.*

6.3 A deliberately partial order-11 census

The following is the campaign’s Round-3 census attempt [13], reported with its exact coverage.

Proposition 6.3. *[computational, partial: 22 of 87 pointed cycle types at order 11; 0 of 174 at order 13] [machine-verified: `problems/etp677/r3f_enum.py`, `r3f_n11_merged.json`] Relabelling an arbitrary element to 0 and conjugating the left-division row P_0 to its standard pointed cycle representative, and using $m(0) \notin \{2, 3\}$ (Theorem 2.15(a)), the order-11 classification splits into*

$$p(10) + p(7) + p(6) + p(5) + p(4) + p(3) + p(2) + p(1) + p(0) = 42 + 15 + 11 + 7 + 5 + 3 + 2 + 1 + 1 = 87$$

*pointed cycle types, which cover every isomorphism class at least once. **22 of these 87 cases have been exhaustively decided, and all 22 are UNSAT.** The completed cases are those with $m(0) = 1$ and remainder cycle type among $1^{10}; 1^8 2; 1^7 3; 1^6 2^2; 1^6 4; 1^5 2 3; 1^5 5; 1^4 2^3; 1^4 2 4; 1^4 3^2; 1^4 6; 1^2 2^2 3; 1^3 2 5; 1^3 3 4; 1^3 7; 1^2 2^4; 1^2 2^2 4; 1^2 2 3^2; 1^2 2 6; 1^2 3^3; 1^2 2^5; 2^5$, totalling 6 592.116 durable solver-seconds with *kissat* 4.0.4.*

This is not a classification of order 11. 65 of the 87 cases are undecided.

Proposition 6.4 (the affine benchmark at orders 11 and 13). *[machine-verified: `problems/etp677/r3f_enum.py`] Over $\mathbb{Z}_{11}$ the affine equations have exactly the four roots $(F, G) = (10, 2), (6, 6), (5, 7), (4, 8)$; the constant multipliers $G^2(F + 1) + G + 1$ are respectively 3, 6, 5, 10, all units, so $c = 0$ in every case. Since $F + G = 1$ in all four, all four are idempotent. Their canonical hashes are distinct, so these are four pairwise non-isomorphic affine isomorphism types: Latin, idempotent Alexander quandles, medial, satisfying E255, each with automorphism group of order 110. At order 13 the affine equations have the single root $G = 11$ with $F = 9$; here $F + G \neq 1$, so translation normalizes c to 0, and the affine benchmark is one class.*

Remark (provisional case). The pointed cycle type $1 \mid (10)$ at order 11 — the case in which the four affine models live — was run far enough to *find* all four, with blocking clauses installed, but the final UNSAT proof excluding a fifth model did not finish. So “exactly four models

of type $1 \mid (10)$ ” is **provisional** and this case is *not* counted among the 22 completed cases. Regression censuses at orders 5 and 7 with the same enumerator returned one class and two classes respectively, matching Theorem 6.1.

6.4 Near-misses and how thin the implication is

Proposition 6.5 (an order-11 near-miss). *[computational: F5 §5.1, `nearmiss.py`] Take the affine E677 quandle of order 11, $x * y = -x + 2y \bmod 11$, and swap the entries $0 * 0$ and $0 * 1$ in row 0. The result is still a left quasigroup, satisfies E677 at 116 of the 121 instances — failing exactly at $(y, x) \in \{(0, 0), (0, 1), (0, 2), (1, 6), (4, 0)\}$ — and violates E255 exactly at $x = 0$, where $c(0) = 2$. Columns 0 and 1 are the only non-permutation columns, as Theorem 2.7 requires. The same recipe (one transposition in one row of an exact model) was checked at the two orders $n = 5$ and $n = 11$, and yielded exactly 5 violated instances at both; no proof for general orders is available, and the pattern is recorded as an observation on those two orders only. In both cases the five violated pairs are*

$$(x_0, x_0), \quad (x_0, x_0 \setminus x_0), \quad (x_0, x_0 * x_0), \quad (x_0 \setminus x_0, x_0 \setminus (x_0 \setminus x_0)), \quad (x_0 * (x_0 * x_0), x_0).$$

Proposition 6.6 (exact robustness profile). *[computational: F5 §5.2, `robust.py` + `kissat`] Encoding “left quasigroup of order n + E255 fails at 0 + at most k instances of E677 violated” with a sequential-counter cardinality constraint:*

n	minimum number of violated E677 instances when E255 fails
4	4 ($k \leq 3$ UNSAT, $k = 4$ SAT; violated set $\{(0, 3), (1, 2), (2, 1), (3, 0)\}$)
5	5 ($k \leq 4$ UNSAT, $k = 5$ SAT; violated set $\{(0, 0), (0, 2), (0, 3), (1, 0), (2, 1)\}$)
6	≥ 4 ($k \leq 3$ UNSAT; $k = 4$ hit the 600 s limit)
11	≤ 5 (explicit witness, Proposition 6.5)

Note that $k = 0$ UNSAT is precisely “no counterexample of order n ”, so this re-derives the non-existence results from scratch. The implication is thin: it is destroyed by 5 broken instances out of n^2 .

Proposition 6.7 (local repair around the order-25 model is excluded to radius 15). *[machine-verified: `problems/etp677/r3a_search.py`] Take the first-branch model $x * y = 2x - y$ on the additive group $\mathbb{F}_5^2$ (order 25; Latin, E677, E255; simultaneously a model on the additive groups of $\mathbb{F}_{25}$ and $\mathbb{Z}_5 \times \mathbb{Z}_5$).*

- (a) *Of the 7 500 single-row transpositions, exactly 24 make column 0 omit 0, and every one of those violates exactly five E677 instances. Of the 115 000 three-cell single-row derangements, exactly 552 make column 0 omit 0, with violation histogram $\{7:24, 8:48, 9:480\}$ — so a third changed cell does not repair the near-miss.*
- (b) *A SAT encoding (one-hot cells, row-Latin, all 625 E677 instances with one-hot intermediates, $\neg X[r, 0, 0]$ for every r (where $X[r, u, v]$ is the cell variable “ $r * u = v$ ”), and a Sinz sequential counter bounding the Hamming distance from the model) returns UNSAT for radius 2, 3, 4, 5, 6, 8, 10, 12 and 15; the radius-15 instance had 56 235 variables and 1 568 729 clauses, and subsumes all smaller radii. **Radius 20 timed out after 180 seconds and is recorded as UNKNOWN, not UNSAT.** No DRAT certificate was requested; these are reproducible solver results, not proof-assistant-certified exclusions.*

Remark (the second linear branch has no small seed). For $Q(T) = T^4 + T^3 + 2T^2 + 2T + 1$, requiring $Q(G) = 0$ with $G + G^3$ invertible, an exhaustive enumeration found *no* solution in any of: scalar $\mathbb{Z}_{15}, \mathbb{Z}_{21}, \mathbb{Z}_{25}, \mathbb{Z}_{27}, \mathbb{Z}_{33}$, the fields $\mathbb{F}_{25} = \mathbb{F}_5[u]/(u^2+2)$ and $\mathbb{F}_{27} = \mathbb{F}_3[u]/(u^3+2u+1)$, the full endomorphism ring $\text{End}(\mathbb{Z}_5 \times \mathbb{Z}_5) = M_2(\mathbb{F}_5)$ (625 candidates), and all 243 endomorphisms of $\mathbb{Z}_3 \times \mathbb{Z}_9$ [machine-verified: `problems/etp677/r3a_search.py`]. Consistently, $Q \equiv (T^2 + 2T + 2)^2 \pmod{3}$ with the quadratic irreducible, and Q is irreducible of degree 4 mod 5. So there is no second-branch seed to perturb at those orders.

6.5 The profile a finite counterexample must match

Collecting the pointwise constraints, we record the target profile. Let M be a finite $E677$ magma violating $E255$, of minimum order n , and let y be a violating element.

1. M is 1-generated as a left quasigroup: the subset generated by y under $*$ and $\backslash$ is closed, is again a finite $E677$ magma, and $E255$ still fails at y there; minimality forces it to be all of M .
2. Column y does not contain the value y (Theorem 2.7); hence R_y is not surjective, hence not injective, so M is not a quasigroup and some value occurs at least twice in column y .
3. y has no left unit, while it does have the always-existing unique right unit $e(y) = \Lambda_y(y)$.
4. $m(y) \geq 4$ (Theorem 2.15); and $m(y) = 4$ automatically violates.
5. $(y * y) * y = \Lambda_y^2(y)$ always, but the R_y -orbit of y is a pure tail that never returns, and y has no R_y -preimage at all (Corollary 2.10).
6. $\Lambda_y^3(y) * y \neq \Lambda_y^4(y)$; $\Lambda_y^4(y) * \Lambda_y^4(y) \neq \Lambda_y^5(y)$; and for *every* $c \in M$, $c * c \neq y \backslash c$ — the squaring map must avoid the entire graph of the permutation Λ_y (Theorem 2.14). These are n simultaneous disequalities and constitute the sharpest single obstruction available.
7. $\sum_w |\text{Fix}(L_w)| < n$: the disjoint sets $\text{Fix}(L_w)$ fail to cover M , and $y \notin \bigcup_w \text{Fix}(L_w)$.
8. M is not affine over any abelian group (Proposition 3.7), and not an affine fibre extension of an $E255$ base (Theorem 4.8).
9. $n \geq 10$ (Theorem 6.1); $n \geq 11$ if the community's unreviewed DRAT results for $n \leq 10$ are accepted. Also $n \neq 8$ since no $E677$ magma of order 8 exists at all.
10. Every L_y and every Λ_y is still a permutation — that is forced; what must fail is only the column structure. Concretely N has all row and column sums n , 0/1 diagonal, $N(t, v) \leq |\text{Im}(R_v)|$, $\sum_t |\text{Im}(R_t)| \geq n^{3/2}$, strongly connected support (Theorem 2.22), and at least one diagonal entry 0.

Item 10 uses two further counting facts we record for completeness.

Lemma 6.8. *[proved in F1 Lemma H, Cor H2, root-verified] With r_t, c_v as in Definition 2.18: $N(t, v) \leq r_v$ for all t, v , and $\sum_{t \in M} r_t \geq n^{3/2}$.*

Proof. Let $a * t = v$. Identity (2.2) with $u := a$, $y := t$ gives $\Lambda_t(a) * (a * t) = \Lambda_t^2(a)$, i.e. $\Lambda_t^2(a) = R_v(\Lambda_t(a)) \in \text{Im}(R_v)$. As a ranges over the $N(t, v)$ solutions of $a * t = v$, the values $\Lambda_t^2(a)$ are $N(t, v)$ distinct elements of $\text{Im}(R_v)$. For the second claim, let $\rho := \sum_t r_t = \sum_v c_v$. From $n = \sum_t N(t, v) \leq c_v r_v$ we get $c_v \geq n/r_v$, so $\rho \geq n \sum_v 1/r_v \geq n \cdot n^2/\rho$ by AM–HM, i.e. $\rho^2 \geq n^3$. $\square$

So a finite $E677$ magma cannot be very far from a quasigroup: its columns carry on average at least $\sqrt{n}$ distinct values. (Not enough to force $r_t = n$; and the order-176 magma has $r_t = 146$ against $n^{1/2} = 13.3$, so this bound is far from binding in practice.)

6.6 Sublemmas that are false

Recorded to save re-derivation. All are refuted by explicit models [computational: F1 §8, R3-B §6].

- “*Finite $E677$ magmas are idempotent.*” False: in $x*y = 4x+y$ over $\mathbb{Z}_7$ only 0 is idempotent.
- “*Every left translation of a finite $E677$ magma has a fixed point.*” False: same model, L_w is fixed-point-free for $w \neq 0$. So Corollary 2.9’s sufficient condition is strictly sufficient.
- “*The squaring map $S(c) = c*c$ is bijective.*” False: for $x*y = 4x+3y$ over $\mathbb{Z}_7$, $S(c) = 7c = 0$ is constant. So Theorem 2.14(3) cannot be closed by invoking bijectivity of S .
- “ *$\delta(y) = (y * y) * y$ is injective.*” False: $\delta \equiv 0$ in $x * y = 4x + y$ over $\mathbb{Z}_7$.
- “ *$E677$ forces $(y * y) * y = y * (y * y)$.*” False whenever $m(y) \nmid 4$.
- “ *$m(y) \in \{2, 3\}$ is a cheap counterexample generator.*” False: those cycle lengths are impossible (Theorem 2.15(a)). $m(y) = 4$ is a valid target since it forces failure, but it is unrealisable for $n \leq 10$.
- “*Every finite $E677$ magma has odd order.*” False (Theorem 3.4).
- “*Every finite $E677$ magma is right-cancellative*” ((Q)). False (Proposition 2.13).
- “ *$N + N^\top = 2J$ ” and “ $N(t, v) + N(v, t) \geq 1$ for $t \neq v$ ”.* Both false (Corollary 4.2).
- “*The right-unit map $e(x) = x \setminus x$ is an endomorphism.*” False: 1050 of 5929 violations in M_{77}^{NT} (Theorem 7.24).
- “ *e is injective.*” False already at order 9, in the unique $E677$ magma of that order, where 0 is a right identity and e is constant (Theorem 7.25). The two failures are independent: e is bijective in M_{77}^{NT} and is an endomorphism in M_9 .
- “*Every non-right-cancellative finite $E677$ magma is idempotent.*” True in every model known before Round 6, and false: M_{77} has 11 idempotents out of 77 (Theorem 7.18).
- “*The blueprint compatibility equations force translation invariance.*” False: M_{77}^{NT} (Theorem 7.20), by the diagonal-decoupling Lemma 7.19.

7 Round 6: self-reference, the order-77 models, and idempotent-free spectra

The sections above report the campaign as it stood when this draft entered adversarial review. A sixth round ran afterwards, in five parallel sub-campaigns [17, 18, 19, 20, 21], and its results are collected here rather than woven into the earlier text, so that a reader can see exactly which statements are new and what each of them cost. Four of them change the earlier text materially.

1. The general-magma analogue of the $\{1, 3\}$ subscript collision is now identified exactly. It is a *self-reduction*: it says that Problem 1.1 holds one level down. In particular Problem 11.8(a) is answered, and the answer is **no** (§7.1).
2. The smallest known non-right-cancellative order drops from 176 to 77, and the order-77 model is the first that is *not* idempotent (§7.3). A second order-77 model is the campaign’s first extension that is not translation-invariant.
3. A new family of counting identities produces an *upper*-bound criterion for E_{255} of exactly the shape Section 5.7 says is missing (§7.2) — together with a proof that no term-level witness for it can exist.
4. A conjecture that Round 6 itself proposed, “ $e(x) = x \setminus x$ is an automorphism”, was refuted twice over, by two independent mechanisms (§7.4). The evidence that had supported it was an artefact of the campaign’s benchmark set, which is the third occurrence of one and the same trap; §7.6 records the pattern as a methodological finding in its own right, and §7.7 gives the resulting model zoo.

7.1 The general-magma $\{1, 3\}$ collision is a self-reduction

Write the E_{677} instance at (y, x) as $x = y * (x * ((y * x) * y))$. Its four multiplication nodes carry the argument pairs

$$P_1 = (y, \Lambda_y x), \quad P_2 = (x, (y * x) * y), \quad P_3 = (y, x), \quad P_4 = (y * x, y),$$

which are exactly the indices $D_1, \dots, D_4$ of Lemma 4.4 in the translation-invariant coordinates. Solving the collision equations gives

$$\begin{aligned} P_1 = P_3 &\iff y * x = x, & P_2 = P_4 &\iff y * x = x \text{ and } x * y = y, \\ P_1 = P_4 &\iff y * x = y \text{ and } y * y = x, & P_2 = P_3 &\iff y = x \text{ and } \delta(x) = x. \end{aligned}$$

Lemma 7.1 (the $\{1, 4\}$ and $\{2, 4\}$ configurations collapse). *[proved in R6-A Lemmas 1.5 and 1.5b, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`] In a finite E_{677} magma: if $y * x = x$ and $x * y = y$ then $x = y$ and x is idempotent; if $y * x = y$ and $y * y = x$ then $x = y$ and x is idempotent.*

Proof. First statement: E_{677} at (y, x) reads $x = y * (x * ((y * x) * y)) = y * (x * (x * y)) = y * (x * y) = y * y$, and symmetrically $y = x * x$. Hence $\delta(x) = (x * x) * x = y * x = x$, so y is a left unit of x ; left units are unique (Theorem 2.7), so $y = \delta(x) = x$ and then $x * x = y = x$. Second statement: $y * x = y$ says $x = \Lambda_y(y) = y_1$ and $y * y = x$ says $x = y_{-1}$; so $\Lambda_y^2(y) = y$, i.e. $m(y) \mid 2$. By Theorem 2.15(a) $m(y) \neq 2$, so $m(y) = 1$. $\square$

So of the three forbidden position pairs $\{1, 3\}$, $\{1, 4\}$, $\{2, 4\}$ of Lemma 4.7, only $\{1, 3\}$ — i.e. $y * x = x$, “ y is a left unit of x ” — is realisable outside the totally degenerate idempotent case, where all four positions coincide. That is the configuration to generalize.

Theorem 7.2 (congruence classes have equal size). *[proved in R6-A Thm 1.1, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`] All classes of a congruence θ on a finite E677 magma M have the same size.*

Proof. $G := M/\theta$ is a finite E677 magma. For a class B and $y \in M$ we have $L_y(B) \subseteq [y] \cdot B$, and $B \mapsto [y] \cdot B$ is the bijection $L_{[y]}$ of the class set. Since L_y is injective, $|L_y(B)| = |B|$, and $\bigsqcup_B L_y(B) \subseteq \bigsqcup_B [y] \cdot B = M$ with both sides of size n ; so $L_y(B) = [y] \cdot B$ and $|B| = |[y] \cdot B|$ for every B . By Theorem 2.22 applied to G , the group generated by the $L_{[y]}$ acts transitively on the classes. $\square$

Theorem 7.3 (the total collision: a class is a smaller E677 magma). *[proved in R6-A Thm 1.2, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`] Let θ be a congruence on a finite E677 magma M and let X be a class with $X * X = X$ (i.e. $[X]$ is idempotent in M/θ). Then*

- (a) X is closed under $*$, and $(X, *)$ is itself a finite E677 magma;
- (b) for $s \in X$: E255 holds at s in M if and only if E255 holds at s in X ;
- (c) for all $t, s \in X$ the four positions $P_1, \dots, P_4$ of the E677 instance at (t, s) all carry the class pair (X, X) — the total $\{1, 2, 3, 4\}$ collision.

Proof. (a) Closure is the hypothesis read elementwise; E677 is an identity, hence inherited, and $L_t|_X$ is an injective self-map of the finite set X . (b) If $a * s = s$ in M then $[a][s] = [s]$ and $[s][s] = [s]$ in M/θ , so uniqueness of left units in M/θ (Theorem 2.7) gives $[a] = [s]$, i.e. $a \in X$; the converse is trivial. (c) $X * X = X$ gives $\Lambda_X X = X$, hence $P_1 = (X, X)$; $(X * X) * X = X$ gives $P_2 = (X, X)$; $P_3 = (X, X)$; and $P_4 = (X * X, X) = (X, X)$. $\square$

Remark (logical status of the collapse: an observation, never a construction premise). The statement “at a diagonal instance all four positions carry the same subscript, so the fibre operation there is itself an E677 magma” appears twice below, and the two readings must be kept apart, because one of them would be circular. This paragraph fixes the logical direction once and for all.

What it is. The collapse is a consequence of idempotency of the base point: if $q \diamond q = q$, then the four subscripts $P_1, \dots, P_4$ of the instance at (q, q) are all equal to (q, q) (Theorem 7.3(c), applied in the quotient). It is thus an observation about the structure of one instance of the system (4.1) — about which subscripts occur in which positions — derived from idempotency, and on its own it asserts nothing about any magma.

The two directions, and where each is used.

- (i) *Restriction.* If a finite E677 magma is given and X is a congruence class with $X * X = X$, then $(X, *)$ inherits E677 (Theorem 7.3(a)). Hypothesis: the large magma; conclusion: the block. This direction is what Corollary 7.8, §7.6 and Theorem 7.28 (hence $\text{FiberSpec}(0) \subseteq S$) use, always with the large magma already in hand.

- (ii) *Construction.* Equation (4.1) is a *hypothesis* imposed on a fibre family, not a consequence of anything; its diagonal instance, written out, is literally the *E677* identity for the single operation $\diamond_{q,q}$. That is an identity between two explicitly given finite systems of equations, and it may be read in either direction without any magma being assumed. It is used in the construction direction — exhibit a $\diamond_{q,q}$, check that it satisfies *E677*, and the diagonal instance of (4.1) is thereby discharged — in Theorems 7.18 and 7.20, and nowhere else.

The circular reading, explicitly disclaimed. It would be circular to infer “ $\diamond_{q,q}$ satisfies *E677*” from “the product magma satisfies *E677*” and then to use that inference as a step towards proving that the product satisfies *E677*. No argument in this paper does so: in (i) the product is a hypothesis and the block is the conclusion; in (ii) neither the product nor the block is assumed — an explicit operation is exhibited and its equations are verified, by hand and by machine. In particular *no theorem in this paper takes the collapse as a premise of its proof*; §7.1 draws from it a *negative* conclusion (Corollary 7.8: the mechanism is a self-reduction, hence cannot be an engine), which is the opposite of using it as one.

The non-idempotent branch, for contrast. If q is *not* idempotent, write x_q for its unique left unit (Theorem 2.7). Then the operation attached to the pair (x_q, q) occurs in exactly three instances of the system: once at positions 1 and 3 simultaneously — the $\{1, 3\}$ collision, which holds because $x_q \diamond q = q$ forces $L_{x_q}^{-1}(q) = q$ — and once each at positions 2 and 4 of two further instances. It is therefore *not* forced to satisfy *E677*, and that is exactly the gap which Theorem 7.5 measures and which Theorem 4.8 closes only under affineness. A later round verified that this three-occurrence picture is independent of the base, on seven different bases [proved in [22], Thm R7-B.0, root-verified]; it is stated in full as Theorem 8.1.

Corollary 7.4 (what Theorem 4.8 really is). *[proved in R6-A Cor 1.2', root-verified] In the translation-invariant family with $c = 0$ the index maps of (4.1) are linear, so all four vanish at $d = 0$ and the $d = 0$ instance of (4.1) reads*

$$s = t \diamond_0 (s \diamond_0 ((t \diamond_0 s) \diamond_0 t)) \quad \text{for all } s, t,$$

*i.e. $(M, \diamond_0)$ is itself an *E677* magma. Consequently $a_0 \neq 0$ is forced (a left-constant operation cannot satisfy *E677* on a set of size ≥ 2), which is Theorem 4.8 at $c = 0$; and *E255* for the extension at (x, s) is equivalent to *E255* for $(M, \diamond_0)$ at s , which holds because $\diamond_0$ is affine and affine *E677* magmas are Latin (Theorem 3.1(d)). For $c \neq 0$ the base is not idempotent, the classes satisfy $X * X \neq X$, and only the partial $\{1, 3\}$ collision survives.*

For the partial collision the conclusion is strictly weaker, and this is the whole story.

Theorem 7.5 (the partial $\{1, 3\}$ collision gives only non-constancy). *[proved in R6-A Thm 1.3, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`] Let Y, X be congruence classes with $Y * X = X$ and $m := |X| = |Y| \geq 2$. Then the block map $Y \times X \rightarrow X$, $(t, s) \mapsto t * s$, is not independent of t .*

Proof. $Y * X = X$ gives $\Lambda_Y X = X$, and *E677* in M/θ gives $X * (X * Y) = \Lambda_Y X = X$. Suppose $t * s = \mu(s)$ for all $t \in Y$, $s \in X$; then $\mu = L_t|_X$ is a bijection of X . *E677* at (t, s) reads $s = t * (s * (\mu(s) * t))$ with $s * (\mu(s) * t) \in X$, so $s = \mu(s * (\mu(s) * t))$. As μ is bijective, $s * (\mu(s) * t)$ is independent of t ; as L_s is injective, $\mu(s) * t$ is independent of t ; but $t \mapsto \mu(s) * t$ is injective on Y . Hence $|Y| = 1$. $\square$

Theorem 7.6 (the block problem is Problem 1.1 verbatim). *[proved in R6-A Thm 1.4, root-verified]* With Y, X, m as above put $\nu(s, s') := \#\{t \in Y : t * s = s'\}$ for $s, s' \in X$. Then all row sums and all column sums of the $m \times m$ matrix ν equal m , and $\nu(s, s) = N(s, s) \leq 1$. Consequently

- $E255$ holds at every point of $X \iff \text{diag } \nu \equiv 1$ — literally (P) for the block;
- the block map is left-injective $\iff \nu \equiv 1$ — literally (Q) for the block.

Proof. Rows: for fixed s , $t \mapsto t * s$ maps Y into X . Columns: for fixed s' and each $t \in Y$ there is exactly one $s \in X$ with $t * s = s'$, because $L_t|_X$ is a bijection of X . Diagonal: $\nu(s, s) \leq N(s, s) \leq 1$ by Lemma 2.6. $\square$

Proposition 7.7 (blocks are arbitrary). *[proved in R6-A Prop 1.6, root-verified] [machine-verified: problems/etp677/R6A_scripts/r6a_blocks.py]* Let X be any finite E677 magma and G any finite E677 magma with an idempotent g . In $X \times G$ the sets $X \times \{h\}$ form a congruence, the class over g satisfies $(X \times \{g\})^2 = X \times \{g\}$, and its block operation is the operation of X . So the block at a total-collision pair can be an arbitrary finite E677 magma.

Corollary 7.8 (answer to Problem 11.8(a)). *[proved in R6-A §1.4–1.6, root-verified]* The affine conclusion does **not** generalize: there are blueprint extensions and pairs y, x with $y \diamond x = x$ at which the fibre operation $\diamond_{y,x}$ is not left-injective. Take the direct product $M_{176} \times G$ with G any E677 magma carrying an idempotent g (Proposition 7.7); the block over g is the order-176 magma, which is not right-cancellative. What does generalize is Theorem 7.5 (non-constancy) — and, by Theorems 7.3 and 7.6, upgrading non-constancy to left-injectivity is (Q) for the block, while the $E255$ statement for the block is (P) for the block.

Remark (why Theorem 4.8 escapes, and what this costs). Theorem 4.8 is true for exactly one reason: in its setting the block is *affine*, and a non-constant affine map over a field is automatically injective. Corollary 7.8 says that this is the only reason. Therefore *the general-magma analogue of the $\{1, 3\}$ self-referential collision is Problem 1.1 itself, one level down*: the mechanism is a fixed point of the problem and cannot resolve it. This is consistent with the other half of the same self-reference, already recorded in §6.5: a minimal counterexample is 1-generated. The campaign’s “single best recommendation for a continuation” of §4.4 is hereby withdrawn in its literal form; what survives of it is Problem 11.8(b), the search for a mechanism that does *not* pass through the block.

7.2 The Ψ identities: an upper-bound criterion, and why it has no term witness

Section 5.7 diagnosed the gap as a direction mismatch: every mechanism in the campaign produces *lower* bounds on the relevant curve mass, while $E255$ needs an *upper* bound. Round 6 produced a statement of exactly that shape.

Theorem 7.9 (column-transport identity). *[proved in R6-A Thm 2.1, root-verified] [machine-verified: problems/etp677/R6A_scripts/r6a_verify.py]* In every finite E677 magma, for every $x \in M$,

$$\sum_{z \in M} N(z * x, z) = n.$$

Proof. $\#\{(a, z) : a * (z * x) = z\} = \sum_a |\text{Fix}(L_a R_x)| = \sum_a N(a, x) = n$, using Lemma 2.19 and the column-sum identity of Theorem 2.20. $\square$

Corollary 7.10. [proved in R6-A Cor 2.1', root-verified] Put $A(x, z) := N(z * x, z)$. Then all row sums and all column sums of A equal n (rows by Theorem 7.9, columns because $x \mapsto z * x$ is a bijection), and $\text{tr } A = \text{tr } N$ (Theorem 7.12 below). Hence

$$E255 \iff \text{tr } A \geq n$$

for an $n \times n$ non-negative integer matrix all of whose line sums are n . This is a genuine constraint linking N to the Cayley table, and it is not inside the marginal package refuted by Proposition 5.3.

Definition 7.11 (the curve masses). For $k \in \mathbb{Z}$ put $\Psi_k := \sum_{x \in M} N(x_k, x)$, where $x_k = \Lambda_x^k(x)$ as in §2. Thus $\Psi_0 = \text{tr } N$, $\Psi_{-1} = \sum_x N(x * x, x)$ and $\Psi_1 = \sum_x N(x \setminus x, x) = \sum_x N(e(x), x)$.

Theorem 7.12 (the one shift that is proved). [proved in R6-A Thm 2.2, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`] $\Psi_{-1} = \Psi_0 = \text{tr } N$, via the explicit bijection

$$\{(t, x) : t * (x * x) = x\} \longrightarrow \{(a, t) : a * t = t\}, \quad (t, x) \mapsto (t * (t * x), t).$$

Proof. By KEY, $\text{Fix } \Theta_t = \{x : (t * x) * t = x\} = \{x : t * (x * x) = x\}$, and $|\text{Fix } \Theta_t| = |\text{Fix}(L_t R_t)| = N(t, t)$ by Lemma 2.19. Composing the two bijections gives the displayed map. $\square$

Theorem 7.13. [proved in R6-A Thm 2.3, root-verified] $\Psi_1 \geq n$, because $x * (x \setminus x) = x$ supplies the witness $a = x$ for every x .

Corollary 7.14 (the missing upper bound). [proved in R6-A Cor 2.4, root-verified] $\Psi_{-1} = \Psi_0 = \text{tr } N \leq n$ and $\Psi_1 \geq n$. Hence

$$\Psi_1 \leq \Psi_0, \quad \text{i.e.} \quad \sum_x N(x \setminus x, x) \leq \sum_x N(x, x) \implies E255,$$

and pointwise $N(x \setminus x, x) \leq N(x, x)$ for a single x already gives E255 at x . More generally, if $v \mapsto N(v, x)$ is constant on the L_x -orbit of x — call this (ORB) — then E255 holds.

Corollary 7.14 is an *upper* bound on a curve mass, it is strictly weaker than (Q) (which implies it), and — unlike the off-diagonal statement (S-off) — it is *not* refuted by the order-176 magma. Three warnings come with it.

Remark (logic correction). [proved in R6-A §7.2, root-verified] $\Psi_1 = n$ alone does **not** imply (P): it says only that each $N(x \setminus x, x) = 1$, which constrains an off-diagonal curve of N and says nothing about $\text{tr } N$. What implies (P) is $\Psi_1 \leq \text{tr } N$. In terms of the sets

$$S_1 := \{(a, x) : a * (x \setminus x) = x\}, \quad S_0 := \{(b, t) : b * t = t\},$$

with $|S_1| = \Psi_1 \geq n$ and $|S_0| = \text{tr } N \leq n$, the prize is an *injection* $S_1 \hookrightarrow S_0$, not a bijection $S_1 \rightarrow M$.

Theorem 7.15 (no term-definable injection). [proved in R6-A Thm 7.4, modulo Proposition 5.1, root-verified] There is no pair of terms (β, τ) in $\{*, \setminus\}$ such that $(a, x) \mapsto (\beta(a, x), \tau(a, x))$ maps S_1 into S_0 injectively in every finite E677 magma.

Proof. S_1 always contains the free diagonal $\{(x, x)\}$, since $x * (x \backslash x) = x$. On it the map is $x \mapsto (b(x), t(x))$ with b, t unary terms, and $b(x) * t(x) = t(x)$ would be an unconditional identity of the variety. By Theorem 2.7, $E255$ holds at $t(x)$ for every x ; injectivity on the diagonal makes t injective, hence surjective, so $E255$ holds everywhere — an equational derivation of $E255$ from $E677$ plus the left-quasigroup axioms, which is what the saturation experiments of Proposition 5.1 failed to find and what Proposition 4.12 rules out for the sibling statement. $\square$

Theorem 7.12 does not contradict this: $S_{-1} = \{(t, x) : t * (x * x) = x\}$ has *no* unconditional elements, so its bijection to S_0 carries no free diagonal. The pattern — *whenever a candidate mapping has an unconditionally non-empty source, it cannot be a term map* — recurs below.

Proposition 7.16 (reformulations). *[proved in R6-A §7.3, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_psi.py`] $\Psi_0 = \#\{(a, p) : R_p(a) = p\}$ and $\Psi_1 = \#\{(a, p) : R_p(a) \in \text{Fix}(R_p)\}$. Since $\text{Fix}(R_p) = e^{-1}(p)$ and $\sum_p |\text{Fix } R_p| = n$ (Proposition 2.11),*

$$\Psi_1 = n \iff \text{for every } p, \text{ every fixed point of } R_p \text{ is its own only } R_p\text{-preimage,}$$

*i.e. $a * (x \backslash x) = x \Rightarrow a = x$: right cancellation restricted to the pairs $(x, e(x))$.*

Theorem 7.17 (no term-level up-step). *[proved in R6-A §3.1–3.3, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`] Full L_x -invariance of column x of N — $N(x * v, x) = N(v, x)$ for all v — is false: 11264 of 30976 pairs violate it in the order-176 magma and 238080 of 246016 in the order-496 magma. Even the support-level statement fails: the number of columns whose support is L_x -invariant is 0 of 176 and 0 of 496 respectively. Consequently there is no term $T(x, a, v)$ in $\{*, \backslash\}$ with*

$$a * (x * v) = x \implies T(x, a, v) * v = x$$

in every finite E677 magma.

Proof of the last statement. If $T(x, \cdot, v)$ is injective for all x, v , the implication injects $R_{x*v}^{-1}(x)$ into $R_v^{-1}(x)$, i.e. $N(x * v, x) \leq N(v, x)$; summing over v gives $n \leq n$ with equality forced termwise, i.e. full L_x -invariance. If it is not injective, the implication still gives $N(x * v, x) > 0 \Rightarrow N(v, x) > 0$, i.e. support L_x -invariance. Both are refuted by the displayed counts. $\square$

So the natural route to (ORB) — start from the free fact $N(x_1, x) \geq 1$ and propagate along the L_x -cycle back to x_0 , a genuinely finite argument not blocked by Proposition 5.1 — is dead at the level of uniform witnesses. Any proof of $\Psi_1 \leq \text{tr } N$ must be non-uniform.

Remark (what the numerics are worth). [computational: R6-A Obs. 2.5 on 13 models; R6-C §6 on the order-77 model; root-verified] $\Psi_k = n$ for all $-6 \leq k \leq 6$ on all models the campaign owns, and the order-77 model of §7.3 gives $\Psi_1 = \text{tr } N = 77$ with equality on its first genuinely non-idempotent test. This looks overwhelming and means very little: §7.6 explains why, and §7.5 records that the one explicit design able to falsify $\Psi_1 = \text{tr } N$ — a fibre family with $a_{-c/B} = 0$ over the exceptional base — provably does not exist for fibre fields of size $q \leq 7$.

7.3 Two E677 magmas of order 77

Section 5.8 left the extension framework with the record that *every* known non-right-cancellative model was idempotent, so that every orbit-type conjecture of §7.2 held vacuously in the entire test bed (§7.6). The remedy was to put a *non-idempotent* affine E677 magma — the Q -branch of Theorem 3.1 — at the zero-offset fibre, where Corollary 7.4 says the compatibility equation is literally E677 for that one operation.

Theorem 7.18 (the order-77 model M_{77}). [proved in R6-C §§1–4, root-verified] [machine-verified: `problems/etp677/R6C_scripts/r6c_verify.py`; Lean 4 kernel certificate `lean/proofenv/M77.lean`] Let the base be $\mathbb{F}_{11}$ with $x \diamond y = 6x + 6y$ (translation-invariant, since $6 + 6 = 1$), let the fibre be $\mathbb{F}_7$, and put $s \diamond_d t = a_d s + b_d t$ with

$$(a_d, b_d) = \begin{cases} (4, 1) & d = 0, \\ (0, 1) & d \in \{1, 3, 4, 5, 9\} \text{ (nonzero squares)}, \\ (3, 5) & d \in \{2, 6, 7, 8, 10\} \text{ (non-squares)}. \end{cases}$$

Then $M_{77} := (\mathbb{F}_{11} \times \mathbb{F}_7, (x, s) * (y, t) = (6x + 6y, a_{y-x}s + b_{y-x}t))$ has order 77 and:

- (a) it satisfies *E677* (zero violations over all 5929 pairs) and every L_Y is a bijection;
- (b) it satisfies *E255* at every element, although it is not idempotent: $X * X = (x, 5s)$, so exactly the 11 elements $(x, 0)$ are idempotent and the other 66 are not;
- (c) it is not right-cancellative — indeed all 77 right translations are non-injective; each row of N has 30 zeros, 42 ones and 5 entries equal to 7, so every column carries exactly 47 distinct values and the global distribution of N is $\{0:2310, 1:3234, 7:385\}$;
- (d) (S-off) nevertheless holds in it (zero violating pairs);
- (e) $\text{tr } N = \Psi_1 = 77$, so Corollary 7.14 holds with equality on the first non-idempotent test available.

Proof sketch. $(F, G) = (4, 1)$ over $\mathbb{F}_7$ satisfies the affine criterion ($1 \cdot 4 + 1 \cdot 4 \cdot 1 = 8 = 1$ and $4 + 16 + 1 = 21 = 0$), so $\diamond_0$ is a non-idempotent affine *E677* magma ($s \diamond_0 s = 5s$); $(F, G) = (6, 6)$ over $\mathbb{F}_{11}$ likewise satisfies it, so the base is a Latin *E677* magma. With $A = B = 6$ and $c = 0$ the index maps of Lemma 4.4 are $D_1 = D_2 = 9d$, $D_3 = 10d$, $D_4 = 6d$; 9 is a square and 10, 6 are non-squares mod 11, so the operation pattern at the four positions is (U, U, V, V) for square d and (V, V, U, U) for non-square d , where $U = (0, 1)$ and $V = (r, r^{-1})$. In both cases (E1) is automatic and (E2) reduces to $r^3 = -1$, whose solutions in $\mathbb{F}_7$ are $r \in \{3, 5, 6\}$; $r = 3$ gives the displayed $(3, 5)$. At $d = 0$ all four positions carry $\diamond_0$ and the two equations are exactly the affine *E677* conditions for $(4, 1)$ (Corollary 7.4). Non-idempotency, *E255* and the failure of right cancellation are then read off: repeated multiplication by the same element stays at offset 0, so $X * X = (x, 5s)$, $(X * X) * X = (x, 0)$ and $((X * X) * X) * X = X$; and at any nonzero square offset the fibre output $0 \cdot s + 1 \cdot t = t$ is independent of s , so seven distinct rows collide in every column. $\square$

Remark (three records, and one thing it does not do). $77 < 176 < 496$: M_{77} is the smallest non-right-cancellative finite *E677* magma known to the campaign or to [14], the first that is not idempotent, and the first non-constant non-idempotent translation-invariant coefficient assignment. It lies outside the θ -family scanned in Proposition 4.11 — its zero-offset fibre has $a_0 + b_0 = 5 \neq 1$ — so the range-limited minimality statement there is unaffected; but the problem of the least non-right-cancellative order now reads “between 10 and 77”. M_{77} satisfies *E255* and is *not* a counterexample to Problem 1.1.

Remark (machine certificate). [machine-verified: `lean/proofenv/M77.lean`] All four properties of Theorem 7.18(a)–(c) are certified in LEAN 4 by kernel-level `decide` — *not* `native_decide` — on the carrier `Fin 77` with the pair (x, s) encoded as $7x + s$. The file carries bridging lemmas

(mulN_enc, offs_sub) certifying that the encoded operation is literally the displayed formula on pairs and that the truncated-subtraction offset is the genuine difference mod 11, so the encoding is not load-bearing. The axiom audit reports {propext, Quot.sound} — strictly inside the permitted set, with no Classical.choice.

The second order-77 model answers a question the framework had never been pushed on: does the compatibility system force translation invariance?

Lemma 7.19 (diagonal decoupling). *[proved in R6-E §2, root-verified] [machine-verified: problems/etp677/R6E_scripts/r6e_verify.py] Work with pair-indexed fibre operations $s \diamond_{x,y} t = a_{x,y}s + b_{x,y}t$, so that (4.1) becomes, for every ordered base pair,*

$$(E1\text{-pair}) \quad b_1(a_2 + b_2a_4b_3) = 1, \quad (E2\text{-pair}) \quad a_1 + b_1b_2(a_4a_3 + b_4) = 0,$$

with subscripts $P_1 = (y, z_2)$, $P_2 = (x, z_4)$, $P_3 = (y, x)$, $P_4 = (z_3, y)$ where $z_3 = y \diamond x$, $z_4 = z_3 \diamond y$, $z_2 = x \diamond z_4$. Over the base $x \diamond y = 6x + 6y$ on $\mathbb{F}_{11}$ the four subscripts have offsets 9d, 9d, 10d, 6d with $d = y - x$; all four multipliers are nonzero. Hence for $x \neq y$ all four P_i are off-diagonal pairs, while for $x = y$ idempotency of the base gives $P_1 = P_2 = P_3 = P_4 = (x, x)$. The eleven diagonal fibre operations may therefore be chosen independently of each other and of the off-diagonal ones, subject only to each being an affine E677 magma.

The collapse at $x = y$ is used here purely in the construction direction of Remark 7.1(ii): two explicit affine operations are exhibited and their E677 equations are checked, so that the corresponding instances of (4.1) are discharged. Nothing about the product magma is assumed in order to obtain them.

Theorem 7.20 (the non-translation-invariant order-77 model M_{77}^{NT}). *[proved in R6-E §§3–5, root-verified] [machine-verified: problems/etp677/R6E_scripts/r6e_verify.py; independently rebuilt from the written specification in R6D_scripts/r6d_m77nt.py] Keep the base $6x + 6y$ on $\mathbb{F}_{11}$ and the fibre $\mathbb{F}_7$, and set*

$$(a_{x,y}, b_{x,y}) = \begin{cases} (4, 3) & x = y = 0, \\ (4, 1) & x = y \neq 0, \\ (0, 1) & x \neq y, \ y - x \text{ a nonzero square}, \\ (3, 5) & x \neq y, \ y - x \text{ a non-square}. \end{cases}$$

Then the resulting order-77 magma satisfies E677 (zero violations over all 5929 pairs), all 77 left translations are permutations, it satisfies E255, has 11 idempotents and 77 non-injective columns — and it is not translation-invariant: the pairs (0,0) and (1,1) have the same offset 0 but different coefficients. It is not a coordinate-gauge artefact either: rescaling a translation-invariant assignment by nonzero fibre scalars μ_x sends $(a_{x,y}, b_{x,y})$ to $(a_{y-x}\mu_x/\mu_{x \diamond y}, b_{y-x}\mu_y/\mu_{x \diamond y})$, and on an idempotent base $x = y$ cancels the scale, leaving every diagonal coefficient unchanged.

Proof sketch. (4,3) over $\mathbb{F}_7$ satisfies the affine E677 equations $(4 \cdot 3 \cdot (1 + 9) = 120 = 1$ and $4 + 9 \cdot 16 + 27 = 175 = 0)$, as does (4,1); by Lemma 7.19 the eleven diagonal instances ask for nothing else. The off-diagonal instances are those of Theorem 7.18 verbatim. All 121 pair-indexed instances therefore hold, and the blueprint extension theorem gives E677. $\square$

7.4 The right-unit map $e(x) = x \backslash x$: a conjecture refuted twice

Recall from §2 that every x has a unique right unit $e(x) = \Lambda_x(x) = x \backslash x$. (This e takes an *element*; the isoperimetric $e(O)$ of Theorem 2.23 takes a *set*. The two are unrelated.) Round 6 began by observing that e is a term operation and that it was an automorphism in every model then available.

Lemma 7.21. *[proved in R6-A §8.1, R6-B §15, R6-D §1, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_e.py`, `R6D_scripts/r6d_check1.py`] $e(w) = w \backslash w = w * ((w * w) * w)$, so e is a term of the language $\{*\}$ alone. Consequently “ e is an endomorphism” is the pure $\{*\}$ -identity $(x * y) * ((x \backslash x) * (y \backslash y)) = x * y$, equivalently: the graph of e is a subalgebra of $M \times M$. Moreover $\text{Fix}(e) = \{x : x * x = x\}$ and $\sum_p |\text{Fix}(R_p)| = n$ (Proposition 2.11), so*

$$e \text{ injective} \iff e \text{ surjective} \iff e \text{ bijective} \iff \text{every } R_p \text{ has exactly one fixed point.}$$

Conjecture (R6-B-2, proposed and then refuted). In every finite E677 magma, e is an automorphism.

It held in 10 of 10 models when proposed, including M_{77} , M_{176} and M_{496} . Both halves are false, for different reasons, and the two refutations are independent: in M_{77}^{NT} the map e is bijective but not a homomorphism, while in the order-9 model below e is a homomorphism (it is constant) but not injective. Both are needed.

Proposition 7.22 (why the evidence was worthless). *[proved in R6-A Prop 8.4, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_e.py`] Let $M = \mathbb{F}_p \times \mathbb{F}_q$ be any translation-invariant blueprint extension with base $x \diamond y = Ax + By + c$, $A + B = 1$, and homogeneous linear fibres $s \diamond_d t = a_d s + b_d t$, $b_d \neq 0$. Put $d_0 := -c/B$ and $\lambda := (1 - a_{d_0})/b_{d_0}$. Then*

$$e((x, s)) = (x - c/B, \lambda s),$$

and e is an endomorphism — an automorphism iff $\lambda \neq 0$ — for every c and every admissible fibre family.

Proof. Base: $Ax + Bw + c = x \iff w = x - c/B$, so the offset from x to its base right unit is the constant d_0 , independent of x . Fibre: $a_{d_0}s + b_{d_0}\sigma = s \iff \sigma = \lambda s$. Endomorphism: $e(X)$ and $e(Y)$ have base offset $(y - c/B) - (x - c/B) = y - x$, the same offset as X, Y , so $e(X) * e(Y) = (x \diamond y - c/B, \lambda(a_d s + b_d t)) = e(X * Y)$, using $A + B = 1$. $\square$

So the entire translation-invariant family — which is every model the campaign owned — satisfies the conjecture for a two-line reason with no E677 content. On M_{77} the prediction is $e((x, s)) = (x, 4s)$, exact on all 77 elements, and M_{77} is neither medial nor left-distributive, so the reason is not mediativity but fibre-constancy.

Proposition 7.23 (not an equational consequence). *[proved in R6-A §8.2, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_e.py`] In the free 677 magma of [2, §13.2] — where E677 holds with zero violations and $w * e(w) = w$ holds with zero violations, so $e(w)$ really is a right unit — the identity $e(x * y) = e(x) * e(y)$ fails at all 4356 of 4356 instances of terms of size ≤ 3 , the smallest being $x = y = a$. Hence no derivation of the conjecture from the E677 identity in the language $\{*\}$ exists. As in Proposition 4.12, the caveat is that this free magma is not a left quasigroup (L_a is not injective there, with an explicit witness), so this closes the equational route only.*

Theorem 7.24 (first refutation: e need not be a homomorphism). [proved in R6-E §5, root-verified] [machine-verified: `problems/etp677/R6E_scripts/r6e_verify.py`; independently `R6D_scripts/r6d_m77nt.py`] In M_{77}^{NT} (Theorem 7.20) the right-unit map is $e((x, s)) = (x, \lambda_x s)$ with $\lambda_0 = 6$ and $\lambda_x = 4$ for $x \neq 0$; both scalars are nonzero, so e is a bijection. It is not a homomorphism: it fails at 1050 of the 5929 ordered pairs. An explicit witness is

$$X = (1, 0), \quad Y = (0, 1) : \quad e(X * Y) = (6, 6) \neq (6, 2) = e(X) * e(Y).$$

Proof. The base is idempotent, so its right-unit map is the identity, and solving $4s + 3t = s$ at base point 0 gives $t = 6s$, while $4s + t = s$ elsewhere gives $t = 4s$. By the criterion of Proposition 7.22 extended to pair-indexed coefficients, e is a homomorphism exactly when $\lambda_{x \diamond y} a_{x,y} = a_{x,y} \lambda_x$ and $\lambda_{x \diamond y} b_{x,y} = b_{x,y} \lambda_y$ for all x, y . At $x = 1, y = 0$ one has $x \diamond y = 6$, $(a_{1,0}, b_{1,0}) = (3, 5)$, and the second equation would require $4 \cdot 5 = 5 \cdot 6$ in $\mathbb{F}_7$, i.e. $6 = 2$. The coefficient obstruction occurs at 25 of the 121 base pairs, and at each the homomorphism defect is a nonzero linear form in the two fibre variables, hence fails on 42 of their 49 values: $25 \times 42 = 1050$. $\square$

Theorem 7.25 (second refutation: e need not even be injective). [proved in R6-D §3, root-verified] [machine-verified: `problems/etp677/R6D_scripts/r6d_min9.py`, `r6d_m9.py`] Let $M_9 := (\mathbb{F}_3^2, x * y = x + Gy)$ with $G = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ over $\mathbb{F}_3$, which satisfies $G + G^3 = I$. Then M_9 is an E677 magma of order 9 (zero violations over all 81 pairs), it is a quasigroup, it satisfies E255, and it has exactly one idempotent. Its coefficient F is I , so 0 is a right identity and $e \equiv 0$ is constant: e is not injective. Moreover 9 is the least order at which this happens.

Proof. $F = I$ is forced by $F = (G + G^3)^{-1} = I$, and $x * 0 = x$ makes 0 a right identity, so $x \setminus x = 0$ for every x . Minimality: orders 2, 3, 4, 6, 8 carry no E677 magma at all and at orders 5, 7 the map e is bijective (Theorem 6.1 and Table 1), so no smaller example exists. Independently, a MACE4 search over “left quasigroup + KEY + $\exists a \neq b, c : a * c = a \wedge b * c = b$ ” terminated with `exit(exhausted)` and zero models at $n = 5, 7$ and returned exactly this model at $n = 9$. $\square$

Remark (the counterexample was in Theorem 6.1 all along). [machine-verified: `problems/etp677/R6_zoo.py`] M_9 is not a new magma. By Theorem 6.1 there is exactly one E677 magma of order 9 up to isomorphism, the affine model over $\mathbb{F}_9 = \mathbb{F}_3[t]/(t^2 + 1)$ with $(F, G) = (1, t + 2)$, and an explicit isomorphism $M_9 \rightarrow (\mathbb{F}_9, x + (t + 2)y)$ is exhibited by `R6_zoo.py` (both have $F = 1$, so both have 0 as a right identity). MACE4 rediscovered it because the campaign’s working benchmark set — cyclic affine models, $\mathbb{F}_{16}$, and translation-invariant extensions — contained no odd-order *non-cyclic* model, $\mathbb{Z}/9$ carrying no affine E677 structure at all. *The refutation of half of a Round-6 conjecture was sitting inside this paper’s own classification table, unused, because the object used for testing conjectures was not the classification but the benchmark set.* See §7.6.

Corollary 7.26 (the affine criterion for bijectivity). [proved in R6-D §6, root-verified] For an affine model $x * y = Fx + Gy + c$ one has $e(x) = G^{-1}((1 - F)x - c)$, so

$$e \text{ is bijective} \iff 1 - F \text{ is invertible} \iff G^3 + G - 1 \text{ is invertible,}$$

using $F = (G + G^3)^{-1}$. M_9 realizes the extreme case $G^3 + G - 1 = 0$.

Remark (what survives, and the new target). [proved in R6-D §5, root-verified] [computational: 13 models; root-verified] $\text{Fix}(e) = \text{Idem}(M)$ is a definitional identity and is untouched. What dies is the route “ $e \in \text{Aut} \Rightarrow \text{Idem}(M) = \text{Fix}(e)$ is a subalgebra $\Rightarrow \langle e \rangle$ -orbit lengths are divisible by 31

$\Rightarrow 31 \mid n$ (Conjecture R6-B-3 of §7.5): the orbit partition needs e to be a permutation, and in M_9 it is constant. The statement “Idem(M) is a subalgebra” is *not* refuted — it holds in all 13 models, including M_{77}^{NT} and M_9 — and it now has a clean equivalent reduction: since $a^2 = a$ and $b^2 = b$ for idempotents,

$$\text{Idem}(M) \text{ is closed} \iff (a * b) * (a * b) = a * b \text{ for all idempotent } a, b,$$

i.e. the square identity $(a * b)^2 = a^2 * b^2$ restricted to idempotent pairs. In M_{77}^{NT} that identity has 1050 violations globally but none on idempotent pairs, which is exactly why Idem is still closed there. This strictly weaker statement is the recommended successor target.

7.5 Idempotent-free magmas and the fibre spectrum

Proposition 5.10 left the extension programme with exactly one habitat: the base $x \diamond y = 5x - 4y + c$ on $\mathbb{F}_{31}$ with $c \neq 0$, the only translation-invariant $E677$ base with no idempotent. Round 6 asked what the fibre sizes over that base can be. Write

$$S := \{ |M| : M \text{ a finite } E677 \text{ magma} \}, \quad S_0 := \{ |M| \in S : M \text{ has no idempotent} \},$$

and $\text{FiberSpec}(c) :=$ the set of fibre sizes m for which the base $5x - 4y + c$ admits a translation-invariant fibre family. The *fibre spectrum conjecture* is $\text{FiberSpec}(1) = S$.

Theorem 7.27 (uniqueness of the exceptional base). *[proved in R6-B Thm 3, root-verified] [machine-verified: `problems/etp677/R6B_tools.py`] Over a commutative ring, $x \diamond y = ax + by + c$ satisfies $E677$ iff $ab(1 + b^2) = 1$, $a + a^2b^2 + b^3 = 0$ and $c(ab^2 + b^2 + b + 1) = 0$; the first two are equivalent to $a = (b + b^3)^{-1}$ and $P(b) = \Phi_{10}(b)Q(b) = 0$; translation invariance $a + b = 1$ is equivalent to $\Phi_{10}(b) = 0$; and, when c is a non-zero-divisor, the third is equivalent to $Q(b) = 0$. Hence a non-idempotent translation-invariant affine base requires $\Phi_{10}(b) = Q(b) = 0$, which forces $248 = 8 \cdot 31 = 0$. Over a field characteristic 2 is excluded, so the characteristic is 31, and then $b = -5/9 = -4$, $a = 5$, with c any nonzero constant (normalizable to 1).*

So the exceptional base is not a special base: it is the unique non-idempotent translation-invariant affine $E677$ base over any field. The whole extension-based counterexample programme rests on that one arithmetic fact.

Theorem 7.28 (fibres over an idempotent are submagmas). *[proved in R6-B Thm 2, root-verified] Let B satisfy $E677$, let $\pi : B \rightarrow C$ be a magma homomorphism and let $e \in C$ be idempotent. Then $\pi^{-1}(e)$ is closed in B , hence is itself an $E677$ magma when nonempty. Consequently $\text{FiberSpec}(0) = S$: over any $c = 0$ translation-invariant base the fibre spectrum is the order spectrum, and the fibre spectrum conjecture is a theorem there.*

This is Theorem 7.3 again, and it explains structurally why $c \neq 0$ is the only escape: E_c has no idempotent because $x \diamond x = x + c$. In the other direction, sufficiency is free and holds over every base.

Theorem 7.29 (sufficiency). *[proved in R6-B Thm 1, root-verified] [machine-verified: `problems/etp677/R6B_tools.py`] If $(M, \circ)$ is any finite $E677$ magma then the constant family $\diamond_\delta \equiv \circ$ satisfies all 31 instances of (4.1) over any base E_c , so $S \subseteq \text{FiberSpec}(c)$ for every c . Verified for $m = 5$ (order 155) and $m = 16$ (order 496), with a negative control (the ω -quandle, which is not $E677$) producing 7440 violations.*

Remark (constant is not the same as rigid). [proved in R6-B Prop 1.1, root-verified] [computational: on the order-496 magma] “All translation-invariant solutions over the exceptional base are constant” is *false*. If $\gamma \in \text{Aut}(M, \circ)$ with $\gamma^{31} = \text{id}$, then $\diamond_\delta(s, t) := \gamma^{4\delta-1}(s) \circ \gamma^{5\delta-1}(t)$ is again a solution (it is the constant solution transported by $\psi(x, s) = (x, \gamma^x(s))$). Taking $(M, \circ)$ to be the order-496 magma and γ a base translation gives a solution whose $\diamond_0$ and $\diamond_1$ differ at all 246016 points. The correct rigidity statement can only be “every solution is gauge-equivalent to a constant one”.

Theorem 7.30 (structural equivalence and the free $\mathbb{Z}_{31}$ -action). [proved in R6-B Thm 4 and Cors. 4.1–4.5, root-verified] $m \in \text{FiberSpec}(c)$ if and only if there is a finite E677 magma B , a surjective homomorphism $\pi : B \rightarrow E_c$ with all fibres of size m , and $\theta \in \text{Aut}(B)$ with $\pi(\theta X) = \pi(X) + 1$; such a θ has order 31 and acts freely. Consequently, for $c \neq 0$:

- (a) B has no idempotent, so $S \subseteq \text{FiberSpec}(1) \subseteq \{m : 31m \in S_0\}$;
- (b) either of “ $31n \in S \Rightarrow n \in S$ ” and “ $S_0 \subseteq 31 \cdot S$ ” implies the necessity half of the fibre spectrum conjecture;
- (c) $8 \in \text{FiberSpec}(1)$ iff there is an E677 magma of order 248 with no idempotent carrying a free $\mathbb{Z}_{31}$ of automorphisms;
- (d) every automorphism-invariant subset of B has size divisible by 31; in particular the set of elements at which E255 fails has size $\equiv 0 \pmod{31}$, so a translation-invariant counterexample cannot break at a single point — it must break on at least one full $\mathbb{Z}_{31}$ -orbit;
- (e) (single-orbit reduction) if a solution violates E255 at $(0, \sigma_0)$ then its restriction to the closure of $\{\sigma_0\}$ under the 31 operations $\diamond_\delta$ is again a solution violating E255 at the same point. So one may assume without loss that the fibre is generated by the violating point.

Theorem 7.31 (the exceptional base blocks every local mechanism). [proved in R6-B Thm 5, root-verified] [machine-verified: `problems/etp677/R6B_tools.py`] For $E_1 = (\mathbb{F}_{31}, 5x - 4y + 1)$:

- (a) E_1 has no proper subalgebra and no nontrivial congruence. (Any nonempty subalgebra is closed under $x \mapsto x \diamond x = x + 1$, hence is everything; for congruences, $x \theta y$ forces $5x + k \theta 5y + k$ for all k , and $5^3 \equiv 1 \pmod{31}$ gives translation invariance of θ , so the classes are cosets of a subgroup H with $5H = H$, and $\mathbb{Z}_{31}$ has no proper subgroup.) So constructions by subalgebra, quotient or preimage produce nothing.
- (b) For $c \neq 0$ no instance of (4.1) degenerates to E677: the most collapsed instance, $d = -c/A$, still involves three distinct operations (for $c = 1$: $d = 6$ with subscripts $(25, 8, 25, 6)$). For $c = 0$ the unique coincident instance is $d = 0$, where all four subscripts agree.
- (c) No θ -twist $X * Y := \theta^a(X \diamond \theta^j Y)$ carries the $c = 1$ system onto a $c' = 0$ system, so Theorem 7.28 cannot be imported.
- (d) The affine symmetry group of the subscript system collapses from $\mathbb{F}_{31}^\times$ (order 30, at $c = 0$) to the trivial group (at $c \neq 0$). The blueprint’s own $\{0, \text{QR}, \text{QNR}\}$ class-function ansatz is a $\mathbb{F}_{31}^\times$ -class function and therefore has no analogue at $c \neq 0$.

Together these say that the necessity half cannot be obtained by any subalgebra, quotient, twist or symmetry argument: it needs a global theorem about S or S_0 . Round 6 obtained that theorem for affine magmas.

Theorem 7.32 (S_0 is an ideal). [proved in R6-B Thm 8, root-verified] [machine-verified: `problems/etp677/R6B_S0.py`] $A \times B$ has an idempotent iff both A and B do. Hence $S_0 \cdot S \subseteq S_0$: S_0 is an ideal of the multiplicative monoid S . Since $31 \in S_0$ (the exceptional model itself), $31 \cdot S \subseteq S_0$. Machine check: $E_{31} \times \{1, \mathbb{F}_5, \mathbb{F}_7, E_{31}\}$ gives orders 31, 155, 217, 961, all with zero idempotents and zero E677 violations.

Theorem 7.33 (the elimination constant is exactly 31). [proved in R6-B Thm 9; Bézout identity re-verified coefficientwise, root-verified] In $\mathbb{Z}[b]$, with $\Phi_{10}(b) = b^4 - b^3 + b^2 - b + 1$ and $Q(b) = b^4 + b^3 + 2b^2 + 2b + 1$ the two factors of P ,

$$\text{Res}(\Phi_{10}, Q) = 31, \quad (18 + 16b + b^2 + 10b^3) \Phi_{10}(b) + (13 - 24b + 19b^2 - 10b^3) Q(b) = 31.$$

So for any commutative ring R and any $b \in R$ the ideal $(\Phi_{10}(b), Q(b))$ contains 31. This is sharper than the “ $248 = 8 \cdot 31 = 0$ ” of Theorem 7.27: the factor 2 is eliminated entirely, because Φ_{10} and Q are coprime mod 2.

Theorem 7.34 (idempotent-free affine orders are exactly $31 \cdot S$). [proved in R6-B Thm 10, root-verified] [machine-verified: `problems/etp677/R6B_S0.py`] Let M be a finite abelian group, $a, b \in \text{End}(M)$ commuting with b invertible, $c \in M$, and suppose $B = (M, x * y = ax + by + c)$ satisfies E677 and has no idempotent. Then $31 \mid |M|$ and $|M|/31 \in S$. Combining with Theorem 7.32, $S_0^{\text{aff}} = 31 \cdot S^{\text{aff}}$.

Proof sketch. From $ab(1+b^2) = 1$, b and $1+b^2$ are units and $a = (b+b^3)^{-1}$; then $u := a+b-1 = \Phi_{10}(b)(b+b^3)^{-1}$, so $\text{Im}(u) = \text{Im}(\Phi_{10}(b))$, and $(1+b^2)(ab^2 + b^2 + b + 1) = Q(b)$, so the constant condition gives $Q(b)c = 0$. Now $x * x = x \iff (a+b-1)x = -c$, so B has no idempotent iff $c \notin \text{Im}(\Phi_{10}(b))$. Evaluating the Bézout identity of Theorem 7.33 at b and applying it to c gives $31c = \Phi_{10}(b)(A(b)c) \in \text{Im}(\Phi_{10}(b))$. If $31 \nmid |M|$ then multiplication by 31 is an automorphism whose inverse is multiplication by an integer, hence commutes with $\Phi_{10}(b)$, giving $c \in \text{Im}(\Phi_{10}(b))$ — a contradiction. So $31 \mid |M|$. Finally decompose $M = \bigoplus_p M_p$ into primary components (each is characteristic, hence a, b -invariant); B is the product of the B_p , each a quotient of B and hence an E677 magma, so $|M_p| \in S$; B has no idempotent iff some B_p has none, and that forces $p = 31$; then $|M|/31 = 31^{k-1} \prod_{p \neq 31} |M_p|$ with every factor in S and S closed under multiplication. $\square$

Corollary 7.35. [proved in R6-B Cors. 10.1–10.2, root-verified] (a) If $m \in \text{FiberSpec}(1)$ and the resulting order- $31m$ magma is affine, then $m \in S$: the fibre spectrum conjecture holds for all extensions whose total magma is affine. (b) There is no idempotent-free affine E677 magma of order 248, because $248/31 = 8 \notin S$ (Theorem 6.1).

A scan over all (a, b, c) on 22 small rings finds idempotent-free affine models at exactly the orders 31, 155, 217, 961 = $31 \cdot \{1, 5, 7, 31\}$, matching Theorem 7.34 exactly [machine-verified: `problems/etp677/R6B_S0.py`]. The general case is open and is the campaign’s cleanest remaining sub-goal:

Conjecture (R6-B). Every idempotent-free finite E677 magma has order divisible by 31; more strongly, $S_0 = 31 \cdot S$. Either this or “ $31n \in S \Rightarrow n \in S$ ” makes the fibre spectrum conjecture a theorem and empties the $m = 8$ search space (Theorem 7.30(b),(c)).

The affine fibre classes, computed. Theorem 7 of [18] reduces a translation-invariant affine fibre family to a system in the ring generated by the coefficients: with $\diamond_\delta(s, t) = a_\delta s + b_\delta t + e_\delta$,

(4.1) at d is equivalent to

$$\begin{aligned} \text{(I)} \quad & b_1 a_2 + b_1 b_2 a_4 b_3 = 1, & \text{(II)} \quad & a_1 + b_1 b_2 a_4 a_3 + b_1 b_2 b_4 = 0, \\ \text{(III)} \quad & b_1 b_2 a_4 e_3 + b_1 b_2 e_4 + b_1 e_2 + e_1 = 0, \end{aligned}$$

and (III) is homogeneous linear in e , so $e \equiv 0$ is always a solution and *existence is decided by (I)–(II) alone*. Since (I)–(II) live in the ring generated by the a_δ, b_δ , one solves once per coefficient ring. A complete depth-first solver [computational: R6-B §8; complete over the stated rings, no SAT] gives: no solution for $m \in \{2, 3, 4, 6\}$ over every possible coefficient ring; exactly one solution for $m = 5$, the constant family $(a, b) \equiv (2, 4)$; and for $m = 8$, no solution over $\mathbb{Z}_8$, $\mathbb{Z}_4 \times \mathbb{Z}_2$ (including the full noncommutative End), or any commutative coefficient ring on $\mathbb{Z}_2^3$ except $\mathbb{F}_8$. *Honest gaps*: $R = \mathbb{F}_8$ (search space 56, terminated by the round’s compute budget) and $M = \mathbb{Z}_2^3$ with noncommuting coefficients ($M_3(\mathbb{F}_2)$, search space 86 016, not attempted); and the whole computation covers *affine* fibres only, so it says nothing about the nonlinear five-operation gadget of §5.8.

Theorem 7.36 (the exceptional base over small fibre fields). *[computational: R6-B §§17–19, complete solution; 1296 algebraic shards, $5.66 \cdot 10^{10}$ nodes] For homogeneous affine fibre families over the exceptional base $5x - 4y + 1$ on $\mathbb{F}_{31}$: there is no solution for $q \in \{2, 3, 4, 6\}$ (and for every coefficient ring of those orders); exactly one for $q = 5$, the constant family $(2, 4)$; and exactly two for $q = 7$, the constant families $(4, 1)$ and $(4, 3)$. The solution space is therefore exactly the set of constant families, one for each affine E677 structure on the fibre — one at $q = 5$, two at $q = 7$. In particular there is no family with $a_8 = 0$, where $8 = -c/B$.*

The last clause matters for §7.2. [17], §7.5–7.6 designed an explicit potential counterexample to $\Psi_1 = \text{tr } N$: over a base *without* idempotents, E255 is protected at the fibre pair $(\delta(x), x)$ while the x -term of Ψ_1 is decided at the pair $(x, e(x))$, and these two coincide *only at idempotent base points*; on E_c the corresponding offsets are c/A and $-c/B$, always distinct, and $-c/B$ avoids all three conflict indices of Lemma 4.7. Any fibre family with $a_{-c/B} = 0$ would give a magma satisfying E255 with $\Psi_1 = nq > n = \text{tr } N$, refuting both $\Psi_1 = \text{tr } N$ and (ORB). Theorem 7.36 shows that no such *affine* family exists for $q \leq 7$; a constant family always has $a = (b + b^3)^{-1}$, a unit, so the design necessarily requires a non-constant — hence, in this range, a non-affine — family. The design and the range limitation are both recorded rather than resolved.

7.6 Methodology: the evolution of the benchmark set

Four times in this campaign a statement was believed on the strength of holding in every available model, and four times the belief was an artefact of what those models had in common. The pattern is worth recording as a result.

1. **Phase 1 — cyclic affine models.** Orders ≤ 9 plus a scan of $ax + by \bmod n$: every model odd, every model a quasigroup. Outcome: the *parity conjecture* (Section 3.2), refuted by the $\mathbb{F}_{16}$ quandle, which is not cyclic; and the pursuit of (Q) for two rounds (§5.2), refuted by the order-496 magma.
2. **Phase 2 — adding M_{176} and M_{496} .** These are non-right-cancellative, and they immediately killed the term $b = (x * x) \backslash a$, which is a perfect “up-step” in *every* Latin model tested (orders 5, 7, 7, 11, 11, 11, 11, 13, 16, 19, 19) and fails in 21120 of 30976 instances in M_{176} [machine-verified: `problems/etp677/R6A_scripts/r6a_verify.py`]. Had they not been

in the set, that term would have looked like a proof of Problem 1.1. The same filter caught, on the day it was installed, a claimed complete proof of Problem 1.1 produced by one of the campaign’s own third-party reasoning models: its key identity $N^2 = nN$ holds automatically in every Latin model and fails on 5390 of 5929 pairs in M_{77} and on 14080 of 30976 in M_{176} [machine-verified: `problems/etp677/R6_zoo.py`] — i.e. the argument had silently assumed (Q), which is false.

3. **Phase 3 — adding M_{77} .** Both M_{176} and M_{496} are *idempotent*, so $\delta = e = \text{id}$ and $x_k = x$ for all k in them: the entire Ψ /orbit programme of §7.2 is *vacuously* true there, and direct products do not help because all these curve masses are multiplicative [proved in R6-A §4.4, root-verified] [machine-verified: `problems/etp677/R6A_scripts/r6a_blocks.py`]. M_{77} was built (§7.3) precisely to be non-idempotent and non-right-cancellative. It still does not discriminate: by Theorem 7.3 the L_x -orbit of x stays inside its congruence class, and that class is a Latin block. Within the $c = 0$ translation-invariant family this is unavoidable, and it is the same self-similarity as §7.1.
4. **Phase 4 — adding M_{77}^{NT} and M_9 .** Every model up to that point was translation-invariant, and Proposition 7.22 shows that in *every* such model the right-unit map is automatically an endomorphism for a two-line linear reason. The conjecture “ $e \in \text{Aut}$ ” was therefore supported by 10 models with zero discriminating power, and fell to the first non-translation-invariant model (Theorem 7.24) and to the order-9 model that the benchmark set had never included (Theorem 7.25).

Two rules were adopted as a result, and are recommended to anyone continuing this work.

- **Filter set.** Every term or identity search must be run against $\{M_9, M_{77}, M_{77}^{\text{NT}}, M_{176}, M_{496}\}$ at minimum: a non-cyclic odd-order model with a right identity, a non-idempotent non-right-cancellative model, a non-translation-invariant one, and two idempotent non-right-cancellative ones. One measured consequence: of 595 unary terms that are endomorphisms in all 11 Phase-2 models, only 32 survive the addition of M_{77}^{NT} , and those 32 are all equal to the identity map [machine-verified: `problems/etp677/R6D_scripts/r6d_survive.py`].
- **The E_{255} blind spot.** *Every* finite E_{677} magma known to anyone satisfies E_{255} . Therefore “holds in all known models” carries **zero** information about whether a statement is a consequence of E_{677} or only of $E_{677} + E_{255}$; a term-mining run on the model set will happily rediscover E_{255} itself as a “law”. No identity obtained by model filtering may be used on the main line before it has been derived from E_{677} independently. This is the exact sense in which the campaign’s numerical evidence for $\Psi_1 = \text{tr } N$ (§7.2) must be discounted.

Remark (a retracted conjecture, recorded). [computational: R6-D §4.3] The $595 \rightarrow 32$ collapse above was itself preceded by a Round-6 conjecture — that *every* unary term is an endomorphism — proposed on the Phase-2 set and withdrawn within the round. It is recorded here for the same reason as the parity conjecture and (Q): the failure mode is the interesting datum.

Remark (what automated theorem provers did and did not do). [computational: R6-D §4.2, Prover9/Mace4 and E 3.1, each run under 300 s and 3 GB] Bounded runs proved the division formula, $\text{KEY} \Rightarrow E_{677}$, $e(y) * (y * y) = (y * y) * y$ and $\delta(x) = x \setminus e(x)$ in seconds (five calibration targets,

all discharged). The same configuration failed to prove “ e is an endomorphism” and the square identity $(x * y)^2 = x^2 * y^2$ from $E677$, with and without $E255$. *Those failures are not evidence of underivability* — see the reading of Proposition 5.1 — but the first of them is now known to be a failure for the right reason (Theorem 7.24).

7.7 The model zoo

Table 1 lists every finite $E677$ magma the campaign can construct, with the properties that decide what each one can test. All columns are recomputed from the models themselves.

model	n	Idem	(Q)	TI	affine	e bij.	what it is good for
$\mathbb{Z}_5, 2x - y$	5	5	yes	—	yes	yes	the unique order-5 model; Takasaki quandle
$\mathbb{Z}_7, 4x + y$	7	1	yes	—	yes	yes	first non-idempotent model; has a left identity
$\mathbb{Z}_7, 4x + 3y$	7	1	yes	—	yes	yes	Q -branch; squaring map constant
$\mathbb{F}_9, x + (t + 2)y$	9	1	yes	—	yes	no	$= M_9$; unique order-9 model; 0 <i>is a right identity</i> , $e \equiv 0$; kills “ e injective”
$\mathbb{Z}_{11}, 4x + 8y$	11	11	yes	—	yes	yes	one of four order-11 classes
$\mathbb{Z}_{13}, 9x + 11y$	13	1	yes	—	yes	yes	unique order-13 affine class
$\mathbb{F}_{16}, (1 + z)x + zy$	16	16	yes	—	yes	yes	smallest even order; killed the parity conjecture
$E_{31}, 5x - 4y + 1$	31	0	yes	—	yes	yes	<i>the</i> exceptional base: the only idempotent-free translation-invariant affine base (Thm 7.27)
M_{77}	77	11	no	yes	fibres	yes	smallest known non-right-cancellative; first non-idempotent such; first non-idempotent test of $\Psi_1 \leq \text{tr } N$
M_{77}^{NT}	77	11	no	no	fibres	yes	first non-translation-invariant extension; e <i>not a homomorphism</i> (1050/5929)
M_{176}	176	176	no	yes	fibres	yes	refutes (S-off) and the isoperimetric threshold; kills term-level up-steps
M_{496}	496	496	no	yes	fibres	yes	the blueprint example; $ F_{pq} $ nearly attains the row-code bound
$E_{31} \times \mathbb{Z}_5$	155	0	yes	—	yes	yes	idempotent-free, order $31 \cdot 5$
$E_{31} \times \mathbb{Z}_7$	217	0	yes	—	yes	yes	idempotent-free, order $31 \cdot 7$
$E_{31} \times E_{31}$	961	0	yes	—	yes	yes	idempotent-free, order 31^2 ; S_0 is an ideal (Thm 7.32)

Table 1: The model zoo. “TI” applies only to blueprint extensions; “affine” means affine over an abelian group, and “fibres” means the fibre operations are affine while the magma itself is an extension, not affine. *Every* entry satisfies $E677$ and $E255$ with zero violations, has all rows permutations, and has $\Psi_1 = \text{tr } N = n$. Idem is closed in every entry *except* M_{77}^D , and the three Round-9 models have genuinely non-affine fibre operations, which is what makes them decisive (Lemma 10.4). The columns |Idem|, (Q), e bijective, e homomorphism, Idem closed, $\text{tr } N$ and Ψ_1 are recomputed by `problems/etp677/R6_zoo.py`, which also exhibits the isomorphism $M_9 \cong (\mathbb{F}_9, x + (t + 2)y)$.

Three features of the table are the campaign’s current position in compressed form. Every entry satisfies $E255$ — that is the blind spot of §7.6. Every entry has $\Psi_1 = \text{tr } N$ — that is the strongest unrefuted criterion (Corollary 7.14), supported by evidence that §7.6 says must

be discounted. And exactly one structural feature separates the idempotent-free entries from the rest: they all have order divisible by 31, which is Conjecture R6-B and, if proved, closes the extension programme.

8 Round 7: the mixed-idempotent base, the cocycle protection theorem, and the quotient theorems

Round 6 ended with one route still open. Corollary 7.8 had shown that the $\{1, 3\}$ collision cannot be upgraded in general, and Proposition 5.10 had shown that an *idempotent* base makes the construction circular. That leaves bases with *some* idempotent element but not all — the “mixed” case — of which the smallest is $\mathbb{F}_7$ with $x \diamond y = 4x + 3y$. Round 7 attacked it [22]. It did not produce a counterexample and it did not prove protection; what it produced is (i) an exact, base-independent description of the local configuration (§8.1), (ii) three strictly stronger no-go theorems, one of which subsumes [2, Lemma 13.4] (§8.2), (iii) a protection theorem for the whole separable/cocycle family (§8.3), and (iv) two structure theorems about *minimal* counterexamples that convert the Round-5 obstacle into a statement about quotients (§8.4).

8.1 The occurrence trichotomy

Remark 7.1 quoted the following statement in passing; here it is with its proof, since the rest of the section is organized around it. For a base $E677$ magma B and $a \in B$ satisfying $E255$, write x_a for the unique left unit of a (Theorem 2.7) and $\star := \diamond_{x_a, a}$ for the fibre operation attached to that pair — the operation that decides $E255$ at level a .

Theorem 8.1 (occurrence trichotomy; base-independent). *[proved in R7-B.0, root-verified] [computational: verified on seven bases, problems/etp677/R7A_scripts/r7b_baseindep.py; root-verified] Let B be any finite $E677$ magma satisfying $E255$ and $a \in B$.*

- (a) *If a is idempotent (equivalently $x_a = a$), the instance of (4.1) at (a, a) has $P_1 = P_2 = P_3 = P_4 = (a, a)$: $\star$ is forced to be an $E677$ magma and $E255$ for $\star$ is the original problem. This is the Round-5 circularity (Proposition 5.10, Theorem 7.3).*
- (b) *If a is not idempotent, then $\star$ occurs in exactly three instances of (4.1): in the instance at (a, x_a) it occupies positions 1 and 3 simultaneously — the $\{1, 3\}$ collision — and it occupies position 2 of one further instance and position 4 of one more.*

Proof. $P_3 = (y, x) = (x_a, a)$ forces $(x, y) = (a, x_a)$. $P_1 = (y, \Lambda_y x) = (x_a, a)$ forces $y = x_a$ and $x = L_{x_a}(a) = x_a \diamond a = a$, the same instance; the coincidence is exactly $L_{x_a}(a) = a$, i.e. the definition of a left unit. For (a), $a \diamond a = a$ collapses all four subscripts. $\square$

base	idempotents	non-idempotent levels with the 3-occurrence pattern
$\mathbb{F}_5(2x - y), \mathbb{F}_{11}(6x + 6y), \mathbb{F}_{31}(5x - 4y)$	all	— (all collapse 4-fold)
$\mathbb{F}_{31}(5x - 4y + 1)$	none	31/31
$\mathbb{F}_7(4x + 3y), \mathbb{F}_7(4x + y), \mathbb{F}_{13}(9x + 11y)$	exactly 1	6/6, 6/6, 12/12

Corollary 8.2 (changing the base does not help). *[proved in R7-B.0, root-verified] The local configuration around the $E255$ -deciding operation is the same for every base: at every non-idempotent level it is the three-instance picture with the $\{1, 3\}$ collision, and at every idempotent level it is*

the circular four-fold collapse. In particular the idempotent-free $\mathbb{F}_{31}$ route of §5.8 and the mixed $\mathbb{F}_7$ route of this section are locally identical; only all-idempotent bases are genuinely different, and those are circular.

8.2 The mixed-idempotent base $\mathbb{F}_7$, $x \diamond y = 4x + 3y$

This base has $x \diamond x = 0$ for every x , hence the unique idempotent 0; for $a \neq 0$ the left unit of a is $3a \neq a$; and $4 + 3 = 0 \neq 1$, so it is *not* translation-invariant. The pair-indexed system (4.1) has $P_1 = (y, 5x + y)$, $P_2 = (x, 5x + 5y)$, $P_3 = (y, x)$, $P_4 = (3x + 4y, y)$; all four are invertible $\mathbb{F}_7$ -linear maps of $\mathbb{F}_7^2$, so each of the 49 pairs occurs exactly once in each position and the system is *square*: 49 operations, 49 equations [computational: R7-A §2, r7a_struct.py]. The instance at $(0, 0)$ collapses four-fold, so $|M|$ must lie in the E677 spectrum.

Theorem 8.3 (occurrence theorem at this base). *[proved in R7-A Thm 1, root-verified] [computational: problems/etp677/R7A_scripts/r7a_base.py] For $a \neq 0$ the operation $\star = \diamond_{3a,a}$ occurs in exactly the three instances*

$$\begin{aligned} I_1 &= (a, 3a) && \text{positions 1 and 3,} && I_2 &= (3a, 0) && \text{position 2,} \\ I_3 &= (2a, a) && \text{position 4.} \end{aligned}$$

Theorem 8.4 (E255 criterion in two operations). *[proved in R7-A Thm 2, root-verified] [computational: 4000 random systems, r7a_criterion.py] For any pair-indexed extension $N = B \times M$, any $a \in \mathbb{F}_7$ and $\sigma \in M$:*

$$E255 \text{ holds at } (a, \sigma) \iff \exists s : \sigma \diamond_{a,4a} (s \diamond_{2a,a} \sigma) = s \iff \text{some left row of } \star \text{ fixes } \sigma.$$

Proof. E255 at Y in N means some X has $X \star Y = Y$ (Theorem 2.7); in the base $x \diamond y = y$ has the unique solution $x = 3y$, so E255 at (a, σ) says $s \diamond_{3a,a} \sigma = \sigma$ is solvable. Substituting into the instance I_3 , whose P_4 is $(3a, a)$, and using that $s \mapsto \sigma \diamond_{a,2a} s$ is a bijection gives the displayed equivalence. $\square$

The next theorem is the round's most reusable output: it strengthens [2, Lemma 13.4] in three directions at once.

Theorem 8.5 (row separation). *[proved in R7-A Thm 3, root-verified] Fix a and put $\star = \diamond_{3a,a}$. If $s \star t = s' \star t$ for all $t \in M$ then $s = s'$: the $|M|$ left rows of $\star$ are pairwise distinct permutations.*

Proof. Put $x = 3a$, $y = a$, so $x \diamond y = y$ and $y \diamond (y \diamond x) = y$. Row equality says $L_{(x,s)}$ and $L_{(x,s')}$ agree on the whole fibre over y . The E677 instance gives $(y, t) = L_{(x,s)} L_{(y,t)} L_{L_{(x,s)}(y,t)}(x, s)$, and $L_{(x,s)}(y, t) = (y, s \star t) = (y, s' \star t) = L_{(x,s')}(y, t)$, so the same identity for s' has the same inner word. Both inner words lie in the fibre over y , where $L_{(x,s)} = L_{(x,s')}$; cancelling $L_{(x,s)}$ and then twice more gives $(x, s) = (x, s')$. $\square$

Corollary 8.6 (separability rigidity, strictly stronger than blueprint Lemma 13.4). *[proved in R7-A Cor 3a, root-verified] Let M carry any group structure, abelian or not. If the single operation $\star = \diamond_{3a,a}$ has the separable form $s \star t = \alpha(s) \cdot \beta(t) \cdot c$ for arbitrary maps $\alpha, \beta : M \rightarrow M$ and a constant c — no homomorphism assumption — then α is injective, hence every column map of $\star$ is a bijection, and E255 holds at (a, σ) for every σ .*

Blueprint Lemma 13.4 is the special case “ M abelian, α, β endomorphisms, and all 49 operations of that shape”; the corollary needs one operation, no abelianness and no homomorphism property. Two further consequences are worth isolating.

Corollary 8.7. [proved in R7-A Cor 3b–3d, root-verified] (a) *E255 fails at (a, σ) only if $\star$ is genuinely two-variable at σ : there are $s \neq s'$ with $s \star \sigma = s' \star \sigma$ but $s \star t \neq s' \star t$ for some t . Every additive, group-affine, Frobenius-twisted or endomorphism-coefficient fibre system is therefore dead.* (b) (fibre-T3) *For $s \neq s'$ the set $\{t : \lambda_s(t) = \lambda_{s'}(t)\}$, $\lambda_s := L_s^*$, contains no two λ_s -consecutive points — Theorem 2.17 applied inside the fibre.* (c) *For each σ at most one s has $\lambda_s(\sigma) = \sigma$; hence E255 holds in N iff $\sum_a \sum_s |\text{Fix}(\lambda_s^{(a)})| = 7|M|$.*

Theorem 8.8 (three-neighbour affine rigidity). [proved in R7-A Thm 4, root-verified] [computational: all 8000 coefficient triples over $\mathbb{F}_5$, `r7a_verify.py V5`] *If the three operations $\diamond_{a,4a}$, $\diamond_{2a,a}$, $\diamond_{a,2a}$ are affine over an abelian group, then $\star = \diamond_{3a,a}$ is affine and, by Theorem 8.5, its left coefficient is invertible; hence E255 holds at level a .* It suffices that 3 of the 49 operations be affine, not all 49.

Remark (the $\{1, 3\}$ collision is *not* a protection here). [computational: R7-A §§6–7, `r7a_local.py`, `r7a_local3.py`, `r7a_global.py`] This is the essential difference from Round 5. At an idempotent level the collapse is four-fold and forces the block to be an *E677* magma (Theorem 8.1(a)) — genuinely circular. Here only positions 1 and 3 collide, and the entire content of that collision is (i) a determination of $\diamond_{a,3a}$ from $\star$ and $\diamond_{a,6a}$, and (ii) the fibre analogue of Theorem 2.17 (Corollary 8.7(b)) — verified to be exactly the same condition, 0 mismatches in 4000 trials. Defective operations can satisfy it: over $|M| = 5$ there is an explicit *ten-operation* assignment solving I_1, I_2, I_3 simultaneously, with all ten operations left-invertible and $\star$ missing two values, so that *E255* would fail at two points; and all 20 single-transposition defective perturbations of the direct-product star extend to solutions of $\{I_1, I_2, I_3\}$. *There is no local protection at this base; any protection theorem must be global.*

Remark (the global wall). [computational: R7-A §8, R7-B §B.6; time-boxed randomised search, no SAT] The 49-operation system is square, with propagation rules determining the operation at P_4 (always) or at P_1 (when the derived map is bijective) from the other three; the minimal propagation seed is 19 operations (P_4 only) or 16 (P_4 and P_1), so blind enumeration is out of reach. An annealer seeded at the direct product with a forced defect reaches 7 violated cells out of 1225 at $|M| = 5$, 6 out of 2401 at $|M| = 7$, and 7 out of 3969 resp. 12544 at $|M| = 9, 16$; the control (same annealer, no defect required) returns to 0 immediately. *In every best state the residual violations are exactly the three star-instances I_1, I_2, I_3 : the other 46 instances are perfectly satisfiable around a defective star.* §8.3 explains why.

8.3 The cocycle protection theorem

Corollary 8.6 kills every $\star$ that is separable over a *group*. The natural escape is to let the fibre be a magma: take $(M, \circ)$ to be itself a finite *E677* magma — for instance one of the non-right-cancellative models of §7.7, where right translations need not be surjective and a defect looks available — and put

$$s \diamond_{x,y} t := \theta_{x,y}(s) \circ \psi_{x,y}(t), \quad \theta, \psi \in \text{Aut}(M, \circ).$$

Matching (4.1) term by term against *E677* for $(M, \circ)$ gives the *sufficient* cocycle conditions

$$(R1) \ \psi_i \theta_j = 1, \quad (R2) \ \psi_j \theta_l \psi_k = \theta_j, \quad (R3) \ \psi_l = \theta_l \theta_k, \quad (R4) \ \psi_i \psi_j \theta_l \theta_k = \theta_i,$$

with i, j, k, l the four positions. The constant solution builds an order-35 extension with zero violations [computational: `r7b_verify2.py S1`].

Theorem 8.9 (cocycle protection). *[proved in R7-B.1; arbitrary, possibly non-abelian, automorphism group; arbitrary base, root-verified] For a term-matched cocycle assignment, (R1)–(R4) force $\psi_{P_3} = \theta_{P_3} \theta_{P_1}^{-1}$. At the collision instance $P_1 = P_3 = \star$ (Theorem 8.1(b)), so $\psi_\star = \text{id}$ and hence $R_\sigma^\star(s) = \theta_\star(s) \circ \sigma$, whose image is $\text{Im}(R_\sigma^\circ)$. Since $(M, \circ)$ satisfies E255 we have $\sigma \in \text{Im}(R_\sigma^\circ)$, so E255 holds in the extension at every point of every non-idempotent level: the construction can produce a counterexample only if the fibre already is one.*

Proof. From (R1), $\theta_j = \psi_i^{-1}$; substituting into (R2) gives $\psi_i \psi_j \theta_l = \psi_k^{-1}$, and putting that into (R4) gives $\psi_k^{-1} \theta_k = \theta_i$, i.e. $\psi_{P_3} = \theta_{P_3} \theta_{P_1}^{-1}$. $\square$

Theorem 8.10 (thinness criterion). *[proved in R7-B.2, root-verified] Let $\star(s, t) = \theta(s) \circ \psi(t)$ with θ, ψ any bijections of a set M carrying a binary operation $\circ$, and let $N_\circ(c, v) := \#\{r : r \circ c = v\}$. Then $\#\{s : s \star \sigma = \sigma\} = N_\circ(\psi\sigma, \sigma)$. Theorem 2.7 applied to the big magma forces this to be ≤ 1 , and E255 fails at (a, σ) exactly when it is 0. Consequently:*

- (a) *if the fibre is right-cancellative then no defect is possible — this re-proves Corollary 8.6 and extends it to every quasigroup fibre with θ, ψ arbitrary bijections;*
- (b) *[computational: r7b_thin.py, bipartite matching] “N-thin with a hole” permutations do exist for all four non-right-cancellative models, so Theorem 2.7 alone does not close the route;*
- (c) *[computational: 0/330] but none of the 330 affine automorphisms of M_{77} is thin with a hole, so even ignoring Theorem 8.9 the automorphism-separable route over M_{77} is closed.*

Remark (honest scope). [computational: R7-B §B.5] The cocycle conditions are sufficient, not necessary. Over the quasigroup fibre $\mathbb{F}_5(2s+4t)$ there are “accidental” solutions of (4.1) violating (R1)–(R4) — 2048 of 2304 at a generic instance, and 84000 of 90000 separable solutions of the collision instance have $\psi_\star \neq \text{id}$. They are harmless there (Theorem 8.10(a) kills any defect anyway). Over M_{77} , where a defect is conceivable, the accidental solutions vanish: of the 36 separable solutions of the collision instance, 0 have $\psi_\star \neq \text{id}$. That last statement is [computational: M_{77} only], not proved in general. This is why every global search of §8.2 failed near the direct product: the seed and all its neighbours lie inside this family, which is now proved defect-free.

8.4 Quotient theorems: what a minimal counterexample cannot be

Round 7 closed by asking for the theorem “every blueprint extension satisfies E255”. The first thing that had to be settled was the statement.

Proposition 8.11. *[proved in R7-C C.1, root-verified] The one-element magma is a finite E677 magma satisfying E255, and a pair-indexed extension of it by a fibre M is a single operation on M whose instance of (4.1) is exactly E677. Hence “every blueprint extension over every base satisfies E255” is Problem 1.1. The statement must be restricted to non-trivial bases.*

Theorem 8.12 (every quotient is a blueprint extension). *[proved in R7-C C.2, root-verified] [computational: r7c_quotient.py on five models] In every E677 magma, E677 in the form $x = L_y L_x L_{L_y x}(y)$ exhibits x in the $\langle L \rangle$ -orbit of y , so $\langle L_b : b \in M \rangle$ is transitive. Consequently, if $\pi : N \rightarrow B$ is a surjective homomorphism of finite E677 magmas then L_Y maps $\pi^{-1}(x)$ bijectively onto $\pi^{-1}(\pi(Y) \diamond x)$, so all fibres are equinumerous, and choosing bijections realises N as a pair-indexed extension of B satisfying (4.1). Therefore*

$$\begin{aligned} & \text{“every blueprint extension over a base with } 1 < |B| < |N| \text{ satisfies E255”} \\ & \iff \text{“a minimal counterexample is simple.”} \end{aligned}$$

Theorem 8.13 (no idempotent quotient). *[proved in R7-C C.3, root-verified] Let N be a counterexample to Problem 1.1 of minimal order and $\pi : N \rightarrow B$ a proper quotient. If $a \in B$ is idempotent then, by Theorem 8.1(a), $\diamond_{a,a}$ is an E677 magma on the fibre and, by Theorem 8.4 specialised to $x_a = a$, E255 at (a, σ) in N is literally E255 for $\diamond_{a,a}$ at σ ; minimality forces that to hold. Hence E255 holds at every point of $\pi^{-1}(a)$, and:*

- the E255-failure set of a minimal counterexample lies over the non-idempotent part of every quotient;
- no quotient of a minimal counterexample is an idempotent magma. *This kills every Alexander-quandle / translation-invariant-with- $c = 0$ quotient outright, and in particular the base shape of M_{77} , M_{77}^{NT} (whose only proper quotient is the fully idempotent $\mathbb{F}_{11}(6x+6y)$ [computational: root-verified]) and of M_{176} , M_{496} (base $\mathbb{F}_{31}(5x-4y)$, fully idempotent). Those models remain legal as fibres.*
- The $\mathbb{F}_7(4x+3y)$ route is not killed: that base has exactly one idempotent, so only the level $a = 0$ is protected — precisely the level §8.2 had already excluded.

This upgrades the Round-5 obstacle (Proposition 5.10) from “this construction fails” to a structure theorem about minimal counterexamples. Combined with Corollary 8.2, the entire remaining extension-type search space is the single uniform three-instance configuration dissected in §8.2.

Proposition 8.14 (unique witness and the Φ -swap). *[proved in R7-C C.4, root-verified] [computational: 200 resp. 50 gauge-twisted extensions, `r7c_witness.py`] At a non-idempotent level, write $F := \diamond_{P_2(I_1)}$ and $G := \diamond_{P_4(I_1)}$. Then the instance I_1 forces the unique candidate*

$$t(\sigma) := (L_\sigma^G)^{-1}((L_\sigma^F)^{-1}(\sigma)), \quad \text{E255 at } (a, \sigma) \iff t(\sigma) \star \sigma = \sigma.$$

Moreover, with $\tau := t(\sigma) \star \sigma$ and $\Phi := L_{t(\sigma)}^* \circ L_\sigma^F \circ R_{t(\sigma)}^G$, substituting $s = \sigma$, $t = t(\sigma)$ into I_1 gives $\Phi(\sigma) = \tau$ and $\Phi(\tau) = \sigma$: a defect at σ forces Φ to be a transposition on $\{\sigma, \tau\}$.

Remark (the residual is not an equational invariant). [computational: R7-C C.5, prover9 saturation] The annealing residual of §8.2 is always exactly three instances, which suggests a conservation law. The natural candidate is

$$(\text{INV}) \quad \text{the E255-failure set } D \text{ is } \langle L \rangle\text{-invariant,}$$

which would be decisive: with transitivity (Theorem 8.12) it gives $D = \emptyset$ or $D = M$, and $D = M$ is impossible once M has an idempotent. Prover9 was run on both translates (“E255 at $x \Rightarrow$ E255 at $z * x$ ” and at $x * z$) and *saturated* in both cases (`exit (sos_empty)`), so (INV) is not a first-order consequence of E677; any such conservation law must use finiteness, exactly as Proposition 5.1 predicts. The residual is instead explained by Theorem 8.9.

Remark (what Round 7 leaves). The named obstacle is (NTS): a *non-separable* star, or an accidental (non-term-matched) separable star over a non-right-cancellative fibre, whose column defect survives the other 46 instances — with the extra requirement of Proposition 8.14 that Φ be a transposition. Everything separable and term-matched is dead (Theorem 8.9); every quasi-group fibre is dead (Theorem 8.10(a)); every group-affine fibre system is dead (Corollary 8.6); every system with three affine neighbours is dead (Theorem 8.8); and the $M_{77}/M_{176}/M_{496}$ base shape is dead (Theorem 8.13). Round 8 appeared to close (NTS) as well, conditionally on one verified but unproved law (Corollary 9.13); **Round 9 refuted that law and reopened (NTS)**, and indeed proved that non-separable fibre operations exist (§10).

9 Round 8: simplicity, the transport law, and two barriers to first-order proof

Theorem 8.12 reduced one half of the programme to a single statement: *a minimal counterexample is simple*. Round 8 [23, 24, 25] attacked the contrapositive,

$$(S) \quad \text{a simple finite E677 magma is right-cancellative,}$$

which together with the already-proved (Q) $\Rightarrow$ E255 (Corollary 2.8) and a closure of (NTS) would give Problem 1.1. (S) is **not** proved. What Round 8 delivered is a dictionary that moves the whole question into permutation group theory, an exact reduction of (S) to three finite statements, a transport law that implies two of them and has a decisive structural consequence, and — the most durable output — *two independent proofs that the remaining statements cannot be reached by first-order reasoning at all*.

Throughout, $F_{ab} = \{t : a * t = b * t\} = \text{Fix}(L_b^{-1}L_a)$ as in Theorem 2.17, $\ker R_t$ is the partition of M by $a \mapsto a * t$, and the *collapse relation* is

$$a \varrho b : \Longleftrightarrow a \neq b \text{ and } F_{ab} \neq \emptyset.$$

(ϱ here is a relation on M ; it is unrelated to the offset reflection ρ of Corollary 4.6.) Thus (Q) says $\varrho = \emptyset$. Write ϱ^* for the transitive closure, $G := \langle L_y : y \in M \rangle \leq \text{Sym}(M)$, and recall that G is transitive (Theorem 8.12).

9.1 The congruence/block dictionary

Theorem 9.1 (dictionary). *[proved in R8-A Thm 1, root-verified] [machine-verified: problems/etp677/R8A_scripts/r8a_verify.py] For an equivalence θ on a finite E677 magma let $K_\theta \leq G$ be the kernel of the action of G on M/θ . Then θ is a congruence if and only if*

(i) θ is G -invariant — a block system of $\langle L \rangle$ — and

(ii) $L_b^{-1}L_a \in K_\theta$ for every $a \theta b$.

Proof. If θ is a congruence then every L_c preserves it, hence so does $L_c^{-1} = L_c^{r-1}$, giving (i); and right compatibility gives $(a * c) \theta (b * c)$, so applying the θ -preserving bijection L_b^{-1} yields $L_b^{-1}L_a(c) \theta c$ for every c , which is (ii). Conversely (i) is left compatibility, and from (ii) $L_b^{-1}(a * c) \theta c$; applying L_b gives $(a * c) \theta (b * c)$. $\square$

Corollary 9.2. *[proved in R8-A C1.1–C1.4, root-verified] (a) Every congruence class is an $\langle L \rangle$ -block, and since G is transitive all classes are equinumerous (a one-line sharpening of Theorem 8.12). (b) $\text{Con}(M)$ embeds as a lattice into the block lattice of the transitive group G , i.e. into the subgroup interval $[G_x, G]$. (c) $\langle L \rangle$ **primitive** $\Rightarrow M$ **simple**. (d) M has no congruence with k classes unless an E677 magma of order k exists; in particular none with 2, 3 or 4 classes (Theorem 6.1).*

Part (c) is unconditional and is what makes the counterexample side concrete: *to refute (S) it suffices to exhibit a non-right-cancellative finite E677 magma whose left-multiplication group is primitive* — a property decidable from the Cayley table in $O(n^3)$. All four known non-right-cancellative models are imprimitive with a unique minimal block system (Table 2).

Theorem 9.3 (column-kernel intersection). *[proved in R8-A Thm 2, root-verified] In every finite E677 magma $\bigcap_t \ker R_t = \Delta$. Consequently, if M is not a quasigroup then $t \mapsto \ker R_t$ is not constant, i.e. the column-kernel relation $\theta_\kappa := \{(t, t') : \ker R_t = \ker R_{t'}\}$ is a proper equivalence.*

Proof. $a \ker R_t b$ for every t says $L_a = L_b$, whence $a = b$ by Lemma 2.16(L1). If all $\ker R_t$ were equal to a common K then $K = \Delta$ and every column is injective. $\square$

This kills unconditionally one half of the escape “every pair collapses *and* all columns look alike”; it is what supplies *properness* in the reduction below.

9.2 The master reduction, and the collapse-count identity

Theorem 9.4 (master reduction). *[proved in R8-A Thm 4, root-verified] [implication intact; leg (A) refuted by M_{77}^D , so this route is dead — it is replaced by Theorem 10.10] Consider*

$$(A) \quad t \varrho t' \implies \ker R_t = \ker R_{t'}, \quad (B) \quad \varrho^* \text{ is a congruence.}$$

If M is a finite E677 magma with $|M| > 1$ satisfying (A) and (B) and M is not right-cancellative, then M is not simple. Hence (A) + (B) $\Rightarrow$ (S).

Proof. Non-Latin gives $\varrho \neq \emptyset$. (A) says $\varrho \subseteq \theta_\kappa$, and θ_κ is an equivalence, so $\varrho^* \subseteq \theta_\kappa \neq \nabla$ by Theorem 9.3. By (B), ϱ^* is a congruence, and it is neither Δ nor ∇ . $\square$

So (A) supplies properness and (B) compatibility. The second half has a purely numerical sufficient condition. Let $W(a, b) := |F_{ab}| = |\text{Fix}(L_b^{-1}L_a)|$ be the *collapse-count matrix*.

Proposition 9.5. *[proved in R8-A Prop 4.1, root-verified] [implication intact; (W) **REFUTED** by M_{77}^D (2112/17 787), having previously passed $\sim 38\,000$ random triples on 10 models with 0 mismatches] Consider*

$$(W) \quad W(c * a, c * b) = W(a, b) \quad \text{and} \quad W(c \setminus a, c \setminus b) = W(a, b) \quad \text{for all } a, b, c,$$

i.e. W is constant on the orbitals of the diagonal $\langle L \rangle$ -action. Then (W) $\Rightarrow$ (B): ϱ becomes a compatible reflexive symmetric relation (a tolerance), and the transitive closure of a tolerance is a congruence.

Remark (the symmetry explanation of (W) is false). [computational: R8-A §4.3; refuted] One would expect (W) to come from a conjugacy $L_{c*b}^{-1}L_{c*a} \sim L_b^{-1}L_a$ or an isotopy $L_{c*a} = A_c L_a B_c$. Both are *refuted*: the two displacements need not even have the same cycle type (871/3949 random triples differ on M_{77} , 1636/3952 on M_{77}^{NT} , 2795/3976 on M_{176} ; they do agree on M_{496}). In M_{77} the two fixed-point sets are the QR- and the QNR-halves of the fibration, equal in size only because $|\text{QR}_{11}| = |\text{QNR}_{11}|$. (W) is a genuine *counting* coincidence, not a symmetry — which already tells us that its proof cannot be structural.

9.3 The transport law (T) and the twisted diagonal

Definition 9.6.

$$(T) \quad a * t = b * t \implies (c * a) \setminus (c * t) = (c * b) \setminus (c * t) \quad \text{for all } c,$$

equivalently $L_c(F_{ab}) \subseteq E_{c*a, c*b}$, where $E_{xy} := \{v : x \setminus v = y \setminus v\} = L_x(F_{xy})$ is the value-side collapse set. (The label (T) is a statement, not the state map T of Theorem 5.2.)

Theorem 9.7. [REFUTED 2026-08-17 by M_{77}^D and M_{385}^{canon} — see Theorem 10.6. The verification record below stands as recorded and is what made the conjecture look safe] (T) **is false in general** (Theorem 10.6); the statement and its consequences are kept because the implications they carry are intact and because the shape of the error is the point (§10.4). As recorded before Round 9: (T) held in every finite E677 magma the campaign could then construct — exhaustively on M_{77} , M_{77}^{NT} (1 245 090 instances each) and M_{176} (14 868 480), plus 200 000 random on M_{496} and exhaustively on $M_{77} \times \mathbb{Z}_5$: 0 violations in 17.3M instances. It is not a triviality of finite left quasigroups: on uniformly random left quasigroups it fails on 75.3% ($n = 5$), 80.5% ($n = 6$), 84.2% ($n = 7$) and 86.2% ($n = 8$) of collapse instances [computational: 200 000 instances each].

Proposition 9.8. [proved in R8-B Prop B.1, root-verified] [implication intact; hypothesis (T) refuted, §10.4] (T) implies the exact transport $F_{c*a, c*b} = \Lambda_{c*a} L_c(F_{ab})$ and $E_{c*a, c*b} = L_c(F_{ab})$; hence (W), hence (B) via Proposition 9.5. It also implies the disjointness law $F_{c*a, c*b} \cap L_c(F_{ab}) = \emptyset$, which is therefore not an independent conjecture but Theorem 2.17 transported.

Proof. $\Lambda_{c*a} L_c$ is a bijection of M carrying F_{ab} into $F_{c*a, c*b}$, so $|F_{ab}| \leq |F_{c*a, c*b}|$; for fixed c the map $(a, b) \mapsto (c * a, c * b)$ is a bijection of M^2 , so summing over all pairs turns the inequality into an equality pointwise, and the inclusion into an equality. Disjointness is then Theorem 2.17 ($F_{xy} \cap L_x(F_{xy}) = \emptyset$) applied to the pair $(c * a, c * b)$. $\square$

Proposition 9.9. [proved in R8-B Prop B.2, from a prover9 saturation, root-verified] (T) is not first-order derivable from E677 plus the left-division axioms: it implies the goal “ $a * t = b * t \rightarrow \exists s((c * a) * s = (c * b) * s)$ ”, which saturates (`exit (sos_empty)`), and (T) itself saturates directly in 0.7 s. So (T) is a finite-only law.

Remark (the positive control that makes the saturations meaningful). [computational: R8-A §6, prover9] Six further goals — transitivity of ϱ , its left and right compatibility, (A), the disjointness law, and (Q) itself — all saturate in the same axiomatisation. That this is genuine negative information and not a broken encoding is established by the *positive control*: Theorem 2.17 is first-order, and prover9 proves it in 13 steps in exactly the same setting.

For $c \in M$ define the *twisted diagonal map*

$$\Delta_c(a, t) := (c * a, (c * a) \setminus (c * t)),$$

a bijection of $M \times M$, and let $K := \{(a, t) : N(t, a * t) \geq 2\}$ be the *collapse locus*.

Proposition 9.10 (flat form). [proved in R8-B, from (T), root-verified] [implication intact; hypothesis refuted] [computational: verified directly on 4 models] (T) implies that K is invariant under every Δ_c , hence a union of orbits of $G_\Delta := \langle \Delta_c : c \in M \rangle$ acting on $M \times M$.

Theorem 9.11 (affine Δ -dichotomy). [proved in R8-B Thm B.3, root-verified] [computational: 9 parameter combinations, `r8b_delta.py`] Let $x \diamond y = Ax + By + c$ over a field, with A, B invertible. Then G_Δ is transitive on $F \times F$ unless $\diamond$ is idempotent ($A + B = 1$ and $c = 0$), in which case the orbits are exactly the $|F|$ difference classes $\{(x, y) : y - x = \text{const}\}$.

Proof. Solving $A(z \diamond x) + B\xi + c = z \diamond y$ gives $\Delta_z = L + T_z$ with the z -free linear part $L(x, y) = (Bx, y - Ax)$ and $T_z = (Az + c, B^{-1}A(1 - A)z - B^{-1}Ac)$. Hence G_Δ contains every $\Delta_z \Delta_{z'}^{-1}$, i.e. all translations along $d := (1, B^{-1}(1 - A))$, and conjugation by Δ_z turns a translation by v into one by Lv . Now Ld is parallel to d iff $A + B = 1$. If $A + B \neq 1$ then d, Ld are independent and G_Δ contains all translations, so it is transitive. If $A + B = 1$ then $d = (1, 1)$ and every Δ_z shifts the difference $y - x$ by the same constant $-cB^{-1}$: for $c = 0$ the orbits are the difference classes, and for $c \neq 0$ the shift generates the additive group, so G_Δ is transitive again. $\square$

Theorem 9.12 (the dichotomy). *[proved in R8-B Thm B.4, root-verified; conditional on (T)]* **[VOIDED 2026-08-17 by M_{77}^D : the implication below is intact, its hypothesis is false — see §10.4]** Void in general. *[v7.2 correction:] a previous version of this note added “true for extensions with clean fibre operations (Lemma 10.4), which is where (T) does hold”. That addition was unproved and its second half is false: M_{385}^{canon} is an extension over a Latin base all of whose fibre operations are clean, and it refutes (T) (Proposition 10.9(c)). The hypothesis of this theorem is available only on the models where it has been checked (§10.4). Let M be a finite E677 magma whose Δ -action is transitive on $M \times M$, and assume (T). Then either M is a quasigroup, or E255 fails at every element of M ($\text{tr } N = 0$). In the second case M has no idempotent, and if M is a minimal counterexample it has no proper subalgebra at all.*

Proof. By Proposition 9.10, K is a union of Δ -orbits, so transitivity gives $K = \emptyset$ or $K = M \times M$. In the first case every column is injective. In the second, $N(t, v) \geq 2$ for every $v \in \text{Im}(R_t)$, while $N(t, t) \leq 1$ by Lemma 2.6; hence $N(t, t) = 0$ for every t , i.e. no t has a left unit. An idempotent would give $N(x, x) \geq 1$. Finally a proper subalgebra of a minimal counterexample is a smaller E677 magma, hence satisfies E255, and E255 at a point is the same statement there and in M . $\square$

Corollary 9.13. *[proved in R8-B Cors. B.4.1–B.4.3, root-verified; conditional on (T)]* **[VOIDED 2026-08-17 by M_{77}^D : the implication below is intact, its hypothesis is false — see §10.4]** **Retracted: the $\mathbb{F}_7(4x+3y)$ base is alive again, and (NTS) is reopened.** *What survives is the implication itself, on any class where (T) is available; [v7.2:] the earlier claim that “extensions with clean fibre operations” is such a class is withdrawn, since M_{385}^{canon} is clean and refutes (T) (Proposition 10.9(c)). (a) A finite E677 magma with transitive Δ and at least one idempotent is a quasigroup. (b) **The $\mathbb{F}_7(4x+3y)$ base dies.** It is non-idempotent, so Δ is transitive on it (Theorem 9.11, and 1 orbit of 49 [computational:]), and it has an idempotent; the collapse pattern of any pair-indexed extension projects to a Δ -invariant subset of $B \times B$, which is therefore empty (the extension is a quasigroup) or everything (making the diagonal fibre operation over the idempotent a smaller non-Latin E677 magma, impossible for the smallest such, and E255-protected by Theorem 8.13 for a minimal counterexample). This closes exactly the hole (NTS) that Round 7 left open — conditionally on (T). (c) What a base may still be: combining with Theorem 8.13, every affine proper quotient of a non-simple minimal counterexample must be idempotent-free, i.e. of the exceptional $\mathbb{F}_{31}(5x-4y+1)$ shape whose orders are pinned to $31 \cdot S$ by Theorem 7.34; and those are Δ -transitive, so Theorem 9.12 then forces E255 to fail everywhere, not merely somewhere.*

Remark (the decisive fork — resolved in §10). Either no non-Latin pair-indexed extension over $\mathbb{F}_7(4x+3y)$ — or over any Δ -transitive base with an idempotent — exists, which closes (NTS); or such an extension exists and *refutes* (T), killing the (W)/(B) route. Both outcomes are decisive, and the object to be searched is the one Round 7 already spent a round on. **Note added: the second branch is the true one** — Theorem 10.6. Consistency check: the bases of all four known non-Latin models are idempotent, hence Δ -imprimitive with the difference classes as orbits — which is exactly what allows their collapse patterns (the QR-classes of M_{77} , the $\{\pm 1\}$ -classes of M_{176}) to exist. Had any known model possessed a Δ -transitive base with an idempotent, (T) would already be refuted.

9.4 The shear decomposition and the counting shadows

Theorem 9.14 (shear decomposition). *[proved in R8-D Thm D.1, root-verified]* *[computational: 0 violations on 10 models, `r8d_shear.py`] Let $\Omega(x, y) := (x, x \setminus y)$ and $\Pi(r, s) := (r * s, r)$, both bijections*

of $M \times M$ (Π is the pair rotation of Proposition 5.6). Then for every c

$$\Delta_c = \Omega \circ (L_c \times L_c), \quad \Pi \circ \Delta_c = (L_c \times L_c) \circ \text{sw}.$$

Unconditionally, Δ_c carries the column fibre $\{(a, t) : a \in M\}$ onto the value fibre $\{(A, s) : A * s = c * t\}$.

So the twisted diagonal action is nothing but the plain diagonal action followed by one *fixed*, c -independent shear. That single observation determines the shape of everything below.

Definition 9.15. Put $\nu(a, t) := N(t, a * t)$ and $A(x, y) := N(x \setminus y, y)$. The *cardinality shadow* of (T) is

$$(T^*) \quad \nu \text{ is } \Delta\text{-invariant, equivalently } A(c * a, c * t) = \nu(a, t) \text{ for all } a, t, c.$$

Proposition 9.16. [proved in R8-D Prop D.4, root-verified] [implication intact; (T*) itself **REFUTED** by M_{77}^D , having previously been exhaustive for every model of order ≤ 77 (456 533 instances each) and 300 000 random on M_{176} , M_{496} with 0 violations] (T) $\Rightarrow$ (T*), and in fact $L_c(S(a, t)) = S(\Delta_c(a, t))$ exactly, where $S(a, t) := \{b : b * t = a * t\}$. Conversely (T) $\iff$ (T*) plus the single inclusion $L_c(S(a, t)) \subseteq S(\Delta_c(a, t))$; so (T*) is strictly the cardinality half, and it is the half a counting proof could reach.

Remark (no invariance explanation exists). [computational: R8-D §D.3; refuted] One might hope that ν (or A) is invariant under the *plain* diagonal action and that Ω preserves it. Both fail massively: $\nu(c * a, c * t) = \nu(a, t)$ fails in 53 820 of 59 319 triples on M_{77} , 16 920/46 656 on M_{176} , 28 830/29 791 on M_{496} , and A behaves identically. So (T*) is a *shear-compensation* identity: the plain diagonal action moves ν a great deal and the fixed shear Ω moves it back exactly. Together with the refutation of conjugacy and isotopy (§9.2), this closes every invariance-style explanation of (T).

Corollary 9.17 (the two marginals). [proved in R8-D Cors. D.6, D.6.1, D.7, root-verified; conditional on (T)] [**VOIDED** 2026-08-17 by M_{77}^D , in which $S_a \in \{247, 257\}$ and $\Sigma/n \notin \mathbb{Z}$: the implication below is intact, its hypothesis is false — see §10.4] In particular “a left identity forces a quasigroup” loses its proof. Write $S_a := \sum_t N(t, a * t) = \sum_b |F_{ab}|$, $H(t) := \sum_v N(t, v)^2$, $G(u) := \sum_w N(w, u)^2$ and $\Sigma := \sum_{t,v} N(t, v)^2$. Assuming (T):

- (a) S_a is constant in a (the t -marginal, plus transitivity of $\langle L \rangle$);
- (b) all or nothing: $S_a \geq n$ always, with equality iff row a collides with no other row, so either every row is collision-free — M is a quasigroup — or no row is. In particular a finite E677 magma with a left identity e has $S_e = \text{tr } N \leq n$, hence $\Sigma \leq n^2$, hence (Cauchy–Schwarz, $\Sigma \geq n^2$ always) it is a quasigroup;
- (c) $G(c * t) = H(t)$, and since $\text{supp } N$ is strongly connected (Theorem 2.22) this forces $H \equiv G \equiv \Sigma/n$ constant.

Proposition 9.18 (the global count is exactly circular). [proved in R8-D Prop D.8, root-verified] [unconditional: neither the statement nor the proof uses (T) or Corollary 9.17] Let $\Xi(c) := |\Delta_c(K) \cap K|$. Then $\Xi(c) \leq |K|$ with equality iff K is Δ_c -invariant, so “ K is Δ -invariant for all c ” is equivalent to $\sum_c \Xi(c) = n|K|$. But $\sum_c \Xi(c) = n|K| - \#\{(c, a, t) : (a, t) \notin K, \Delta_c(a, t) \in K\}$, so that identity is literally the statement being proved. Global double counting on K therefore carries no information beyond the pointwise statement.

Remark (the marginal comparison, restated without the voided corollary — v7.2). The last clause of this proposition previously read “...and the two marginals of Corollary 9.17 are strictly weaker”, which compares against a corollary Round 9 has since voided. The comparison is restated here so that it stands on its own. The two objects being compared are the *statistics*

$$(\mathbf{M1}) \ a \mapsto S_a \text{ is constant,} \quad (\mathbf{M2}) \ H \equiv G \equiv \Sigma/n,$$

which are defined in every finite magma, whatever the status of (T). Unconditionally, (M1) and (M2) fix only the t -sum and the a -sum of $\nu \circ \Delta_c$ and say nothing about its distribution, so neither can imply Δ -invariance of K . That they are *strictly* weaker than (T) was in v7 an expectation; it is now a fact witnessed by a model: M_{385}^{canon} satisfies (M1) and (M2) exactly ($S_a \equiv H \equiv G \equiv \Sigma/n = 595$) and refutes (T) (Theorem 10.8, stratum 3). What Corollary 9.17 supplied — and no longer does — is the converse direction, that (T) implies (M1) and (M2); the two filters of the remark below are therefore *necessary conditions for (T) in the class where the corollary’s hypothesis holds*, and not a test for (T) anywhere.

Remark (two cheap falsification filters — and their blind spot). [proved in R8-D §D.6, root-verified; conditional on (T)] **[VOIDED 2026-08-17 by M_{385}^{canon} , which refutes (T) while passing both filters exactly: the implication below is intact, its hypothesis is false — see §10.4]** Corollary 9.17 converts into two $O(n^2)$ table statistics that any candidate model must pass:

$$\mathbf{Filter\ 1:} \ S_a = \sum_t N(t, a * t) \text{ independent of } a; \quad \mathbf{Filter\ 2:} \ H(t) = G(u) = \Sigma/n \text{ for all } t, u.$$

A candidate violating either *refutes* (T) *outright*, hence the whole (W)/(B) route, without any inspection of collapse sets. Both filters are part of Table 2.

9.5 Blocks versus congruences: the statement C1

Corollary 9.2(c) gives “primitive $\Rightarrow$ simple”. The converse needs the extra statement

$$(C1) \quad \text{every } \langle L \rangle\text{-block system of a finite } E677 \text{ magma is a congruence,}$$

which would upgrade “a minimal counterexample is simple” (Theorem 8.12) to “ $\langle L \rangle$ is primitive” and hand the problem to O’Nan–Scott. Round 8 did not settle (C1) but made it decidable, proved it for two large classes, and found no counterexample [24].

Proposition 9.19 (decidability). [proved in R8-C §1, root-verified] [machine-verified: `problems/etp677/R8C_scripts/r8c_blocks.py`] *A partition is an $\langle L \rangle$ -block system iff it is closed under $x \sim y \Rightarrow c * x \sim c * y$, and a congruence iff it is closed under that and under $x \sim y \Rightarrow x * c \sim y * c$. Every block system is the join of the principal ones $\beta(0, b)$ (using transitivity), and joins of congruences are congruences. Hence (C1) holds for M iff $\text{leftclosure}(0, b) = \text{fullclosure}(0, b)$ for every b — a test costing $O(n^3)$.*

Theorem 9.20 ((C1) for affine models). [proved in R8-C T-A, root-verified] *Let $x \diamond y = Fx + Gy + c$ be an affine $E677$ magma on a finite abelian group A . Then every $\langle L \rangle$ -block system is a congruence.*

Proof. $L_y L_0^{-1}$ is translation by Fy , and F is onto, so $\langle L \rangle$ contains all translations and $\langle L \rangle = A \rtimes \langle G \rangle$ with A regular; the blocks through 0 are the G -invariant subgroups S , and the coset partition by S is a congruence iff $FS \subseteq S$ and $GS \subseteq S$. Now $F = (G + G^3)^{-1}$ and $u := G + G^3$ lies in the finite commutative ring $\mathbb{Z}[G] \leq \text{End}(A)$; u acts invertibly, so it is not a zero divisor of $\mathbb{Z}[G]$, and a non-zero-divisor of a finite commutative ring is a unit. Hence $F = u^{-1} \in \mathbb{Z}[G]$ and every G -invariant subgroup is F -invariant. $\square$

Theorem 9.21 (no 2-block system). *[proved in R8-C T-B, root-verified] [machine-verified: problems/etp677/R8C_scripts/r8c_proofs.py] No finite E677 magma has an $\langle L \rangle$ -block system with exactly two blocks.*

Proof. Let $\pi : M \rightarrow \mathbb{Z}_2$ be the block map and $f(c) \in \mathbb{Z}_2$ the indicator of “ L_c swaps the blocks”, so $\pi(c * x) = \pi(x) + f(c)$. Applying π to KEY in the form $L_{y*x}(y) = \Lambda_x \Lambda_y(x)$ gives $f(y * x) = \pi(x) + \pi(y) + f(x) + f(y)$. Hence $\lambda(x) := (\pi(x), f(x))$ is a magma homomorphism onto its image in $(\mathbb{Z}_2^2, \bullet)$ with $(p, q) \bullet (r, s) = (r + q, p + q + r + s)$, and the image has at least two elements because π is onto. But $(\mathbb{Z}_2^2, \bullet)$ has no subalgebra of order ≥ 2 satisfying E677 — by direct enumeration of all 15 subsets its only proper subalgebras are two singletons, and the whole 4-element algebra fails E677. $\square$

This is self-contained: it needs only the 4-element label table, not the exhaustive small-order searches, and it strengthens Corollary 9.2(d), which only said that a 2-block system could not be a congruence.

Theorem 9.22 (refinement is a congruence). *[proved in R8-A Thm 5 and R8-C T-C, root-verified] Project KEY through a block system θ with block set B , $k = |B|$, and put $\Phi(x) := \bar{L}_x \in \bar{G} := G/K_\theta$. Then $\Phi(y * x)(\bar{y}) = \Phi(x)^{-1} \Phi(y)^{-1}(\bar{x})$, and θ is a congruence iff Φ is constant on blocks. If $\bar{G}$ acts regularly on B , then $\lambda(x) := (\bar{x}, \Phi(x))$ is a magma homomorphism, so $\ker \lambda$ is a congruence with at most k^2 classes; hence either θ is a congruence or it is strictly refined by a congruence with $\leq k^2$ classes.*

For general $\bar{G}$ the method stalls for a precise reason: the projected identity pins $\Phi(y * x)$ down at the *single* point $\bar{y}$, and a permutation of $k > 2$ points is not determined by one value.

Theorem 9.23 (exact reformulation, and where it becomes circular). *[proved in R8-C T-D, root-verified] (C1) is equivalent to “every left-compatible equivalence of a finite E677 magma is right-compatible”. Moreover, for any left-compatible θ and any $a \theta b$,*

$$L_b^{-1} L_a \in K_\theta \iff \Theta_a(x) \theta \Theta_b(x) \forall x \iff \Lambda_a(x) \theta \Lambda_b(x) \forall x \iff L_b L_a^{-1} \in K_\theta,$$

and the first and last are equivalent for free because $K_\theta \trianglelefteq G$. So the natural term-level attack — rewrite $L_b^{-1} L_a$ through KEY and push it through θ — provably closes into a tautology: KEY carries the kernel condition onto its own Λ -mirror.

Remark (status of (C1)). [computational: R8-C §§2, 7; 31 models, orders 5–496, 0 separations] (C1) was verified on 31 models — the zoo, several direct products, $M_{77} \times \mathbb{Z}_5$ of order 385, M_{496} , and ten *new* non-affine translation-invariant gadget extensions built for the purpose — with no separation between the block lattice and the congruence lattice. Every observed block count lies in the E677 spectrum, as (C1) requires. A bounded first-order search timed out rather than saturating, with a passing positive control, so (C1) is recorded as **conjectured true**, not as shown-underivable: the honest statement is that the natural term route is provably circular (Theorem 9.23) and no derivation was found within the budget.

9.6 The two legs, and why there are only two

Status note (added in v7.2). Every statement in this subsection is made under (T), or under (T) + (A), and Round 9 refuted both (Theorem 10.6). The implications proved here are intact and are used below only as implications; *no conclusion of this subsection may be read as an*

unconditional fact about finite E677 magmas, and each statement now carries that qualification locally as well as in the retraction table of §10.4. The one statement of the subsection that is not known to be false is (R) itself, which remains unrefuted.

Under (T) — false in general, see the status note — (W) holds, hence ϱ^* is an $\langle L \rangle$ -block system. By Theorem 9.1 it is a congruence iff

$$(R) \quad a \varrho b \text{ and } a * u = b * w \implies u \varrho^* w,$$

verified exhaustively with 0 violations in 2.1M instances over the four non-Latin models then known [conjectured; exhaustive 4/4 at the time; [v7.2] (R) was re-verified in Round 9 on the three new models as well, 0/17787 in M_{77}^D and 0/444675 in each order-385 object, so the count is now 7/7]. Round 8 then showed that (R) is not an independent third leg [25].

Theorem 9.24 (mirror forms). *[proved in R8-E E1, E5, root-verified] [computational: identical instance counts on three models] (R) is equivalent to each of (R^Λ) : $a \varrho b \implies a \setminus v \varrho^* b \setminus v$ for every v , and (R^Θ) : $\Theta_a(v) \varrho^* \Theta_b(v)$ for every v . Moreover ϱ is self-mirror: $E_{ab} = L_a(F_{ab}) = L_b(F_{ab})$, so $a \varrho b$ iff $a \setminus u = b \setminus u$ for some u .*

Theorem 9.25 ((R) is the (C1)-instance at ϱ^*). *[proved in R8-E E2, root-verified] [implication intact, hypothesis refuted in Round 9: the hypothesis (T) is refuted, and so is (A) in the displayed critical lemma — see §10.4. Nothing below may be read as an unconditional statement about finite E677 magmas] (T) + (C1) $\implies$ (R). Conversely (R) is the kernel condition at the single partition ϱ^* , so it inherits every (C1) result — in particular it holds in every affine model (Theorem 9.20), and ϱ^* never has exactly two classes (Theorem 9.21). Under those hypotheses the critical lemma therefore reads*

$$(A) + (T) + (C1) \implies (S).$$

[v7.2:] two of the three hypotheses are now known false, so the display is an implication with no available antecedent; it is superseded by Theorem 10.10, which reaches (S) from (B) + (Prop) instead. What survives here unconditionally is the second sentence: (R) is an instance of (C1), and every (C1) theorem applies to it.

Theorem 9.26 (separation criterion). *[proved in R8-E E3, root-verified] [computational: 4/4 models, r8e_sep.py] [implication intact, hypothesis refuted in Round 9: both hypotheses, (T) and (A), are refuted — see §10.4. Nothing below may be read as an unconditional statement about finite E677 magmas] Assume (T) and (A), so ϱ^* is a block system with k blocks of size m and induced group $\bar{G} \leq \text{Sym}(B)$. If $\text{fix}(\bar{G}) := \max\{|\text{Fix}(h)| : 1 \neq h \in \bar{G}\} < |F_{ab}|/m$ for every ϱ -pair, then (R) holds. In M_{77} , M_{77}^{NT} , M_{176} and M_{496} the group $\bar{G}$ is $\text{AGL}(1, 11)$ resp. $31 : \mathbb{Z}_{10}$ — Frobenius, so $\text{fix}(\bar{G}) = 1$ — while $|F_{ab}|/m = 5, 5, 2, 15$. Hence in those four models, whose hypotheses were checked one by one, (R) is a theorem and not an assumption. [v7.2 scope correction:] this was stated in v7 as “in every known non-Latin model”. It is not: the criterion needs (T) and (A), which M_{77}^D refutes, and (T), which M_{385}^{canon} refutes, so it says nothing about the three models added in Round 9. (R) nevertheless still holds in all of them — 0/17787 in M_{77}^D and 0/444675 in each order-385 object [machine-verified: problems/etp677/R9_audit.py] — but there by verification, not by this criterion.*

Proof. For $t \in F_{ab}$ we have $\Phi(a)(\bar{t}) = \Phi(b)(\bar{t})$. By (A), F_{ab} is ϱ^* -saturated, hence a union of $|F_{ab}|/m$ blocks, so $\Phi(b)^{-1}\Phi(a)$ fixes at least that many blocks; the hypothesis forces it to be the identity, which is (R) via Theorem 9.1. $\square$

Theorem 9.27 (the mechanism behind (A)). [proved in R8-E E4, root-verified] [computational: 0/2541 and 0/14520 partial fixed sets] **[implication intact, hypothesis refuted in Round 9: the hypothesis (T) is refuted, and the subject (A) is false in general as well — see §10.4. Nothing below may be read as an unconditional statement about finite E677 magmas]** Under (T) + (R): (A) holds iff every displacement $L_b^{-1}L_a$ restricted to every ϱ^* -block is the identity or fixed-point-free; it suffices that K_{ϱ^*} act semiregularly on each block. In the known models that inner group is the translation group of the fibre, regular of order m — which is exactly why (A) costs nothing there. Slogan: (R) is about the outer action on the blocks (Frobenius); (A) is about the inner action inside a block (semiregular). [v7.2:] read as an implication only. Its hypothesis (T) and its subject (A) are both refuted in general (Theorem 10.6); what the theorem still does is explain why (A) costs nothing in the models where it does hold, which Proposition 10.9(a) now upgrades to a proof for the whole clean class.

Remark (failure blueprints: (A) and (T) need the same object). [proved in R8-E §7, R8-B §B.6, root-verified] Let the base of a non-simple M be Latin, so $\varrho \subseteq \varrho^* \subseteq$ the fibration. When ϱ^* equals the fibration — as in all four known models — (A) says exactly: *every fibre operation must have all of its columns with the same kernel*. For affine fibres the column kernel is $\ker \alpha_{x,y}$, independent of the column, so (A) is free. A refutation therefore needs a *non-affine* fibre operation, degenerate at one fibre point and injective at another — which is *verbatim* the requirement for refuting (T). So the two open legs share one minimal failure structure, and one search object decides both. (R), by contrast, can fail only if ϱ^* is strictly finer than the fibration and $\text{fix}(\bar{G}) \geq |F_{ab}|/m$ — the opposite of the Frobenius behaviour every known quotient has.

CORRECTED IN ROUND 9 (v7.1 local marker): Theorem 10.8(ii) shows this remark is only *half* right — the shared failure structure kills (T), but only its *unbalanced* version also kills (A): the canonical order-385 object refutes (T) while satisfying (A) exactly. “One search object decides both” is retracted; the two laws are decoupled.

Remark (the zoo has almost no discriminating power here). [computational: R8-A §7; WARN-2 discipline of §7.6] All four known non-Latin models are pair-indexed extensions with *affine* fibre operations, in which a column collapses iff $\alpha_{x,y} = 0$ — a condition depending only on the base coordinates, which is (A) for free. So (A), (B) and (W) holding on the zoo is close to *zero* evidence. The same applies to the prior “all known non-Latin models are non-simple”: they are non-simple *by construction*, and by Theorem 8.12 every non-simple magma is an extension. The one genuinely unconditional fact in support of (S) is Theorem 9.3. This is the fifth occurrence of the trap catalogued in §7.6.

9.7 Two barriers: existential and injective

Corollary 9.17(a) suggests proving S_a constant *unconditionally*, which would make the “all or nothing” dichotomy and “a left identity forces a quasigroup” unconditional. Round 8 mapped that route completely, and found two distinct reasons why the campaign’s proof technique cannot reach it.

Theorem 9.28 (the existential barrier). [proved in R8-F Thm F.2, root-verified] [computational: exhaustive over 6 739 216 depth- ≤ 2 term pairs] Let $X_a := \{(u, t) : u * t = a * t\}$, so $|X_a| = S_a$. Exactly eight pairs of depth- ≤ 2 terms define a bijection $X_a \rightarrow X_{c*a}$, all of the form $(u, t) \mapsto (c * u, \gamma)$ with

$$\begin{aligned} \gamma \in \{ & c * (a \setminus t), \quad c * (u \setminus t), \quad (c * a) \setminus (c * t), \quad (c * u) \setminus (c * t), \\ & (c * c) * (a \setminus t), \quad (c * c) * (u \setminus t), \quad (c \setminus c) * (a \setminus t), \quad (c \setminus c) * (u \setminus t) \}, \end{aligned}$$

the third being (T) itself. But for any term γ the implication $u * t = a * t \Rightarrow (c * u) * \gamma = (c * a) * \gamma$ supplies a witness for the saturating goal of Proposition 9.9, so every such statement is finite-only. Hence the technique of Theorems 7.9 and 7.12 — a term-defined bijection whose correctness is a first-order consequence — cannot prove S -constancy through any of them, and at depth ≤ 2 there is no bijection with a different first component.

Remark (honest nuance). S -constancy is a cardinality statement and does not itself imply that goal, so the target is not logically blocked — only this route to it is. A proof must be a global counting argument that never exhibits a pointwise correspondence. Consistently, the group-like explanation is refuted: if the displacement multiset $\{L_b^{-1}L_a : b\}$ were independent of a then S would be constant [proved in R8-F §F.3, root-verified], but that hypothesis [computational: holds in all six Latin benchmarks and] *fails in all four non-Latin models*.

The companion identity is sharper and cleaner.

Theorem 9.29 (the Q -matrix). [proved in R8-G Thm G.1, root-verified] [machine-verified: `problems/etp677/R8_invariants.py`] Let $Q(a, x) := \#\{z : a * z = z * x\} = |\text{Fix}(L_a^{-1}R_x)|$. Then all row sums of Q equal n — for each z the equation $a * z = z * x$ determines $x = z \setminus (a * z)$ uniquely — so Q has total n^2 , and

$$(S') \quad S'_x := \sum_z N(z, z * x) = \sum_a Q(a, x) = n \quad \text{for every } x$$

is exactly the statement that Q has constant column sums. Equivalently, (S') is Theorem 7.9 with L_a replaced by L_a^{-1} .

Theorem 9.30 (the injectivity barrier). [the barrier is **[computed]** and stands; (S') itself is **REFUTED** 2026-08-17 by M_{77}^D , where $S'_x \in \{42, 77, 112, 147\}$ — it had been exact, value n , in 10/10 models, with $\approx 100\%$ failure on random left quasigroups] [computational: R8-G §§G.2–G.4; 12 prover9 saturations] The proof of Theorem 7.9 does not transpose: the identity $|\text{Fix}(L_a R_x)| = N(a, x)$ has no analogue for Q (on M_{77} the entries of N lie in $\{0, 1, 7\}$ and those of Q in $\{0, 11\}$); the uniqueness mechanism “exactly one z with $b * z = z * x$ ” is refuted (M_{77} : $\{0 : 5390, 11 : 539\}$; M_{176} : $\{0 : 5280, 1 : 25344, 16 : 352\}$; it survives in the Latin models and, curiously, in M_{496}); and the opposite magma is not an $E677$ magma, so there is no duality. Since $\sum_x S'_x = n^2$ unconditionally, (S') follows from an injection $\Psi_x := \{(a, z) : a * z = z * x\} \rightarrow M$, and exactly 21 depth- ≤ 2 terms are injective on every Ψ_x of every one of the ten models. **None of those injectivity statements is first-order derivable**: twelve of them were put to prover9 and all twelve saturated.

Remark (what this upgrades). Injectivity has no existential in its conclusion, so the barrier of Theorem 9.28 does not apply to these statements — yet they are finite-only too. This is an independent second barrier, and it upgrades the campaign’s reading of Proposition 5.1: it is not only witness/existential statements about the collapse structure that need finiteness, *plain injectivity statements do as well*. Every known route to an exact $E677$ counting identity beyond Theorems 7.9 and 7.12 now passes through a non-first-order step.

Proposition 9.31 (where (S') would inject). [proved in R8-G Prop G.5, root-verified; conditional on (S')] [**VOIDED** 2026-08-17 by M_{77}^D : the implication below is intact, its hypothesis is false — see §10.4] (S') implies that every finite $E677$ magma with a right identity satisfies $E255$: if $z * x_0 = z$ for all z then $S'_{x_0} = \sum_z N(z, z) = \text{tr } N$, so (S') gives $\text{tr } N = n$, which is $E255$ by Theorem 2.7. M_9 is such a magma (Theorem 7.25), with $\text{tr } N = 9$. This is the companion of Corollary 9.17(b), “a left identity forces a quasigroup”: the two shadows of (T) each kill one identity type. Beyond this, (S') does not inject into (A), (T) or (S); it is a technique target.

9.8 Status, the decisive object, and the Round-9 outlook

statement	status (as of the end of Round 8; see §10.4)
(S)	open
(A) + (T) + (C1) $\Rightarrow$ (S)	proved (Thms. 9.4, 9.25); <i>route dead in Round 9</i>
(T) $\Rightarrow$ (W) $\Rightarrow$ (B), (T) $\Rightarrow$ (T*)	proved (implications intact)
(T), (T*), (A), (S')	refuted in Round 9 (Thm. 10.6); each had been exhaustively verified on every model then available, and none is first-order
(R)	conjectured, exhaustively verified, not first-order; survives, and is redundant in every known model (Thm. 9.26)
(C1)	conjectured true; proved for affine models; survives

Remark (the decisive object). Three independent lines — the (NTS) hole of Round 7, the failure blueprint for (T) (§9.3), and the failure blueprint for (A) (§9.6) — converge on *one* object:

a finite $E677$ magma given as a pair-indexed extension whose fibre operation at some base pair is **non-affine and genuinely two-variable**, degenerate at one fibre point and injective at another, over a base whose Δ -orbit structure permits a partial collapse pattern.

Such an object refutes (T) and (A) simultaneously, or — if it provably does not exist — closes (NTS) and, with (C1), gives (S). **Note added: it exists** (Theorem 10.5), and Theorem 10.8 shows that the two halves of the prediction come apart: a τ -dependent object always refutes (T), but only an *unbalanced* one also refutes (A). The advice to pre-screen candidates with Filters 1 and 2 was *wrong* — M_{385}^{canon} passes both and still refutes (T). What survives of this paragraph is the last clause: by Corollary 9.2(c) the strongest form of the search is for a non-Latin model with *primitive* $\langle L \rangle$.

Remark (cross-vendor result, not independently verified). [claimed in [26]; source: an external reasoning model; *not* re-derived or machine-checked by this campaign] A parallel line of the campaign put the same target to a third-party reasoning model. It reports (i) two near-misses — an order-13 table failing $E677$ at 6 of 169 instances, and an order-23 construction $a * b = a + p(b - a)$ with p a derangement (so that *no* element has a left unit) failing at 46 of 529 pairs — and (ii) an impossibility argument for the narrowed route above: if the other fibre operations are held at a fixed cocycle completion, then one of the three star-instances determines the left-unit-pair operation $\star$ *uniquely*, and the unique solution is the original, separable one; hence no transposition Φ can be implemented and a single-operation defect cannot exist. If correct, this is consistent with, and sharper than, Theorem 8.9: it says a counterexample must perturb at least one *other* fibre operation away from the cocycle family at the same time. **We record it as a claim.** It has not been re-derived here, its machine check has not been run in our pipeline, and nothing in §§8–9.8 depends on it. **Note added:** Round 9 proved it, in the stronger form Theorem 10.3, and the same source went on to produce one of the two decisive objects (§10.3); both are now verified here.

Table 2 is the Round-7/8 companion of Table 1, and it is consistent with it: M_9 appears in both, in Table 1 as the model that refutes injectivity of e and here as a primitive quasigroup with a single Δ -orbit. The two filters are constant in every row, as they must be if (T) is true; the only genuinely new row is $M_{77} \times \mathbb{Z}_5$, whose values are the multiplicative products of those of its factors — another instance of the multiplicativity that makes direct products useless as tests (§7.6).

model	n	(Q)	ϱ^*	prim.	(C1)	$\#\Delta\text{-orb.}$	$ K $	S_a	Σ	$H=G$	(T)
$\mathbb{Z}_5, 2x - y$	5	yes	5	yes	yes	5	0	5	25	5	0
$\mathbb{Z}_7, 4x + y$	7	yes	7	yes	yes	1	0	7	49	7	0
$\mathbb{Z}_7, 4x + 3y$	7	yes	7	yes	yes	1	0	7	49	7	0
$\mathbb{F}_9 = M_9$	9	yes	9	yes	yes	1	0	9	81	9	0
$\mathbb{Z}_{11}, 4x + 8y$	11	yes	11	yes	yes	11	0	11	121	11	0
$\mathbb{F}_{16}$ quandle	16	yes	16	yes	yes	16	0	16	256	16	0
E_{31}	31	yes	31	yes	yes	1	0	31	961	31	0
M_{77}	77	no	11	no	yes	11	2695	287	22099	287	0
M_{77}^{NT}	77	no	11	no	yes	11	2695	287	22099	287	0
M_{176}	176	no	11	no	yes	12	5632	656	115456	656	0
$M_{77} \times \mathbb{Z}_5$	385	no	55	no	yes	55	67375	1435	552475	1435	0

Table 2: Round-7/8 invariants, recomputed by `problems/etp677/R8_invariants.py`. ϱ^* = number of classes of the transitive closure of the collapse relation; “prim.” = $\langle L \rangle$ primitive; $\#\Delta\text{-orb.}$ = orbits of the twisted diagonal action; K = collapse locus; S_a and $H=G$ are the two filters of §9.4 (a single value means constant, as (T) predicts); (T) = violations of the transport law, exhaustive for $n \leq 77$ and 200 000 random above. **Note added:** (T) is false in general (Theorem 10.6); a zero in that column means only that the law holds *in that model*. Every model here does have clean fibre operations (Lemma 10.4), which is a common feature of the rows and not an explanation of them: [v7.2] cleanness does *not* imply (T) (Proposition 10.9(c)). The Q -matrix row sums (Theorem 9.29) and $S'_x = n$ were verified for every row. M_{496} is omitted — its Δ -orbit computation is $O(n^3) = 1.2 \cdot 10^8$ — and its figures (31 ϱ^* -classes, $\Sigma = 2\,031\,616$, $S_a = 4096$, (T) verified on 200 000 random instances) are quoted from [23].

10 Round 9: the decisive object, and a retraction chain

Section 9.8 ended with a specification rather than a theorem: three independent lines converged on one hypothetical object — a pair-indexed extension with a *non-affine* fibre operation whose degeneracy depends on the fibre point — and the fork was that such an object either does not exist, closing (NTS), or exists and refutes the transport law (T).

It exists. Round 9 [27] constructed two of them independently, and the second branch of the fork is the true one. This section reports the objects, the theorems that made them constructible, and — at equal length, because it is the more useful half — the resulting *retraction chain*: every statement of Sections 8 and 9 that rested on (T) is now void in general. Following the convention of this paper, nothing is deleted: the voided statements are kept, with their proofs, and marked. A retracted conditional theorem still records what *is* true — the implication — and the record of which hypotheses died is the campaign’s most transferable output.

10.1 The general extension equation and three structural theorems

Everything below drops the affine ansatz. Let $(B, \diamond)$ be a *Latin* finite $E677$ magma (by Theorem 8.12 a non-simple $E677$ magma is such an extension, and a non-Latin base would be a smaller counterexample), let V be a q -set, and for each ordered pair $(x, y) \in B^2$ let $\diamond_{x,y}$ be a left quasigroup operation on V — its *rows* $\lambda_s^{x,y} : t \mapsto s \diamond_{x,y} t$ are permutations, its *columns* $\rho_t^{x,y} : s \mapsto s \diamond_{x,y} t$ are arbitrary maps. Put $(x, s) * (y, t) := (x \diamond y, s \diamond_{x,y} t)$, and with $z_3 = y \diamond x$, $z_4 = z_3 \diamond y$, $z_2 = x \diamond z_4$ set $P_1 = (y, z_2)$, $P_2 = (x, z_4)$, $P_3 = (y, x)$, $P_4 = (z_3, y)$ as before.

Theorem 10.1 (the general extension equation). *[proved in R9-A Thm 1, root-verified] [computational: agrees with the full-table E677 check on every model] $N = B \times V$ satisfies E677 if and only if for all $x, y \in B$ and all $s, t \in V$*

$$t \diamond_{P_1} \left(s \diamond_{P_2} \left((t \diamond_{P_3} s) \diamond_{P_4} t \right) \right) = s.$$

The two scalar equations of Lemma 7.19 are its affine specialisation. Moreover (the collapse dictionary) the extension fails right cancellation iff some column $\rho_\tau^{x,y}$ is non-injective, and all of F_{ab} , $\ker R_t$, the multiplicity matrix, S_a , H and G are assembled fibrewise from the column-kernel data of the $\diamond_{x,y}$, by the formulas displayed in the proof.

Proof. Write $X = (x, s)$ and $Y = (y, t)$. Then

$$Y * X = (z_3, t \diamond_{P_3} s), \quad (Y * X) * Y = (z_4, (t \diamond_{P_3} s) \diamond_{P_4} t), \quad X * ((Y * X) * Y) = (z_2, s \diamond_{P_2} (\cdots)),$$

using at each step that the fibre operation used in a product is indexed by the ordered pair of base coordinates of its two factors: those pairs are (y, x) , (z_3, y) and (x, z_4) , that is, P_3 , P_4 and P_2 . One more product gives $Y * (X * ((Y * X) * Y)) = (y \diamond z_2, t \diamond_{P_1} (\cdots))$ with index $P_1 = (y, z_2)$. The base coordinate is $y \diamond z_2 = y \diamond (x \diamond z_4) = x$, since $(B, \diamond)$ itself satisfies E677. So the E677 instance at (X, Y) holds iff its fibre coordinate equals s , which is the displayed equation; quantifying over $x, y \in B$ and $s, t \in V$ gives the equivalence. (Every ordered pair (x, y) arises, so no instance is missed.)

For the dictionary, note first that $(x, s) * (y, \tau) = (x', s') * (y, \tau)$ forces $x \diamond y = x' \diamond y$, hence $x = x'$ because B is Latin, and then $\rho_\tau^{x,y}(s) = \rho_\tau^{x',y}(s')$. Consequently, writing elements of N as pairs:

$$\ker R_{(y,\tau)} = \bigsqcup_{x \in B} \{x\} \times \ker \rho_\tau^{x,y}, \quad F_{(x,s)(x',s')} = \begin{cases} \{(y, \tau) : \rho_\tau^{x,y}(s) = \rho_\tau^{x',y}(s')\}, & x = x', \\ \emptyset, & x \neq x', \end{cases}$$

and, for the multiplicity matrix of the extension, $N((y, \tau), (w, v)) = |(\rho_\tau^{x,y})^{-1}(v)|$ where x is the unique base element with $x \diamond y = w$. In particular a column of the Cayley table of N is non-injective iff one of the $\rho_\tau^{x,y}$ is, which is the right-cancellation statement. Since $S_a = \sum_t N(t, a * t)$, $H(t) = \sum_v N(t, v)^2$, $G(u) = \sum_w N(w, u)^2$ and $\Sigma = \sum_{t,v} N(t, v)^2$ are sums of these entries, each of them is a sum over $x \in B$ of a quantity depending only on the column kernels of $\diamond_{x,y}$; the explicit weighted form, for translation-invariant extensions, is Definition 10.7(iii). $\square$

Theorem 10.2 (degeneracy transfer). *[proved in R9-A Thm 2, root-verified] Under the equation of Theorem 10.1, for every t*

$$\rho_t^{P_4} \circ \lambda_t^{P_3} = \Theta_t, \quad \Theta_t(s) := s \setminus_{P_2} (t \setminus_{P_1} s).$$

Hence $\ker \rho_t^{P_4} = \lambda_t^{P_3}(\ker \Theta_t)$: the entire column-collapse structure of the fibre operation at P_4 — including how it depends on the fibre point t — is determined by $\diamond_{P_1}$ and $\diamond_{P_2}$ alone; $\diamond_{P_3}$ only relabels. Consequently, writing $N_Q(t, w) := |\{u \in V : u \diamond_Q t = w\}|$ for the multiplicity matrix of a single fibre operation,

$$N_{P_4}(t, w) = |\text{Fix}(\lambda_t^{P_1} \circ \rho_w^{P_2})|.$$

This is the pair-indexed analogue of Lemma 2.19, derived below from the extension equation; [v7.2:] it is not an instance of that lemma, which is a statement about a whole E677 magma and about the operation's own left division, and the fibre operations $\diamond_Q$ are not E677 magmas.

Proof. Read the equation as $\lambda_t^{P_1} \lambda_s^{P_2} \rho_t^{P_4} \lambda_t^{P_3}(s) = s$; since $\lambda_t^{P_1}$ and $\lambda_s^{P_2}$ are permutations, $\rho_t^{P_4} \lambda_t^{P_3}(s) = s \setminus_{P_2} (t \setminus_{P_1} s)$, which is the displayed identity, and applying it to kernels gives $\ker \rho_t^{P_4} = \lambda_t^{P_3}(\ker \Theta_t)$ because $\lambda_t^{P_3}$ is a bijection.

[v7.2, the step the review found missing.] Fix t and w . Substituting $u = \lambda_t^{P_3}(s)$ — legitimate because $\lambda_t^{P_3}$ is a permutation of V — turns $\{u : u \diamond_{P_4} t = w\}$ into $\{s : \rho_t^{P_4} \lambda_t^{P_3}(s) = w\} = \{s : \Theta_t(s) = w\}$, so $N_{P_4}(t, w) = |\Theta_t^{-1}(w)|$. Now unfold Θ_t : by definition of left division,

$$\Theta_t(s) = w \iff s \diamond_{P_2} w = t \setminus_{P_1} s \iff \rho_w^{P_2}(s) = (\lambda_t^{P_1})^{-1}(s) \iff \lambda_t^{P_1}(\rho_w^{P_2}(s)) = s,$$

the last step because $\lambda_t^{P_1}$ is a permutation. Hence $\Theta_t^{-1}(w) = \text{Fix}(\lambda_t^{P_1} \circ \rho_w^{P_2})$, which is the claim. $\square$

Theorem 10.3 (rigidity). *[proved in R9-A Thm 3, root-verified] If the base is Latin then $(x, y) \mapsto P_4(x, y) = ((y \diamond x) \diamond y, y)$ is a bijection of B^2 . Hence every fibre operation is the P_4 of exactly one base pair and satisfies, at that pair,*

$$u \diamond_{P_4} t = \sigma \setminus_{P_2} (t \setminus_{P_1} \sigma), \quad \sigma = t \setminus_{P_3} u.$$

[v7.1 scope correction:] when $P_4 \notin \{P_1, P_2, P_3\}$ this display uniquely determines $\diamond_{P_4}$ from the three partner operations, and such a fibre operation cannot be perturbed in isolation. When the pair is self-referential — $P_4 \in \{P_1, P_2, P_3\}$, as happens for diagonal pairs over idempotent base points, where all four coincide — the display involves left divisions in $\diamond_{P_4}$ itself and is a self-consistency constraint, not an explicit determination; uniqueness is not claimed there, and an isolated perturbation is constrained but not excluded by this theorem alone.

Remark (the cross-vendor claim of §9.8, now verified and generalized). The remark closing §9.8 recorded a third-party impossibility argument [26] — “once the other 48 fibre operations are fixed, one instance forces the left-unit-pair operation to equal the original one, hence separable” — explicitly as an unverified claim. Theorem 10.3 *proves it for every non-self-referential pair*, over every Latin base and every fibre simultaneously — not a peculiarity of order 77 or of the two order-7 bases. [v7.1:] for self-referential (e.g. idempotent-diagonal) pairs the theorem yields only a self-consistency constraint, so the third-party instance is subsumed exactly when its pair is non-self-referential. Its corollary — that a defect must be spread over several fibre operations — is exactly how the object below was found: *both* the square-offset and the non-square-offset operations differ from those of M_{77} . The claim is upgraded from “recorded” to “confirmed for non-self-referential pairs and generalized”.

Lemma 10.4 (dichotomy, and constant-column absorption). *[proved in R9-A §4 and Thm 4, root-verified] [computational: all fibre operations of M_{77} , M_{77}^{NT} , M_{176} , M_{496} are clean] Call a fibre operation clean if every column is either injective or constant. Then $\diamond_Q$ is clean iff it is Latin or constant-column ($s \diamond_Q t = c(t)$ with c a permutation). Anything else is a decisive fibre operation: either some column is injective and another is not (τ -dependent degeneracy), or some column has image size strictly between 1 and q . Moreover, if $\diamond_{P_2}$ is constant-column then $\diamond_{P_4}$ is Latin, and if $\diamond_{P_1}$ is constant-column then Θ_t is independent of t , so $\diamond_{P_4}$ has no τ -dependence: a τ -dependent operation requires both $\diamond_{P_1}$ and $\diamond_{P_2}$ to be non-constant-column.*

Affine fibre operations $s \diamond t = \alpha s + \beta t$ over a field are Latin when $\alpha \neq 0$ and constant-column when $\alpha = 0$ — always clean. That is why (A) cost the old zoo nothing — on the clean class (A) is a theorem, Proposition 10.9(a) — and it is the fifth and sharpest confirmation of the

benchmark trap of §7.6. [v7.2:] v7 said the same of (T) and (W). For (W) cleanness reduces the law to a counting condition on the base but does not settle it, and for (T) the explanation is simply wrong: a clean extension can refute (T) (Proposition 10.9(c)). What the zoo’s affine fibres do guarantee is (A); that (T) also held in all of them remains, on the present evidence, unexplained.

10.2 The decisive object M_{77}^D

Theorem 10.5. *[proved in R9-A §6, root-verified] [machine-verified: `problems/etp677/R9_audit.py`, an independent audit written for this paper reading only the archived table `R9A_scripts/r9a_DO_table.txt`] Let the base be $\mathbb{F}_{11}$ with $x \diamond y = 6x + 6y$ and the fibre $V = \{0, \dots, 6\}$, and take $\diamond_{x,y}$ to be $D : s \diamond t = 4s + t$ at offset $d = y - x = 0$, the table U at nonzero quadratic-residue offsets and the table W at non-residue offsets, where*

U	0	1	2	3	4	5	6	W	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6	0	5	3	4	1	6	0	2
1	0	1	2	3	4	6	5	1	3	1	2	0	4	5	6
2	0	1	2	3	4	5	6	2	6	2	1	5	0	4	3
3	0	1	2	3	4	5	6	3	1	0	6	3	5	2	4
4	0	1	2	3	4	5	6	4	2	4	5	6	1	3	0
5	0	6	2	3	4	5	1	5	0	1	3	4	2	5	6
6	0	5	2	3	4	1	6	6	4	1	0	2	3	5	6

Then $M_{77}^D := (\mathbb{F}_{11} \times V, *)$ is a finite E677 magma of order 77: zero E677 violations over all 5929 pairs, all left translations bijective, E255 holds everywhere, and it is not right-cancellative. Its column profiles are

$$U : (7) (5, 1, 1) (7) (7) (5, 1, 1) (5, 1, 1), \quad W : (1^7) (3, 1^4) (1^7) (1^7) (3, 1^4) (3, 1^4),$$

so W is injective at $t \in \{0, 2, 3, 4\}$ and degenerate at $t \in \{1, 5, 6\}$: τ -dependent degeneracy, and U has partial degeneracy. Neither U nor W is affine, and by Lemma 10.4 neither is separable.

Remark (verification protocol). The construction was verified through four mutually independent code paths — the pair equation of Theorem 10.1, a self-contained PYTHON re-implementation, the *previous round’s* checker `r8a_lib.py`, and a standalone C program reading only the stored table — plus a **negative control** (swapping two table cells produces 8 E677 failures). For this paper the table was audited once more, by code written from the definitions alone (`R9_audit.py`), reproducing every figure below. *The decisive control is that M_{77}^D satisfies every statement that is a genuine first-order consequence of E677:* Theorem 2.17 and its refinement, $N(v, v) \leq 1$, the right-unit count of Proposition 2.11, and the column-transport identity $\sum_z N(z * x, z) = n$ of Theorem 7.9 — all with zero violations. It fails exactly the statements that were only ever conjectured from the affine zoo.

Theorem 10.6 (what M_{77}^D refutes). *[machine-verified: `problems/etp677/R9_audit.py`] In M_{77}^D :*

<i>statement</i>	<i>where</i>	<i>violations</i>
(T) <i>transport law</i>	Def. 9.6	174 900/1 041 810 (<i>exhaustive</i>)
(T*), (T- <i>flat</i>)	Def. 9.15, Prop. 9.10	<i>fail</i>
(W) <i>collapse-count invariance</i>	Prop. 9.5	2112/17 787
(D) <i>disjointness of the transported collapse</i>	Prop. 9.8	528/2541
(A) $t \varrho t' \Rightarrow \ker R_t = \ker R_{t'}$	Thm. 9.4	165/231
Filter 1: S_a <i>constant</i>	§9.4	$S_a \in \{247, 257\}$
Filter 2: $H \equiv G \equiv \Sigma/n$	§9.4	$H, G \in \{207, 287\}, \Sigma/n = 19459/77 \notin \mathbb{Z}$
(S') $\sum_z N(z, z * x) = n$	Thm. 9.29	$S'_x \in \{42, 77, 112, 147\}$
Idem(M) <i>is a subalgebra</i>	Prob. 11.6	55/121 <i>idempotent pairs</i>
θ_κ <i>is a congruence</i>	Thm. 9.3	44 <i>classes, not a congruence</i>

Surviving in it: E255, (R) (0/17 787), (B) (ϱ^* *is a congruence with 11 classes*), (C1), $\theta_\kappa \neq \nabla$ (44 *distinct column kernels, as Theorem 9.3 requires*), $\Psi_1 = \text{tr } N = 77$ (Corollary 7.14 *is untouched*), and every first-order item. It is not simple, so (S) is not refuted.

Status of this table, and of Theorem 10.5 (v7.2). *Both are statements about one explicit 77×77 Cayley table, and nothing in either is proved in-text. What is proved in-text is only that the data displayed in Theorem 10.5 — base, fibre, three 7×7 tables and the offset rule — determine that Cayley table, and that E677 for it is equivalent to the finitely many instances of the equation of Theorem 10.1. Everything else is a finite computation: the E677, E255 and cancellation status, the column profiles, and every row of the table above were computed by `R9_audit.py`, written from the definitions for this paper and reading only the archived table `R9A_scripts/r9a_DQ_table.txt`, and independently by the round's own `r9a_verify.c` (a standalone C program), by `r8a_lib.py` (the previous round's checker, which knows nothing of Round 9) and by a self-contained PYTHON re-implementation; the negative control is in the remark above. Three entries are hand-checkable and are checked in full here: the (T) witness in the remark below, and the (S') and Idem-closure witnesses in the paragraph after it. The reader who wants a proof rather than an audit should re-run the scripts; the paper claims no more than that.*

The last two rows are new here: the reports of [27] did not test (S') or the closure of Idem, and the audit written for this paper found that the same object refutes both. Two hand-checkable witnesses: $\sum_z N(z, z * x)$ already differs from 77 at $x = 0$; and $(0, 0)$ and $(2, 0)$ are both idempotent while $(0, 0) * (2, 0) = (1, 5)$ is not.

Remark (a hand-checkable refutation of (T)). Take $a = (2, 0)$, $b = (2, 1)$, $t = (0, 0)$, $c = (0, 2)$. Both $a * t$ and $b * t$ have base coordinate $6 \cdot 2 + 6 \cdot 0 = 1$ and offset $0 - 2 = 9 \in QR_{11}$, so both use U , and $U[0][0] = U[1][0] = 0$: the columns collapse, $a * t = b * t = (1, 0)$. Now $c * a = (1, W[2][0]) = (1, 6)$ and $c * b = (1, W[2][1]) = (1, 2)$, while $c * t = (0, D[2][0]) = (0, 1)$. Solving $(1, 6) * (y, \tau) = (0, 1)$ gives $y = 10$, offset $9 \in QR$, $U[6][\tau] = 1$, i.e. $\tau = 5$; solving $(1, 2) * (y, \tau) = (0, 1)$ gives $U[2][\tau] = 1$, i.e. $\tau = 1$. So $(c * a) \setminus (c * t) = (10, 5) \neq (10, 1) = (c * b) \setminus (c * t)$.

Remark (the family, and where the object first appears). [computational: R9-A §8; complete DFS validated against the known counts 6 at $n = 5$ and 1680 at $n = 7$] Twenty-two distinct decisive objects were harvested at order 77; all satisfy E677 and E255, all fail both filters and (A), and all have ϱ^* equal to the 11-class fibration. Exhaustive searches show none exists in the analogous families at order 25 (exactly 30 solutions, all with Latin fibre operations) or at order 55 (no τ -dependent object, four conjugacy cases). *Within the searched families the decisive object first appears at order 77; no minimality claim is made outside them.*

10.3 The order-385 objects and the four-stratum theorem

A second decisive object arrived the same evening from an independent source: a third-party reasoning model produced an order-385 magma [28], which was rebuilt inside the framework above and verified here. It is a pair-indexed extension with base $\mathbb{F}_{11} \times \mathbb{F}_5$ (order 55, Latin, idempotent) and fibre $\mathbb{F}_7$, in which the sign of the selector is flipped on the $\mathbb{F}_5$ -diagonal at one coordinate; because that coordinate is part of τ , the coarse column kernel again depends on τ . A deliberate variant M_{385}^R flips the sign on the *offset* instead, which is τ -independent. Both are genuine *E677* magmas satisfying *E255*, and they separate the strata.

The four strata are named by properties of the fibre operations. Those names were used in v7 without definitions — a gap the adversarial review was right to flag — so they are fixed here, in the form they have in [27, §B.2].

Definition 10.7 (the vocabulary of the strata). Let $N = B \times V$ be a pair-indexed extension as in §10.1, with $|V| = q$, and let $\diamond_Q$ be one of its fibre operations, with columns ρ_τ^Q .

- (i) $\diamond_Q$ is *separable* if there is a group structure on V , maps $\alpha, \beta : V \rightarrow V$ and a constant c with $s \diamond_Q t = \alpha(s) \beta(t) c$ — the hypothesis of Corollary 8.6, and the notion whose negation appears in (NTS). It is *clean* if every column is injective or every column is constant, equivalently (Lemma 10.4) if it is Latin or constant-column. *Non-separable* in the table below means: not of that form. An affine operation $s \diamond_Q t = \alpha s + \beta t$ over a field is separable and clean; a separable operation has $\ker \rho_\tau^Q = \ker \alpha$ for *every* τ , so it is clean exactly when α is injective or constant, and in the Round-7 setting of Corollary 8.6 the equation forces the first alternative. The non-separability of the specific operations named below is taken from [27], not re-derived here.
- (ii) $\diamond_Q$ has *τ -independent kernel* if the partition $\ker \rho_\tau^Q$ of V is one and the same for every $\tau \in V$, and *τ -dependent kernel* otherwise. Clean $\Rightarrow$ τ -independent (all kernels discrete, resp. all full), and separable $\Rightarrow$ τ -independent; the converse of the first implication fails, since an operation whose columns all have one and the same kernel with image size strictly between 1 and q is τ -independent and not clean. That is what stratum 2 below exhibits.
- (iii) Suppose N is *translation-invariant* over an affine base $x \diamond y = Fx + Gy$: the operation $\diamond_{x,y}$ depends only on the difference $y - x$. Group the differences into the classes O carrying one and the same fibre operation $\diamond_O$, let w_O be the size of the class, and put

$$N_O(\tau, v) := |(\rho_\tau^O)^{-1}(v)|, \quad f_O(s) := \sum_{\tau \in V} N_O(\tau, s \diamond_O \tau), \quad H_O(\tau) := \sum_{v \in V} N_O(\tau, v)^2.$$

The extension is *balanced* if the two weighted sums $\sum_O w_O f_O(s)$ and $\sum_O w_O H_O(\tau)$ are constant — independent of s , resp. of τ — and *unbalanced* otherwise.

- (iv) All four notions are relative to a chosen presentation of the model as an extension. The same magma can be clean over one base and decisive over another: M_{385}^{canon} is clean over its order-55 base (Proposition 10.9(c)) and τ -dependent over its order-11 base, which is the presentation used in the table.

Remark (why those are the right weighted sums, and what balance means). [machine-verified: problems/etp677/R9_clean_check.py, step 7] By the collapse dictionary in the proof of Theorem 10.1, the two sums of Definition 10.7(iii) are the two filter statistics themselves:

$$S_a = \sum_O w_O f_O(s) \quad \text{for } a = (\cdot, s), \quad H(t) = \sum_O w_O H_O(\tau) \quad \text{for } t = (\cdot, \tau).$$

So Filter 1 is exactly the constancy of the first and Filter 2 that of the second, and inside the translation-invariant family “balanced” and “passes both filters” are the *same* statement — which is why the informative columns of the table below are the (T) and (A) ones. On M_{77}^D , where $w_D = 1$ and $w_U = w_W = 5$, the two sums evaluate to $\{247, 257\}$ and $\{207, 287\}$: the numbers quoted in Theorem 10.6, recovered from the three 7×7 tables alone.

Theorem 10.8 (four strata). *[computational: R9-B §B.2.1: four witness models, instance counts as recorded there] [machine-verified: problems/etp677/R9_audit.py; every row re-derived here from the archived tables]*

	<i>fibre operation</i>	(T)	(A)	<i>Filters 1, 2</i>	<i>witness</i>
1	<i>affine (clean)</i>	<i>holds</i>	<i>holds</i>	<i>pass</i>	$M_{77}, M_{77}^{\text{NT}}, M_{176}, M_{496}$
2	<i>non-separable, τ-independent kernel</i>	<i>holds</i>	<i>holds</i>	<i>pass</i> (1435)	M_{385}^R
3	<i>τ-dependent, balanced</i>	FAILS	<i>holds</i>	pass (595)	M_{385}^{canon}
4	<i>τ-dependent, unbalanced</i>	FAILS	FAILS	fail	M_{77}^D

What this table is, and what it is not (v7.2). Every entry is a finite computation on an archived Cayley table — re-derived for this paper by `R9_audit.py` and, for the two order-385 objects, by the independent C checker `r9b_verify.c` — and the marker has been corrected from “proved” to “computational” accordingly. Three consequences of that:

- (a) *No row is an implication.* The table does not assert “fibre operations of type $X \Rightarrow (T)$ ”; it records what holds in four named models. The only implication of that shape proved anywhere in this paper is Proposition 10.9(a), for (A) on the clean class — and stratum 3 shows that no such implication is available for (T) from cleanness, since M_{385}^{canon} is clean over its own base.
- (b) *The four rows are not a partition.* They are the four types realized so far by the nine benchmark models of §10.6. A clean but non-affine extension, or a separable τ -dependent one, matches no row, and nothing here excludes further strata; *no exhaustiveness is claimed, and none is proved in [27]*. Read the table as a separation result — the four cells it does occupy are pairwise distinguished by the columns — not as a classification theorem.
- (c) *The filter columns for strata 3 and 4 are definitional,* by Definition 10.7(iii) — whose labels presuppose the translation-invariant presentation, which both witnesses do have, over $\mathbb{F}_{11}$ with fibres of size 7 resp. 35. The content of those two rows is that a model exists in each cell, and in particular that a (T)-counterexample can be balanced.

Three corrections follow, each overturning a statement made earlier in this paper.

- (i) **The two filters have a genuine blind spot.** Stratum 3 refutes (T) while passing Filter 1 and Filter 2 *exactly* ($S_a \equiv H \equiv G \equiv \Sigma/n = 595$). The filters are strictly weaker than (T), precisely as Proposition 9.18 predicted for marginals. *A candidate that passes both filters proves nothing.*
- (ii) **(A) and (T) are decoupled.** Stratum 3 refutes (T) but satisfies (A) (0/1155). The claim of §9.6 that the two legs share their minimal failure structure is therefore only half right: the shared structure kills (T), and only the *unbalanced* version also kills (A). M_{77}^D remains the unique known refutation of (A).

- (iii) **Non-separability is not the operative property.** Stratum 2 is non-separable — its coarse column images have sizes 11 and 29, neither dividing 35 — and every conjecture still holds in it. τ -dependence of $\ker \rho_\tau$ is the operative property, exactly as Theorem 10.2 predicts.

10.4 The retraction chain

Every statement below is *kept* in this paper, with its proof, and marked VOIDED. In each case the implication proved there remains true; what has died is a hypothesis.

statement	status after Round 9
(T), Theorem 9.7	REFUTED by M_{77}^D and M_{385}^{canon}
(T*), (T-flat), (W), (D), (A)	REFUTED by M_{77}^D
Propositions 9.8, 9.10, 9.16	implications intact, <i>hypothesis false</i>
Theorem 9.12 (the dichotomy) and Corollary 9.13	VOID in general. In particular “the $\mathbb{F}_7(4x + 3y)$ base is dead” is retracted — that base is alive again, and (NTS) is reopened
Corollary 9.17 and the two filters	VOID in general. In particular “a left identity forces a quasigroup” loses its proof
Theorem 9.4, route (A) + (B) $\Rightarrow$ (S)	implication intact, <i>leg (A) false</i> , so this route is dead
Theorems 9.25, 9.26, 9.27	implications intact, <i>hypotheses unavailable</i>
(S'), Theorem 9.29; Proposition 9.31	REFUTED by M_{77}^D (new here); the row-sum half of Theorem 9.29 is unconditional and survives
Idem(M) a subalgebra (Problem 11.6)	REFUTED by M_{77}^D (new here)
θ_κ a congruence	REFUTED by M_{77}^D
(B), (R), (C1), (S), Theorem 9.3, all first-order items	survive
The benchmark warning of §7.6	vindicated in full

Restricted survivors — rewritten in v7.2. None of the voided results is worthless: each holds verbatim on the class where its hypothesis is verified. The question is which class that is, and v7 answered it wrongly. It asserted that, since every fibre operation of a separable or affine extension is clean (Lemma 10.4), (T), (W), (A) and both filters hold throughout strata 1 and 2, so that Theorem 9.12 and Corollaries 9.13, 9.17 become true statements about extensions with clean fibre operations. **No such implication was ever proved — neither in [27] nor here — and for (T) it is false.** The following proposition, which is the one piece of new mathematics in this revision, separates what cleanness gives from what it does not.

Proposition 10.9 (what cleanness does and does not give). *[proved in this revision (v7.2), root-verified] [machine-verified: problems/etp677/R9_clean_check.py, steps 1-6] Let $N = B \times V$ be a pair-indexed extension over a Latin base, with $|V| = q$, all of whose fibre operations are clean, and put $C_x := \{y \in B : \diamond_{x,y} \text{ is constant-column}\}$. Then:*

- (a) $\ker R_{(y,\tau)}$ depends only on y and not on τ ; hence (A) holds in N .
- (b) $F_{ab} = \emptyset$ unless $a = (x, s)$ and $b = (x, s')$ lie in one fibre, in which case $F_{ab} = C_x \times V$. So $|F_{ab}| = |C_x|q$ depends only on the base coordinate, and (W) holds in N if and only if $x \mapsto |C_x|$ is constant on B .
- (c) (T) does **not** follow. M_{385}^{canon} is such an extension — base $\mathbb{F}_{11} \times \mathbb{F}_5$ with $(c, a) \diamond (d, b) = (6c + 6d, 2a + 4b)$, Latin of order 55; fibre $\mathbb{F}_7$; and each of its 55^2 fibre operations is one of

$s \diamond t = 4s + t$, $s \circ t = t$, $s \circ t = 3s + 5t$, hence *Latin or constant-column* — and it refutes (T), with the explicit witness $(a, b, t, c) = (70, 71, 0, 14)$ in the labelling $(c, a, s) \mapsto (5c + a) \cdot 7 + s$. In it $|C_x| = 5$ for every x , so $|F_{ab}| \equiv 35$ and both (A) and (W) do hold, exactly as (a) and (b) predict.

Proof. By the collapse dictionary in the proof of Theorem 10.1, two elements are identified by $R_{(y,\tau)}$ iff they lie in one fibre, over some x , and are identified by $\rho_\tau^{x,y}$. A clean $\diamond_{x,y}$ is Latin or constant-column (Lemma 10.4), so $\ker \rho_\tau^{x,y}$ is the discrete partition for every τ , or the full partition for every τ ; in both cases it does not depend on τ , and neither does $\ker R_{(y,\tau)} = \bigsqcup_x \{x\} \times \ker \rho_\tau^{x,y}$. For (A), note that $t \varrho t'$ means $t * u = t' * u$ for some u , which forces t and t' into one fibre by the same dictionary; their $\ker R$ then agree by what was just proved. That is (a). The same case distinction gives $F_{(x,s)(x,s')} = \{(y, \tau) : \rho_\tau^{x,y}(s) = \rho_\tau^{x,y}(s')\} = C_x \times V$ for $s \neq s'$, and $\emptyset$ across fibres, which is the first half of (b). For the second half: $c * a$ and $c * b$ lie in the fibre over $\gamma \diamond x$ when c lies over γ and a, b over x , so (W) reads $|C_{\gamma \diamond x}| = |C_x|$ for all γ, x ; since the base is Latin, $\gamma \mapsto \gamma \diamond x$ is onto B , so this holds for all γ, x iff $|C_\cdot|$ is constant. The left-division half of (W) is the same computation with $\gamma \setminus x$. Part (c) is a finite verification, performed independently of the Round-9 scripts by `R9_clean_check.py`: it rebuilds the order-385 table from the selector rule, confirms 0 *E677* violations in 148 225 instances, that the base is Latin, that all 3025 fibre operations are clean, that the displayed witness violates (T) — with 711 480 violations over a 1-in-7 stride in c — and that (A) holds (0/1155) with $|F_{ab}| \equiv 35$. $\square$

Consequently the honest restriction is the following, and no more.

- (1) On the clean class, (A) is a *theorem* (Proposition 10.9(a)), and (W) reduces to a counting condition on the base. This is the mechanism Theorem 9.27 was pointing at, now proved rather than observed.
- (2) On the clean class, (T), (T*), (T-flat) and the two filters have exactly the status they had before Round 9: *computational observations on named models*. They are verified in M_{77} , M_{77}^{NT} , M_{176} and M_{496} (exhaustively for $n \leq 77$, 200 000 random instances on M_{496} ; `R8_invariants.py`) and in M_{385}^{R} (`R9_audit.py`, `r9b_verify.c`) — and (T) is *refuted* in the clean object M_{385}^{canon} . Theorem 9.12 and Corollaries 9.13, 9.17 therefore survive as statements about the models in which their hypothesis has been checked, and not about any structurally delimited class. The sharpest form of the point: over the *same* Latin base of order 55 and with the *same* three clean fibre operations $4s + t$, t , $3s + 5t$, the two order-385 objects differ only in *which pair gets which operation* — and one satisfies (T) while the other refutes it. Whatever governs (T) is a property of the assignment $(x, y) \mapsto \diamond_{x,y}$, not of the individual operations.
- (3) Cleanness is in any case relative to a presentation (Definition 10.7(iv)): M_{385}^{canon} is clean over its order-55 base and τ -dependent over its order-11 base. Any future “restricted survival” claim must name the presentation.
- (4) The current candidate for a structural class on which the transport law holds is τ -independence of the coarse column kernels, in the sense of Theorem 10.8(iii) — a property of the assignment, which is what (2) says is needed. [conjectured; unproved in either direction; the evidence is the five models M_{77} , M_{77}^{NT} , M_{176} , M_{496} , M_{385}^{R} on one side and the two objects M_{77}^{D} , M_{385}^{canon} on the other]

What is lost is exactly the extrapolation from the checked models to all finite *E677* magmas — the extrapolation §7.6 had already flagged as unsupported. The v7 wording of this paragraph is itself the sixth occurrence of that trap, and the first to be caught by an adversarial review rather than by a model.

10.5 The corrected skeleton, and the state of (S)

Theorem 10.10 (the chain has two legs, not three). *[proved in R9-B B.1.1, root-verified]* Let M be a finite *E677* magma, $|M| > 1$, non-Latin. If (B) ϱ^* is a congruence and

$$(\text{Prop}) \quad \varrho^* \neq \nabla,$$

then M is not simple. Hence (B) + (Prop) $\Rightarrow$ (S).

Proof. Non-Latin means some column is non-injective, so $\varrho \supsetneq \Delta$ and a fortiori $\varrho^* \supsetneq \Delta$; with (B) and $\varrho^* \neq \nabla$ the closure is a congruence strictly between Δ and ∇ . $\square$

Remark (the existential weakening (Prop_∃); v7.1 correction). Write

$$(\text{Prop}_{\exists}) \quad \exists a \neq b : F_{ab} = \emptyset, \text{ i.e. some } L_b^{-1}L_a \text{ is fixed-point-free.}$$

Then (Prop) $\Rightarrow$ (Prop_∃): if $\varrho^* \neq \nabla$, pick a, b in different ϱ^* -classes; a direct collapse witness $t \in F_{ab}$ would put them in one class. *The converse fails in general:* $\varrho^* = \nabla$ needs only *connectivity* of the collapse graph, while $\neg(\text{Prop}_{\exists})$ is its *completeness*. A previous version of this theorem glossed the two as equivalent; the gloss was wrong and is hereby retracted (the theorem itself is unaffected, since its hypothesis is $\varrho^* \neq \nabla$). Consequences: (Prop_∃) is a *necessary* condition, so refuting it — proving every pair collapses directly, $\varrho = \nabla$ — refutes (Prop); but establishing (Prop_∃) does *not* establish (Prop). Theorem 10.13 and the design-theoretic reading below are statements about (Prop_∃) and about $\varrho = \nabla$, and are re-anchored accordingly.

(R) is *not* a third leg: it entered only as the kernel half of the dictionary criterion (Theorem 9.1), whose other half — that ϱ^* is a block system — came from (T). And (A) is gone, its role having been to supply properness, which (Prop) now names directly.

Remark ((B) has lost its proof). (B) had exactly two derivations: (W) $\Rightarrow$ (B) (Proposition 9.5) and (T) + (R) $\Rightarrow$ (B) (§9.6), with a third, (T) + (C1) $\Rightarrow$ (R), feeding the second. All three hypotheses are refuted. (B) *is therefore an open conjecture with no surviving proof route*, and its model evidence is the same trap once more: in every known non-Latin model — M_{77} , M_{77}^{NT} , M_{176} , M_{496} , M_{77}^{D} , the twenty-two-member family, both order-385 objects — the magma is a pair-indexed extension over a Latin base, so ϱ is contained in the fibration and ϱ^* is the fibration by construction, hence a congruence for free. The evidence for (B) is close to zero.

Proposition 10.11 ((Prop) is automatic for extensions). *[proved in R9-B B.3.2, root-verified]* In any pair-indexed extension over a Latin base, $(x, s) * (y, t) = (x', s') * (y, t)$ forces $x = x'$, so ϱ is contained in the fibration, hence so is its transitive closure, and $\varrho^* \neq \nabla$. Hence no pair-indexed extension over a Latin base violates (Prop), and no extension-based model can supply evidence for or against it. *[v7.1 scope correction:] a violator of (Prop) must lie outside this extension class; the further step — that it may be taken simple non-Latin, the counterexample target of Corollary 9.2(c) — is available only under the minimal-counterexample reduction, i.e. conditional on the quotient theorem’s hypotheses (NTS/extension-closure), not unconditionally.*

Theorem 10.12 (refined T3). *[proved in R9-B B.3.1, root-verified] [machine-verified: problems/etp677/R9_audit.py] For $a \neq b$, $|F_{ab}| \leq (n - \text{odd}(L_a))/2$ where $\text{odd}(L_a)$ is the number of odd-length cycles of L_a ; in particular $F_{ab} \cap \text{Fix}(L_a) = \emptyset$, and $|F_{ab}| \leq (n - 1)/2$ for odd n .*

Proof. Theorem 2.17 says F_{ab} is an independent set in the functional digraph of L_a , a disjoint union of cycles of lengths ℓ_i ; a cycle of length ℓ has independence number $\lfloor \ell/2 \rfloor$ and $\sum_i \lfloor \ell_i/2 \rfloor = (n - \text{odd}(L_a))/2$. A fixed point is a 1-cycle. $\square$

Theorem 10.13 (why the primitive-group route is silent). *[proved in R9-B B.4.1–B.4.2, root-verified] Put $D := \{L_b^{-1}L_a : a \neq b\} = S^{-1}S \setminus \{1\}$, where $S = \{L_a : a \in M\}$. Then $(\text{Prop}_{\exists})$ — the necessary condition of (Prop), see the v7.1 remark after Theorem 10.10 — holds iff D contains a derangement, iff D is not contained in the union of point stabilisers. Jordan’s theorem and the Fein–Kantor–Schacher theorem produce derangements in the group $\langle L \rangle$, not in the subset D , which is not a subgroup. Worse, by Theorem 10.12 every element of D with a fixed point moves at least $n/2$ points, so D lies exactly in the large-support regime where Jordan-type “small support $\Rightarrow$ contains A_n ” theorems are silent. [v7.2:] the first two sentences are proved; the last is a methodological observation about the reach of those theorems — it says that the hypotheses of the Jordan-type results are provably unavailable here, not that no group-theoretic argument can exist. [computational: on M_{77}^D the fixed-point distribution over $a \neq b$ is $\{0:5390, 25:264, 35:198\}$]*

Remark (the recommended replacement: design theory). *[proved in R9-B B.4, root-verified] $\varrho = \nabla$ says the n partitions $\{\ker R_t\}_{t \in M}$ cover every pair of points: a *resolvable pairwise covering design* on n points with exactly n parallel classes. Immediately $\Sigma \geq 2n^2 - n$; and if moreover every $|F_{ab}| = 1$ the design is a resolvable linear space with $S_a = 2n - 1$ for every a . A linear space on n points whose lines split into n parallel classes is a very rigid object, so de Bruijn–Erdős and Fisher-type inequalities — with $N(v, v) \leq 1$, Proposition 2.11 and Theorem 10.12 as side conditions — are the natural tool, in place of O’Nan–Scott. Note that (Prop) is *existential*, so no ATP can be run on it at all, consistent with §9.7.*

Corrected skeleton.

$$(P) \Leftarrow [(NTS) \text{ closed}] + [(B)] + [(Prop)] + [(Q) \Rightarrow E255 \checkmark]$$

Three open legs. $(Q) \Rightarrow E255$ is Corollary 2.8; (B) newly lost its proof; (Prop) is untestable on any extension; and (NTS) moved *away*, because non-separable fibre operations are now known to exist. The Round-7 reassessment that the extension class was turning towards (P) must be down-weighted accordingly.

10.6 The nine-model benchmark set

Remark (test order matters). The filter set of §7.6 is superseded. It is now the nine models

$$\{M_5, M_9, M_{77}, M_{77}^{NT}, M_{176}, M_{496}, M_{77}^D, M_{385}^{\text{canon}}, M_{385}^R\},$$

tested in this order: M_{77}^D first, since it alone kills (T), (T^*) , (W), (D), (A), (S') , Idem-closure and both filters; then M_{385}^{canon} , which kills (T) while *passing* both filters and (A) — without it one would wrongly conclude that the filters detect (T)-failure; then M_{385}^R , the τ -independent non-separable control, which must *not* be killed. Any claim verified on the other six carries no evidence at all. Table 1 has been extended with the three new models, and Table 2 must be read with the (T) column understood as “holds in this model”, never as evidence for the law.

Remark (what Round 9 did not do). M_{77}^D and both order-385 objects satisfy $E255$. They are not counterexamples to Problem 1.1, and they do not refute (S) — all of them are extensions, hence non-simple. What they refute is a family of conjectures the campaign had erected *about* the route to (P). The genuinely new positive knowledge of this round is Theorems 10.1–10.3 and Lemma 10.4, which for the first time make non-affine models constructible at will: “build a finite $E677$ magma with prescribed collapse behaviour” is now a small constraint problem on $|B|^2$ fibre tables. That machinery, not the refutations, is what should be pointed at (Prop), at (C1) and at the search for a non-Latin magma with primitive $\langle L \rangle$.

11 Open problems and the state of the routes

11.1 The main problem

Conjecture 11.1 ((P); = Problem 1.1). Every element of a finite $E677$ magma has a left unit; equivalently $\sum_w |\text{Fix}(L_w)| = n$; equivalently $\text{tr } N = n$; equivalently every $\Theta_t = R_t L_t$ has a fixed point; equivalently every finite $E677$ magma satisfies $E255$.

The campaign’s own view of this shifted twice and is recorded honestly. After Rounds 1–2, with every known model an affine quasigroup and orders ≤ 9 exhausted, the orchestrating agent judged the evidence to point strongly to “true” — against the community’s expectation [1, 3] that a counterexample exists. That judgement was based on a biased sample (small orders, all affine). The discovery that (Q) is false — the blueprint’s own order-496 example, then our order-176 one — corrected it: the escape mechanism in those models is idempotency of the base, which routes $E255$ through a trivial channel, and it is not obviously unavoidable. The current assessment is *balanced, with a slight tilt toward a counterexample existing*, consistent with the community’s. We state this because a paper that reported only the final assessment would be concealing how the evidence moved.

11.2 Specific open questions

Problem 11.2 (smallest non-right-cancellative order). What is the least order of a finite $E677$ magma that is not right-cancellative? It is at most 77 (Theorem 7.18; it was 176 before Round 6) and at least 10: every $E677$ magma of order ≤ 9 is a quasigroup (Theorem 6.1), and orders 2, 3, 4, 6, 8 carry no model at all. (The targeted small-order SAT search of `soff_sat.py` independently returns UNSAT for $n \leq 7$.) A non-translation-invariant fibre assignment over $\mathbb{F}_5$ would give order 80 and was never searched; Lemma 7.19 now makes such assignments routine to construct, so that is the cheapest available test.

Problem 11.3 (must affine coefficients commute?). Theorem 3.1(a) is a pair of *noncommutative* equations $GF + G^2FG = 1$, $F + G^2F^2 + G^3 = 0$ for $F, G \in \text{End}(A)$. Every solution known to us — here, in [2], and in [4] — has $FG = GF$, equivalently $F = (G + G^3)^{-1}$, and only for those does $P(G) = 0$ and hence the R -module spectrum of Theorem 3.2 apply. Does a solution with $FG \neq GF$ exist? If not, the classification of affine models and the even-order spectrum become unconditional; if one exists, it could realize an order outside the $\mathbb{Z}[u]/(P)$ -module spectrum, and Theorem 3.4(b) would not apply to it. The cheapest test is a search over $\text{End}(A)$ for small non-cyclic A (e.g. $M_2(\mathbb{F}_p)$, $M_3(\mathbb{F}_2)$).

Problem 11.4 (the exceptional base). Does the exceptional translation-invariant base $x * y = 5x - 4y + c$ on $\mathbb{F}_{31}$ ($c \neq 0$) — which by Theorem 7.27 is the unique non-idempotent translation-invariant affine $E677$ base over any field — carry a fibre extension violating $E255$? Fibre size $m = 3$ is impossible even as an extension; $m = 5$ is the first size with a nonempty solution space; $m = 5$ and $m = 7$ are unclassified for general fibre tables. For *affine* fibres the answer is now complete for $m \leq 7$ (Theorem 7.36): the solution space is exactly the set of constant families. By Theorem 7.30(d) any counterexample built here must fail $E255$ on at least 31 elements, and by (e) one may assume the fibre is generated by a single violating point.

Problem 11.5 (the fibre spectrum, and idempotent-free orders). Is $S_0 = 31 \cdot S$ — equivalently, does every idempotent-free finite $E677$ magma have order divisible by 31 (Conjecture R6-B)? Or, weaker in a different direction, is $31n \in S \Rightarrow n \in S$? Either statement makes the fibre spectrum conjecture $\text{FiberSpec}(1) = S$ a theorem and empties the fibre size $m = 8$ (Theorem 7.30(b),(c)). Theorem 7.34 settles the affine case; the open part is a hypothetical idempotent-free magma that is neither affine nor an extension of the exceptional base.

Problem 11.6 (is $\text{Idem}(M)$ a subalgebra? — **answered: no**). **Refuted** by M_{77}^D : $(0, 0)$ and $(2, 0)$ are idempotent while $(0, 0) * (2, 0) = (1, 5)$ is not, and closure fails at 55 of the 121 idempotent pairs (Theorem 10.6). Equivalently the square identity fails on idempotent pairs. This was the surviving fragment of the route through Conjecture R6-B-2 (§7.4) and the successor target recommended there; it is now closed, and with it the last carrier of Conjecture R6-B-3. $\text{Fix}(e) = \text{Idem}(M)$ (Lemma 7.21) is unaffected — it is a definitional identity.

Problem 11.7 ($\Psi_1 \leq \text{tr } N$). Is $\sum_x N(x \setminus x, x) \leq \sum_x N(x, x)$ in every finite $E677$ magma? By Corollary 7.14 this implies Conjecture 11.1; it is implied by (Q), hence strictly weaker; it holds with equality in all 16 models of Table 1; and by Theorem 7.15 no proof of it can be witnessed by a term map, so any proof must be a counting argument. Its strongest available falsification design — a fibre family with $a_{-c/B} = 0$ over the exceptional base — provably does not exist for affine fibres with $q \leq 7$ (Theorem 7.36), so the search must be non-affine.

Problem 11.8 (the general-table $\{1, 3\}$ collision). Theorem 4.8 protects the diagonal by a subscript collision $D_1 = D_3$: positions 1 and 3 of (4.1) are the pairs $(y, \Lambda_y x)$ and (y, x) , and these coincide precisely at a left unit. The subscript coincidence is general; the conclusion drawn from it is not. Two questions, in increasing generality.

- (a) **Answered in Round 6, negatively.** For a *general* fibre table $\diamond_{y,x}$ — not assumed affine — does $y \diamond x = x$ force $\diamond_{y,x}$ to be left-injective? **No**: Corollary 7.8. What survives is non-constancy (Theorem 7.5), and upgrading it to left-injectivity is (Q) for a block that can be an arbitrary finite $E677$ magma (Proposition 7.7). The question is therefore not merely open, it is a restatement of the problem one level down.
- (b) Find the analogue of this self-referential instance of $E677$ at a left unit for a general finite magma, *without* passing through a congruence block — since by §7.1 every route through the block is circular. Rounds 7 and 8 have since mapped this precisely: Theorem 8.1 shows the local picture is the same over every base, Theorem 8.9 closes the entire separable family, and Corollary 9.13 appeared to close the last surviving base conditionally on (T) — but (T) is false (Theorem 10.6), so **that base is alive again** and this problem is fully open. The decisive object of §9.8 exists (§10.2) and the machinery of §10.1 is the tool to point at it.

Problem 11.9 (the transport law — **answered: false**). (T) was the campaign’s single recommended target at the end of Round 8. It is **refuted** (Theorem 10.6). The question it leaves is the one that replaced it: *what actually protects E255?* Both decisive objects satisfy *E255* although every conjecture erected to explain that has now failed, so the protection mechanism in the τ -dependent stratum is unknown. The three-instance analysis of §8.2 is independent of (T) and remains the best available handle.

Problem 11.10 (blocks versus congruences). Prove or refute (C1): every $\langle L \rangle$ -block system of a finite *E677* magma is a congruence. It holds for all affine models (Theorem 9.20) and on 31 tested models; it would upgrade “a minimal counterexample is simple” (Theorem 8.12) to “ $\langle L \rangle$ is primitive” and hand the problem to the O’Nan–Scott classification. A separating model must be non-affine, have at least 5 blocks (Theorem 9.21) and a non-regular induced action (Theorem 9.22).

Problem 11.11 (the critical lemma). Is a simple finite *E677* magma right-cancellative? Equivalently (Theorem 8.12), is a minimal counterexample to Problem 1.1 simple? The Round-8 route (A)+(T)+(C1) is dead (§10.4); the current reduction is (B)+(Prop), Theorem 10.10. To *refute* (S) it suffices to exhibit a non-right-cancellative finite *E677* magma whose left-multiplication group is primitive (Corollary 9.2(c)) — an $O(n^3)$ -checkable property, and one the construction machinery of §10.1 can now be aimed at.

Problem 11.12 (properness). (Prop): in a finite *E677* magma with $|M| > 1$, is $\varrho^* \neq \nabla$ — does the direct-collapse graph fail to be *connected*? With (B) this gives (S) (Theorem 10.10). [v7.1:] its existential weakening (Prop $_{\exists}$) — some displacement $L_b^{-1}L_a$ ($a \neq b$) fixed-point-free, i.e. the collapse graph fails to be *complete* — is necessary but not sufficient. Both are automatic for every extension (Proposition 10.11), so no model in the zoo carries any evidence; under the minimal-counterexample reduction (conditional on NTS) a violator may be taken simple non-Latin. The permutation-group route is provably silent already for (Prop $_{\exists}$) (Theorem 10.13). The design-theoretic reading of §10.5 addresses the *refutation* side: exhibiting a resolvable pairwise covering design of M by its n column kernels proves $\varrho = \nabla$, killing both readings; ruling the design out establishes only (Prop $_{\exists}$), and closing the gap from completeness to connectivity is part of the problem. Note both readings are existential over finite structure, so no ATP applies.

Problem 11.13 ((B), now without a proof route). Is ϱ^* — the transitive closure of the collapse relation — always a congruence? Both derivations died in Round 9 (§10.5), and its model evidence is worthless: in every known non-Latin magma ϱ^* is the fibration *by construction*. This is now the weakest-supported hypothesis in the whole programme, and the first thing the machinery of §10.1 should be pointed at — a τ -dependent object whose ϱ^* is strictly finer than its fibration would refute it.

Problem 11.14 (the transposed counting identity — **answered: false**). (S') $\sum_z N(z, z * x) = n$ is **refuted**: in M_{77}^D the sum takes the values $\{42, 77, 112, 147\}$ (Theorem 10.6). Its proved companion, the row-sum half of Theorem 9.29, is unconditional and survives, as does Theorem 7.9. What remains open is the question behind the target: *is there any exact E677 counting identity of the global-cancellation type beyond Theorems 7.9 and 7.12?* Both barriers of §9.7 still stand, and the refutation of (S') shows the campaign had no way to tell a true one from a false one on the evidence it had.

Problem 11.15 (localized ideal propagation). Implication (2.5) — any set closed under all left translations is closed under all right translations — is the only genuinely global, non-tautological

consequence of KEY found. Is there a localized or weighted version, e.g. one applying to a weighting rather than to a set, which would produce an *upper* bound on $e(O)$ for a restricted family of O ? By Corollary 4.2(c) it cannot be uniform over a size class.

Problem 11.16 (order 10, 12, 14; the small even orders). Does a non-affine $E677$ magma of order 10, 12 or 14 exist? Theorem 3.4 excludes commuting-coefficient affine ones (a hypothetical noncommuting affine model is not covered; none was found in the spot search over $M_2(\mathbb{F}_2)$, $M_2(\mathbb{F}_3)$, $M_2(\mathbb{F}_5)$, $M_3(\mathbb{F}_2)$). Order 10 is within reach of the symmetry-broken decomposition of §6.3.

Problem 11.17 (cycle lengths 4 and 5). Is $m(y) \in \{4, 5\}$ realizable in some finite $E677$ magma? $m(y) \in \{2, 3\}$ is impossible (Theorem 2.15(a)); $m(y) = 4$ forces $E255$ to fail, so a realization at $m = 4$ would be a counterexample; $m = 5$ has never been observed. The exclusions have *different* ranges: $m(y) = 4$ is excluded for $n \leq 10$ (orders ≤ 9 by Theorem 6.1, and the order-10 sub-case $m(0) = 4$ was exhausted, Proposition 6.2(b)), whereas $m(y) = 5$ is excluded only for $n \leq 9$ — the order-10 sub-cases $m(0) = 5, \dots, 10$ never finished and were discontinued.

Problem 11.18 (mediality without (Q)). Every commuting-coefficient affine $E677$ model is medial (Proposition 3.6), and a medial $E677$ quasigroup is affine by Toyoda’s theorem, hence satisfies $E255$. Since (Q) is false, that route cannot be completed as originally planned. Is mediality derivable from $E677$ plus left cancellation? A bounded rewrite saturation reaching 168 182 terms per side did not close it [computational, partial: R3-F].

Problem 11.19 (quantitative robustness). Is it true that in any finite left quasigroup of order $n \geq 5$, if $E255$ fails then at least 5 instances of $E677$ fail, and 5 is attained (Proposition 6.6)? A proof of the “ ≥ 1 ” case is Conjecture 11.1 itself; a proof of “ ≥ 5 ” would presumably expose the mechanism.

Problem 11.20 (an infinite left-cancellative counterexample). Proposition 5.1 predicts that $E677$ plus left cancellation does not imply $E255$, which would be witnessed by an *infinite left-cancellative* counterexample. The known infinite counterexamples (greedy, free) are not left quasigroups. Constructing one would pin down exactly what finiteness contributes.

11.3 Closed routes

Section 5.9 tabulates them, with their exact strengths. In summary, a proof of Conjecture 11.1 may **not** be: a surjectivity/injectivity argument for the state map; a marginal count on N ; a local amplification of the row-code distance bound; a transport-tree or pair-rotation pigeonhole; a parity, sign, 2-adic or Sylow-2 argument; an off-diagonal constraint $N(t, v) + N(v, t) \geq k$; a uniform bound $e(O) \leq |O||O^c|$ over a size class; or a reduction through (Q), which is false. Each of those is closed by a proof or by an explicit model. Separately, and *on weaker grounds*, an equational or quasi-equational derivation from $E677$ plus left cancellation resisted every saturation experiment we ran, but is not proved impossible. The experiments therefore *suggest* — without proving — that a proof will be global and insensitive to the parity of n , and that it will bound $e(O)$ from above on a family of sets that is not a full size class. The route that earlier drafts recommended in the same breath — “find the general-magma analogue of the $\{1, 3\}$ collision” — is now known to be circular (Corollary 7.8), and the criterion of Corollary 7.14 has taken its place as the campaign’s best-shaped unrefuted target. (Round 9 left Corollary 7.14 untouched — $\Psi_1 = \text{tr } N$ holds in all three decisive objects — while refuting the (T) programme that had overtaken it; see §10.4.)

Remark (superseded in Round 6). The record “176 is the smallest known non-right-cancellative order” stood only until Round 6: Theorem 7.18 exhibits one of order 77, which is moreover the first non-idempotent example and is certified in LEAN 4. It lies outside the θ -family scanned in §4.5 (its zero-offset fibre has $a_0 + b_0 \neq 1$), so the scoped minimality statement there is unaffected, and the order-176 magma remains the witness for everything in §4: the order-77 model satisfies (S-off) and therefore does not replace it.

12 Verification artefacts

All paths are relative to the campaign repository root. Python was run with a project virtualenv; SAT solving used `kissat` 4.0.4 [30] unless noted.

artefact	what it establishes
<code>problems/etp677/F1_tools/cp677.c</code>	constraint-propagation enumerator; complete classification $n \leq 9$ (Theorem 6.1)
<code>problems/etp677/F1_tools/verify.py</code> , <code>verify2.py</code> , <code>an9.py</code> , <code>gen2.py</code>	verification of Lemmas 2.4, 2.5, 2.6, Theorems 2.7, 2.14 and Proposition 2.11 on 18 concrete models of orders 5, 7, 7, 11, 13, 19, 31, 37, 35, 49, 55 including direct products; no violations
<code>problems/etp677/F5_scripts/enum677.py</code>	propagating backtracking enumerator
<code>problems/etp677/F5_scripts/sat677.py</code> , <code>sat_queries.py</code> , <code>sat2.py</code> , <code>sat3.py</code>	<code>kissat</code> encodings: existence; <code>not255</code> ; <code>notlatin</code> ; <code>noco10</code> with symmetry breaking; with the proved KEY and Lemma 2.6 clauses as redundant constraints
<code>problems/etp677/F5_scripts/enum_iso.py</code>	SAT enumeration up to isomorphism (blocking whole isomorphism orbits) plus per-model structure report
<code>problems/etp677/F5_scripts/saturate*.py</code>	congruence closure for the free left quasigroup + <i>E677</i> (Proposition 5.1)
<code>problems/etp677/F5_scripts/nearmiss.py</code> , <code>robust.py</code>	Propositions 6.5, 6.6
<code>problems/etp677/R3B_scripts/final_verify.py</code> , <code>constc.py</code>	affine criterion versus brute force: 20 539 cases over $\mathbb{Z}_m$, $m < 40$, all (F, G) ; 105 624 cases over $\mathbb{Z}_m$, $m < 26$, all (F, G, c) — 0 mismatches (Theorem 3.1)
<code>problems/etp677/R3B_scripts/paranoia.py</code>	re-parses the <i>printed</i> order-16 Cayley table and evaluates the raw <i>E677</i> term on all 256 pairs: 0 violations (Theorem 3.4(c))
<code>problems/etp677/R3B_scripts/extra.py</code>	order-25 model on $(\mathbb{Z}/5)^2$; the $\mathbb{F}_9$ order-9 models; exactly 2 isomorphism classes at order 16
<code>problems/etp677/R3B_scripts/analysis.py</code> , <code>mvals.py</code>	sign identity (Proposition 5.8), cycle types, idempotent counts, m -multisets, $\sum_z \text{Fix}(L_z) $

artefact	what it establishes
<code>problems/etp677/R3B_scripts/search.c</code>	independent C enumerator with KEY propagation; re-confirms $n = 2, 3, 4, 6, 8$ empty and $n = 5, 7, 9$ nonempty
<code>problems/etp677/r3a_search.py</code>	Theorem 5.7 (all 84 words of length ≤ 3); second-branch enumeration; Proposition 6.7
<code>problems/etp677/r3f_enum.py</code> , <code>r3f_n11_merged.json</code>	order-11 partial census ledger (Proposition 6.3); affine catalogue; mediality checks
<code>problems/etp677/r3f_saturation.py</code>	bounded mediality rewrite saturation
<code>problems/etp677/R5C_scripts/verify176.py</code> , <code>m176.json</code>	independent from-scratch verification of Theorem 4.1(a)–(e) and the full Cayley table: order, E_{677} over all 176^2 pairs, rows permutations, N margins, E_{255} , the value set of N , the (S-off) pairs and the off-diagonal $N(t, v) + N(v, t)$ distribution. It computes <i>no</i> subset, support-size or connectivity quantity — for those see <code>struct_check.py</code> and <code>isoperim_check.py</code>
<code>problems/etp677/R5C_scripts/blueprint_search.py</code>	$\text{GF}(p^k)$, translation-invariant base enumeration, the index formulas <code>quads</code> of Lemma 4.4, magma builder and checkers
<code>problems/etp677/R5C_scripts/bp_sat.py</code>	the SAT search that found the order-176 magma
<code>problems/etp677/R5C_scripts/minimal_scan.py</code>	Proposition 4.11 (the <i>range-limited</i> minimality scan)
<code>problems/etp677/R5C_scripts/conflict_check.py</code>	Lemma 4.7 over $\text{GF}(3), \text{GF}(4), \text{GF}(5), \text{GF}(8)$; the index coincidence $D_1(-c/A) = D_3(-c/A) = c/A$ of Theorem 4.8 for $p \in \{5, 11, 31, 41, 61, 71\}$
<code>problems/etp677/R5C_scripts/m496.py</code>	rebuild and N -analysis of the blueprint order-496 model (Corollary 4.10)
<code>problems/etp677/R5C_scripts/term_search.py</code>	<i>search</i> for the term witness in the order-496 magma, on a deterministic random sample (default 1200) of the forced pairs; it does not verify the winner on the full domain
<code>problems/etp677/R5C_scripts/term_full_check.py</code>	full-domain verification of the term witness on all 111 600 forced ordered pairs of the order-496 magma, plus the complete truth table of $w(t, v) * t = v$ by pair class
<code>problems/etp677/R5C_scripts/isoperim_check.py</code>	Corollary 4.2(c): $e(O)$ recomputed from <code>m176.json</code> ; exact maxima of $e(O) - O O^c $ at $ O = 1, 2, 3$; explicit witnesses with $e(O) > O O^c $ for every size $2 \leq s \leq 174$
<code>problems/etp677/R5C_scripts/struct_check.py</code>	the order-176 statistics that <code>verify176.py</code> does not compute: $r_t = c_v = 146$, strong connectivity of $\text{supp } N$, diameter 2, $Q = \sum N^2 = 115456$, the $ F_{pq} $ distribution, $\#\text{zeros}(N)$

artefact	what it establishes
problems/etp677/R5C_scripts/soff_sat.py	small-order falsification search for (S-off): UNSAT for $n = 3, 4, 5, 6, 7$
problems/etp677/R5C_scripts/freemagma.py	free 677 magma: E_{677} check and the failure of (S-off) at distinct generators (Proposition 4.12)
problems/etp677/r5a_search.py	the extension encoder; selftest rebuilds the order-496 model with zero violations of (4.1) and zero E_{255} failures
problems/etp677/r5a_audit.py	independent adversarial audit of the index formulas, the E_{255} -violation reduction and the derangement reduction; reported PASS
problems/etp677/r5a_bt.c	specialized backtracking solver for the extension search (searches discontinued; no classification claimed)
<i>Round 6</i>	
problems/etp677/R6A_scripts/r6a_verify.py	the whole of §7.1 and §7.2 on 13 models (orders 5, 7, 7, 11, 11, 11, 11, 13, 16, 19, 19, 176, 496): Theorems 7.2, 7.3, 7.5, Lemma 7.1, Theorems 7.9, 7.12, 7.13, the Ψ_k and BIRT_k tables, and the refutations of Theorem 7.17
problems/etp677/R6A_scripts/r6a_blocks.py	Proposition 7.7 (blocks are arbitrary) and multiplicativity of the curve masses over direct products
problems/etp677/R6A_scripts/r6a_psi.py	Proposition 7.16; the Ψ battery on the order-77 model; the $c \neq 0$ offset analysis behind Theorem 7.36
problems/etp677/R6A_scripts/r6a_e.py	Propositions 7.22 and 7.23: the closed form of e in the translation-invariant family, and the free-magma refutation (4356/4356)
problems/etp677/R6B_tools.py, R6B_qsystem.py, R6B_affine.c	Theorems 7.27, 7.29, 7.31; the fibre-ansatz reductions; the complete affine-fibre solver (deterministic DFS, no SAT)
problems/etp677/R6B_S0.py	Theorems 7.32, 7.33 (Bézout identity, coefficient-wise) and 7.34; the scan producing idempotent-free affine orders $31 \cdot \{1, 5, 7, 31\}$
problems/etp677/R6B_z7_driver.py	Theorem 7.36: the 1296-shard complete solution over $\mathbb{F}_7$ ($5.66 \cdot 10^{10}$ nodes)
problems/etp677/R6C_scripts/r6c_verify.py	Theorem 7.18: the order-77 model, built from the closed formulas and checked exhaustively (no solver, no search)
lean/proofenv/M77.lean	kernel-checked Lean 4 certificate for Theorem 7.18(a)–(c): decide only, axiom footprint $\{\text{propext}, \text{Quot.sound}\}$, with bridging lemmas certifying the encoding

artefact	what it establishes
problems/etp677/R6E_scripts/ r6e_verify.py	Lemma 7.19, Theorems 7.20 and 7.24: the 121 pair-indexed instances, the full table, and the 1050 homomorphism failures
problems/etp677/R6D_scripts/ r6d_m77nt.py, r6d_min9.py, r6d_survive.py problems/etp677/R6D_scripts/ c*.p, cm*.in	independent rebuild of M_{77}^{NT} from its written specification (no code reuse); Theorem 7.25; the $595 \rightarrow 32$ unary-term collapse of §7.6 Prover9/Mace4 and E 3.1 inputs: five discharged calibration targets, the exhausted MACE4 searches at $n = 5, 7, 9$, and the (inconclusive) failed proof attempts
problems/etp677/R6_zoo.py	Table 1: every column recomputed for all 16 models, plus the isomorphism $M_9 \cong (\mathbb{F}_9, x + (t + 2)y)$
<i>Rounds 7 and 8</i>	
problems/etp677/R7A_scripts/ r7a_*.py	the $\mathbb{F}_7(4x + 3y)$ instance system; Theorems 8.3, 8.4, 8.8; the ten-operation defective local solution and the 20/20 extensions; the global annealer and its residual diagnosis (time-boxed randomised search, no SAT)
problems/etp677/R7A_scripts/ r7b_baseindep.py, r7b_cocycle.py, r7b_thin.py problems/etp677/R7A_scripts/ r7c_quotient.py, r7c_witness.py	Theorem 8.1 on seven bases; the 196×98 cocycle system ($B_\star \equiv 0 \pmod{\text{seven primes}}$); Theorem 8.10(b),(c) $\langle L \rangle$ -transitivity and the quotient structure (Theorem 8.12); the unique-witness formula and the Φ -swap (Proposition 8.14)
problems/etp677/R8A_scripts/ r8a_*.py problems/etp677/R8A_scripts/ r8b_transport.py, r8b_delta.py, r8d_shear.py	Theorems 9.1, 9.3, 9.4, the (W) scan and the refuted conjugacy explanation; the term-witness scan (T) exhaustively to $n = 176$ plus the random-left-quasigroup control; Δ -orbits and the affine dichotomy; the shear decomposition, (T*) and the two marginals
problems/etp677/R8A_scripts/ r8f_bijsearch.py, r8f_shadow.py, r8g_sprime.py	the exhaustive $\text{depth} \leq 2$ term-pair search (6 739 216 pairs, exactly 8 bijections); (DS); the Q -matrix, the refutation of the uniqueness mechanism, and the 21 injective terms
problems/etp677/R8A_scripts/ atp/g*.in, h1..h12.in	the prover9 inputs: six saturations for the collapse statements, one for (T), twelve for the injectivity goals — and g7, the positive control , in which Theorem 2.17 is proved in 13 steps
problems/etp677/R8C_scripts/ r8c_*.py	Proposition 9.19 and the 31-model sweep (including ten new non-affine gadget extensions); machine checks for Theorems 9.20, 9.21, 9.23
problems/etp677/R8E_scripts/ r8e_*.py	Theorems 9.24, 9.26, 9.27 and the verification table of (A), (R), (R^Λ), (R^Θ), (T)

artefact	what it establishes
<code>problems/etp677/R8_invariants.py</code>	Table 2: collapse classes, primitivity, (C1), Δ -orbits, $ K $, both filters, Q -row sums and (T), recomputed for eleven models
<i>Round 9</i>	
<code>problems/etp677/R9A_scripts/r9a_D0.py,</code> <code>r9a_D0_table.txt, r9a_verify.c</code>	the decisive object M_{77}^D (Theorem 10.5): self-contained build, the stored 77×77 table, an independent C verifier reading only that table, and the negative control
<code>problems/etp677/R9A_scripts/r9a_dfs.c,</code> <code>r9a_anneal.c, r9a_family.py</code>	the exhaustive DFS at orders 25 and 55 (validated against the known counts 6 and 1680), the directed annealer that found the object, and the battery over the 22-member family
<code>problems/etp677/R9A_scripts/r9b_canon.py,</code> <code>r9b_m385R.py, r9b_verify.c</code> <code>problems/etp677/R9A_scripts/r9b_prop.py,</code> <code>r9b_strat.py</code> <code>problems/etp677/R9_audit.py</code>	the two order-385 objects and a general table checker; <code>R9A_scripts_gpt/magma677_order385.py</code> is the third-party generator refined $T3$, (Prop) probes, θ_κ , and the filter-weight stratification
<code>problems/etp677/R9_clean_check.py,</code> <code>R9_clean_check.out</code>	independent audit written for this paper: reads only the three archived tables and re-derives $E677$, $E255$, every first-order item, and every refutation of Theorem 10.6 — including the two ((S') and Idem-closure) that the reports had not tested written for the v7.2 revision: Proposition 10.9 — that M_{385}^{canon} is a clean extension over a Latin order-55 base which nevertheless refutes (T), while satisfying (A) and $ F_{ab} \equiv C_x q$ — together with the weighted-statistic identities of Definition 10.7(iii) on M_{77}^D

Acknowledgements and division of labour

The attack was organized as a set of independent “approach families” launched in parallel rounds, with the orchestrating agent forbidden to disclose one family’s content to another within a round, so that convergent discoveries would constitute cross-validation rather than contamination. Theorem 2.7 was found independently by three families before being matched against [2, Lemma 13.1(ii)]; the affine spectrum $P = \Phi_{10} \cdot Q$ was found independently and then matched against the blueprint’s Type 1/Type 2 dichotomy. The order-176 magma (Theorem 4.1) was found by a Claude Fable 5 agent using a SAT search inside the extension framework, and verified by a from-scratch re-implementation. Theorem 5.7 and Propositions 6.3, 6.4 were produced by GPT-5.6 through `codex`. Theorem 2.20 and Propositions 5.10, 5.11 were produced by the orchestrating agent. The refutation of the parity conjecture (Theorem 3.4) was produced by a Claude Fable 5 agent that had been tasked with *proving* it. One GPT-5.6 “Pro” web session ran

for 89 minutes on (Q) without producing usable output and was terminated; this is recorded because the cost of unproductive runs is part of an honest account of the method.

References

- [1] J. Bolan, J. Breitner, F. Carlini, T. Tao, et al., *The Equational Theories Project: Advancing Collaborative Mathematical Research at Scale*, arXiv:2512.07087, 2025. <https://arxiv.org/abs/2512.07087>. The finite implication $E677 \Rightarrow E255$ is recorded there as the single remaining open finite implication, with the authors tentatively conjecturing it to be false.
- [2] Equational Theories Project, *Blueprint, Chapter 13: Equation 677*. https://teorth.github.io/equational_theories/blueprint/677-chapter.html. Lemma 13.1 (basic properties, including the left-unit criterion), Lemma 13.2 (seven equivalent pointwise conditions for $E255$ at x), Lemma 13.3 (no linear counterexamples), Lemma 13.4 (no counterexamples via linear extension), §13.1 (the order-496 non-right-cancellative example), §13.2 (the free 677 magma and its violation of $E255$).
- [3] Google DeepMind, *formal-conjectures*, file `FormalConjectures/Other/EquationalTheories_677_255.lean`. <https://github.com/google-deepmind/formal-conjectures>. The file states both directions with marginal notes indicating that the existence of a finite counterexample is expected.
- [4] NSF Institute for Computer-Aided Reasoning in Mathematics, *eq677 model database*. <https://eq677.icarm.cloud>. Built on the model searcher of <https://github.com/memoryleak47/eq677>. As of consultation: every known example also satisfies Equation 255.
- [5] G. Pochuev, *finite-magma-e677-to-e255*. <https://github.com/Grisha-Pochuev/finite-magma-e677-to-e255>. An independent project on the same problem in combinatorial language; it likewise establishes that all rows are permutations and is stalled at localized obstructions.
- [6] AutoMath campaign, internal report `problems/etp677/F1_structural_report.md` (structural algebra of finite 677 magmas).
- [7] AutoMath campaign, internal report `problems/etp677/F4_finitization_report.md` (finitization of the greedy counterexample; Propositions 5 and 6).
- [8] AutoMath campaign, internal report `problems/etp677/F5_counting_report.md` (counting, parity and orbit obstructions; the linear spectrum; the saturation experiments).
- [9] AutoMath campaign, internal report `problems/etp677/R2_Qprime_codex_report.md` (the state map and its circularity; the doubly-stochastic formulation).
- [10] AutoMath campaign, internal report `problems/etp677/R3A_codex_report.md` (amplification of the row-distance bound; second-branch enumeration; perturbation search).
- [11] AutoMath campaign, internal report `problems/etp677/R3B_parity_report.md` (refutation of the parity conjecture; the affine criterion in endomorphism form; the even-order spectrum; eight auxiliary lemmas).

- [12] AutoMath campaign, internal report `problems/etp677/R3C_linking_report.md` (transport tree and pair rotation; Theorems A and B; the isoperimetric reformulation).
- [13] AutoMath campaign, internal report `problems/etp677/R3F_codex_report.md` (partial order-11 census; affine benchmark; mediality).
- [14] AutoMath campaign, internal report `problems/etp677/R4_literature_recon.md` (literature reconnaissance).
- [15] AutoMath campaign, internal report `problems/etp677/R5A_codex_report.md` (adversarial audit of the extension encoder; specialized backtracking solver).
- [16] AutoMath campaign, internal report `problems/etp677/R5C_soff_report.md` (the order-176 magma; structure theory of the translation-invariant extension family with affine fibres; diagonal protection). *Note:* the extremal isoperimetric table of its §4.2 does not list maxima (Corollary 4.2 and the remark following it), and its §5 concluding prose omits the affine-fibre hypothesis under which Theorem 5.1 there is proved.
- [17] AutoMath campaign, internal report `problems/etp677/R6A_selfref_report.md` (the general-magma analogue of the $\{1, 3\}$ collision; the Ψ identities; the Ψ_1 addendum; the e -endomorphism addendum).
- [18] AutoMath campaign, internal report `problems/etp677/R6B_spectrum_report.md` (the fibre spectrum conjecture; uniqueness of the exceptional base; the blocking theorem; S_0 as an ideal, $\text{Res}(\Phi_{10}, Q) = 31$ and $S_0^{\text{aff}} = 31 \cdot S$; the complete $\mathbb{F}_7$ solution).
- [19] AutoMath campaign, internal report `problems/etp677/R6C_codex_report.md` (the order-77 non-idempotent non-right-cancellative model).
- [20] AutoMath campaign, internal report `problems/etp677/R6D_eaut_report.md` (refutation of both halves of “ e is an automorphism”; the order-9 counterexample; the benchmark-set warning).
- [21] AutoMath campaign, internal report `problems/etp677/R6E_codex_report.md` (the diagonal-decoupling lemma; the first non-translation-invariant extension; refutation of the e -homomorphism half).
- [22] AutoMath campaign, internal report `problems/etp677/R7A_report.md` (three chapters: the mixed-idempotent base $\mathbb{F}_7$, $x \diamond y = 4x + 3y$, its occurrence theorem and the generalized Lemma 13.4; the base-independent occurrence trichotomy and the cocycle protection theorem; the quotient theorems and the non-first-order status of the conservation law).
- [23] AutoMath campaign, internal report `problems/etp677/R8A_simple_report.md` (five chapters: the congruence/block dictionary and the master reduction; the transport law (T) and the twisted-diagonal dichotomy; the shear decomposition, the cardinality shadow and the two falsification filters; the existential barrier; the Q -matrix and the injectivity barrier).
- [24] AutoMath campaign, internal report `problems/etp677/R8C_blocks_report.md` ((C1): decidability, the affine case, the non-existence of 2-block systems, and the exact circularity of the term-level route).

- [25] AutoMath campaign, internal report `problems/etp677/R8E_AR_report.md` (the legs (A) and (R): mirror forms, the merger of (R) into (C1), the Frobenius separation criterion, and the shared failure blueprint).
- [26] AutoMath campaign, harvested transcript `problems/etp677/R6_gpt_counterexample.md` (a third-party reasoning model’s counterexample search: two near-misses and a claimed impossibility argument for the single-operation defect route). *Not independently verified by this campaign*; see the remark in §9.8.
- [27] AutoMath campaign, internal report `problems/etp677/R9A_D0_report.md` (two chapters: the general extension equation, the degeneracy-transfer and rigidity theorems, and the decisive object M_{77}^D ; the audit of the repaired chain, the four-stratum theorem, refined $T3$ and the group-theoretic anatomy of (Prop)).
- [28] AutoMath campaign, harvested artefact `problems/etp677/R9_gpt_D0.md` and generator `R9A_scripts_gpt/magma677_order385.py` (a third-party reasoning model’s order-385 object, rebuilt and verified inside the framework of §10.1).
- [29] AutoMath campaign, LEAN 4 certificate `lean/proofenv/M77.lean` ($E677$, $E255$, non-idempotency and failure of right cancellation for the order-77 model, by kernel `decide`; axiom footprint `{propext, Quot.sound}`).
- [30] A. Biere et al., *Kissat SAT solver*, version 4.0.4. <https://github.com/arminbiere/kissat>.